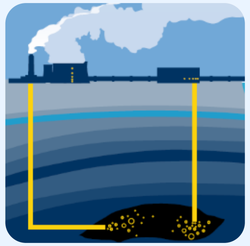

Charge Collective
Final Report

UK Power Networks

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elementenergy
an ERM Group company

Alessandro Zanre
Daisy Tyrer
Sarah Clements
Jake Howe
Celine Cluzel

Charge Collective – an assessment of flexibility services from smart on-street charging

Report Overview

- This Charge Collective report for UK Power Networks evaluates the value that flexibility services could bring to a range of stakeholders from smart on-street charge points
- It covers a qualitative assessment of barriers and opportunities of smart on-street charging, through the completion of literature review, focus groups with consumers and discussions with key stakeholders (Task 1)
- Real on-street charging profile data has been used to model the smart charging potential from on-street charge points (Task 2)
- A key part of the work involved the modelling of a selection UK Power Networks' primary and secondary substations to identify the flexibility opportunities that smart on-street charging can bring to future constrained substations (Task 3)
- For valuable opportunities identified, development of commercial models has been completed (Task 4)

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The Charge Collective report concludes that there are real opportunities for smart on-street charging to bring benefits to both the electricity network and consumers, but wide scale and reliable charge point deployment is required in order to realise these benefits

Executive Summary

Task 1: Evaluating Barriers and Opportunities

Task 2: Analysis of Charging Behaviour + Substation Selection

Task 3: Assessment of Flexibility Opportunities

Task 4: Development of Use Cases and Commercial Models

Report Conclusions

Appendix

The Charge Collective project is investigating strategies to overcome barriers to investment in public charging infrastructure – this specific study sets out use cases where on-street flexibility could be trialled

Overview of study

- UK Power Networks’ Charge Collective project aims to develop a framework to overcome barriers to investment in public charging infrastructure
- This part of the project informs the overall Charge Collective project by undertaking the following activities:
    1. Analysing barriers and opportunities for flexibility services from public charging infrastructure
    2. Analysing customer on-street charging patterns
    3. Assessing opportunities for flexibility from public charging infrastructure
    4. Selecting use cases and commercial models for on-street flexibility services
- Viable use cases identified in the study may be further assessed through real-world trials by UK Power Networks

Approach diagram

Task 1 – Evaluating barriers and opportunities
Aim: review literature on flexibility services and conduct customer research to understand attitudes towards flexible on-street charging

Task 2 – Analysis of on-street charging behaviour
Aim: review and analyse existing datasets of on-street charging events

Task 3 – Assessment of flexibility opportunities
Aim: analyse opportunities to decide which are most likely to bring network benefits, considering qualitative data from customer engagement

Task 4 – Development of Use Cases
Aim: develop a set of use cases and commercial models which could be tested in a real-world trial

Final Report

Published findings from global trials suggest that smart charging at public EVCPs will have very little impact on user experience, but it is unclear if it adds any value to CPOs

Task 1

Smart public charge points likely to have no impact on experience unless additional effort required, or insufficient information given

No impact on user experience

- No difference reported by users and no change in the energy charged in 91% of charging sessions at public on-street EVCPs in Amsterdam

Positive network impact motivates users

- EV drivers in the Netherlands ranked optimal use of sustainable energy and stability of grid as more important than financial advantages

Savings can be passed onto consumers

- Financial advantages can feasibly be offered to smart charge point users
- Further motivates EV drivers to smart charge

Mitigable risk of range anxiety

- Low charging speed caused range anxiety
- More information on charging speed reduces range anxiety for smart EVCP users in Amsterdam

Additional steps inconvenience users

- Charge point users in California were reluctant to take additional steps to participate in smart charging

Smart charging may improve the commercial case of charge point networks, but exact value is unclear

Possibility of additional economic value

- Monetisation of savings from smart charging or flexibility services may improve commercial case
- Maximising consumption of local renewables observed to lower energy costs

Unnecessary to pass savings onto consumers

- Charge point users may not need financial incentives to smart charge, improving the business case for CPOs

Limited modelling of economic value

- Value to CPOs not directly modelled or analysed in the reviewed literature

No improvement in CP occupancy

- No improvement in occupancy ratio demonstrated, which reduces the value of smart charging to CPOs

Potential increase in costs

- Smart charging observed to increase connection costs for CPOs (only when charging power increased to reduce renewable curtailment)

The literature review covered 14 reports, 7 of which included information on consumer experience, and 7 the value to CPOs. See page 22 for the full list of papers reviewed. Links to summaries of relevant reports included, other reports may be relevant to each point. EVCP = Electric vehicle charge point, DNO = Distribution Network Operator

Reviewed literature indicates that, although the high volume of data sharing results in a complex system, smart public EVCPs could offer flexibility services to the DNO

Task 1

With sufficient participation, smart public charging can offer flexibility services to a DNO

Flexibility services feasible	• Achievable including integration with local flexibility market
High suitability of public charging behaviour	• Long plug-in sessions with low charging time over times of peak demand observed, particularly in urban residential areas
Low-cost flexibility	• Cost of EV flexibility observed to be lower than that of stationary storage or renewables curtailment
High participation necessary	• Substantial proportion of EV drivers in a local area must participate for sufficient flexibility availability • Low participation also issue for forecasting
Congestion from rebound peak unlikely	• Load shifting risks a rebound peak in demand • Demonstrated that additional network congestion not caused and overall network demand smoothed

The system required for charge points to offer flexibility services is complex and requires high volume of data sharing

Technically feasible	• Demonstrated the application of a static smart charging profile • Shown integration with a local flexibility market via an aggregator, alongside multiple flexibility sources
Complex system with many stakeholders	• High volume of data must be shared between many stakeholders - open communication protocols exist, but may have to be expanded
Market framework demonstrated	• Market frameworks created and demonstrated • Observed difficulty exchanging messages despite single aggregator
Accurate load forecasting difficult	• Complex system reliant on load forecasting • Difficulty forecasting observed, particularly with low participation or few EVCPs at LV feeder level
High suitability of public charge points	• Public charging behaviour is suitable for participation in flexibility services • In some cases, charge point hardware or software upgrades may be required, as well as grid connection

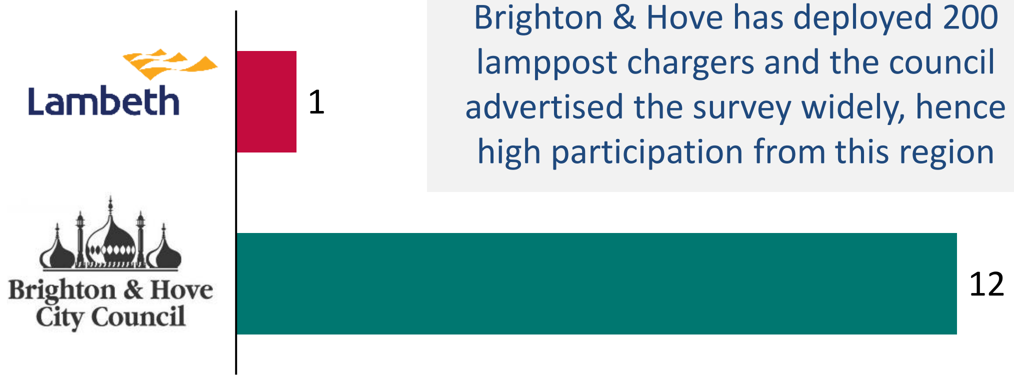
The literature review covered 14 reports, 10 of which included information on value to DNOs, and 11 practicalities and standards. See page 22 for the full list of papers reviewed.
Links to summaries of relevant reports included, may be other reports relevant to each point. Rebound peak = Peak in demand after avoided demand period

The majority of participants in the Charge Collective focus groups were: from Brighton, drive an EV and charge their vehicle at a public on-street charge point

Profile of Focus Group Participants

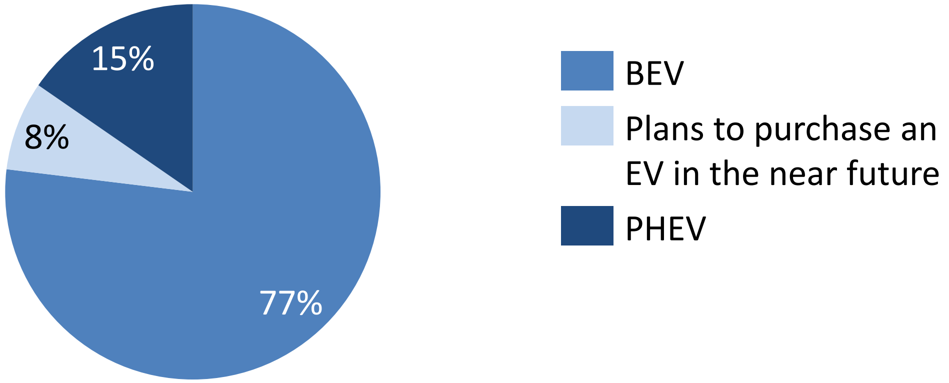
- **Three consumer focus groups** were carried out as part of the project with a total of **13 participants** from **Lambeth** or **Brighton & Hove**
- The majority of participants (10) drive a **BEV**, while 2 own a PHEV and one participant does not yet have an EV
- Most participants (8) use **public on-street charge points** for the majority of their EV charging
- No participants use a charge point in a residential car park*
- Given the status of the market, participants can all be considered 'early adopters' so might not be representative of all future EV drivers

Local Authority

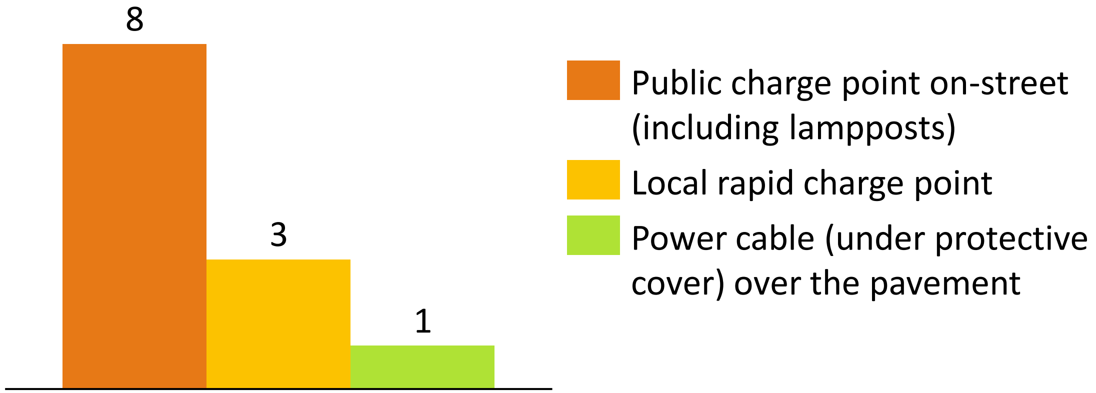


Brighton & Hove has deployed 200 lamppost chargers and the council advertised the survey widely, hence high participation from this region

EV Ownership



Main EV charging method



*The scope of UK Power Networks' research includes public, residential car parks. We anticipate concerns to be similar to those of on-street charge point users.

If public charging infrastructure were plentiful and reliable, users would be willing to smart charge provided they have sufficient information and control

Task 1

This research has provided useful insight into consumer attitudes towards smart charging at public, on-street charge points* but also an understanding of the barriers faced by current users of on-street charging infrastructure, which is still in its infancy.

Existing on-street charging infrastructure is not trusted	• Many participants had experienced failure of on-street charge points while charging • Most participants were frustrated by the low power and limited availability of on-street charge points
Smart charging exacerbates existing concerns	• Biggest concern about public smart charging was the possibility of the battery not being charged when needed and long charging times, and most participants linked this to the low power and availability of on-street charge points • If on-street charging infrastructure was plentiful, reliable and of sufficient power, users would be happy to smart charge
On-street EVCP users want information about their smart charging session	• Most participants viewed practical information about their smart charging session, such as charging time, as essential • Other information, such as quantification of renewable energy consumed, were viewed as interesting but not essential
Level of control and assurance needed at smart public EVCPs	• All participants were willing to smart charge at public charge points, as long as they could either opt in or opt out • A guarantee of a minimum percentage charge from smart charging session was a prerequisite for participating in public smart charging
Public EVCP users motivated by renewable energy	• Participants consider optimal use of renewable energy as a more important advantage of smart charging than cost savings • Short term cost savings were considered more important than long-term electricity price reduction

* The scope of UK Power Networks' research includes public, residential car parks. We anticipate concerns to be similar to those of on-street charge point users. Two participants discussed their use of local, destination car parks. It was negatively viewed, largely due to parking restrictions.

Network Operators should plan for flexibility services from public chargers, but CPO engagement is required to ensure infrastructure is sufficient to allow for smart charging

Task 1

- Smart charging is deemed **advantageous** by public charge point users, and it is likely that smart public charge points will be used by customers
- UK Power Networks can plan for **flexibility services from public charging infrastructure** reducing the impact of charging on the grid in the future

- **Existing problems with on-street charging infrastructure** must be solved before consumers will be willing to smart charge
- **Communication with charge point operators** is needed from the DNOs to ensure that charging infrastructure is sufficient and reliable, and payment systems are clear before smart charging is integrated

- On-street charge point users want charging infrastructure capable of **sufficient power** for charging in reasonable timeframes (generally 11kW or above)
- This will in turn put **additional pressure on the network** and increase the need for smart charging from public charging infrastructure

DNO = Distribution Network Operator

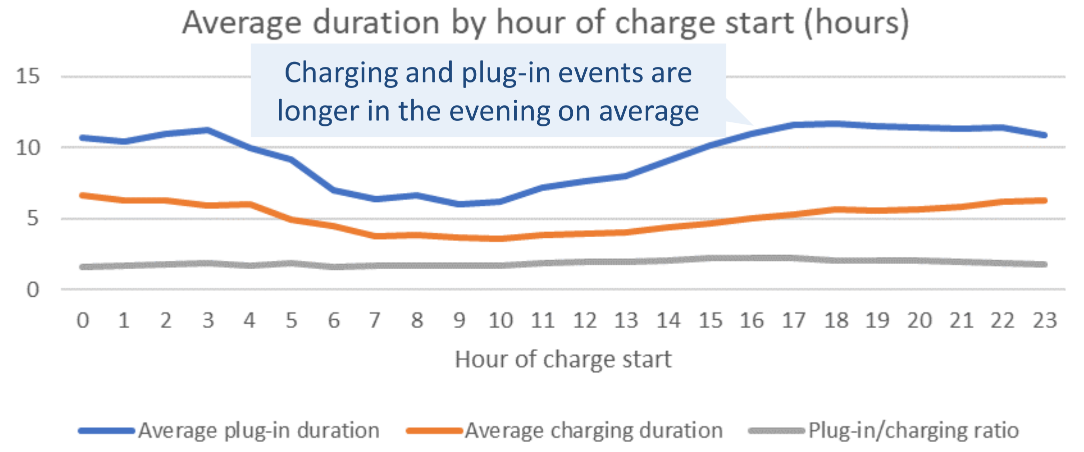
Analysis of charging event data suggests there is potential for demand to be shifted using on-street smart charging

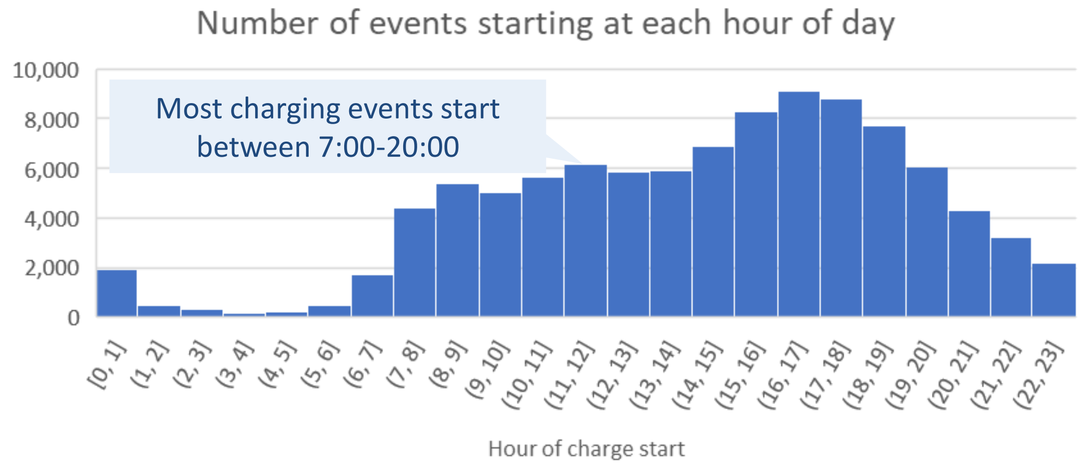
Task 2

Findings from charge event data

- A sample of charging event data containing 100,000 events from 1,300 charge points from a large charge point operator has been provided for March 2019 – March 2021
- The most common times for charging events to start are between 14:00 and 20:00
- Events starting between these times have an average plug-in duration of > 10 hours, while charging duration is ~5 hours
- Underlying domestic load on the distribution network tends to peak in the early evening – therefore there is potential to shift on-street charging demand away from this time and into the overnight period to minimise peak demand
- On-street charging demand shifting may not be as effective on assets dominated by industrial and commercial loads, which tend to peak during the day, while domestic EV charging loads peak in the evening

Variation in charging behaviour by hour of day





Study substations have been selected based on several criteria of interest, including high predicted on-street charging demand and capacity breaches in next 10 years

Task 2

Substation selection process

- A pair of primary and secondary substations from 5 distinct use cases were analysed in this study
- Use cases were selected to be representative of the range of EV uptake, reliance on on-street charging, and network loading conditions across UKPN's network. Factors captured included:
  - Level of off-street parking access
  - Rural, suburban, and urban locations
  - Substations with predominantly domestic and predominantly I&C customers
- Within each of these use cases, substations were selected which are expected to show constraints in the next 10 years and with high levels of on-street residential charging demand

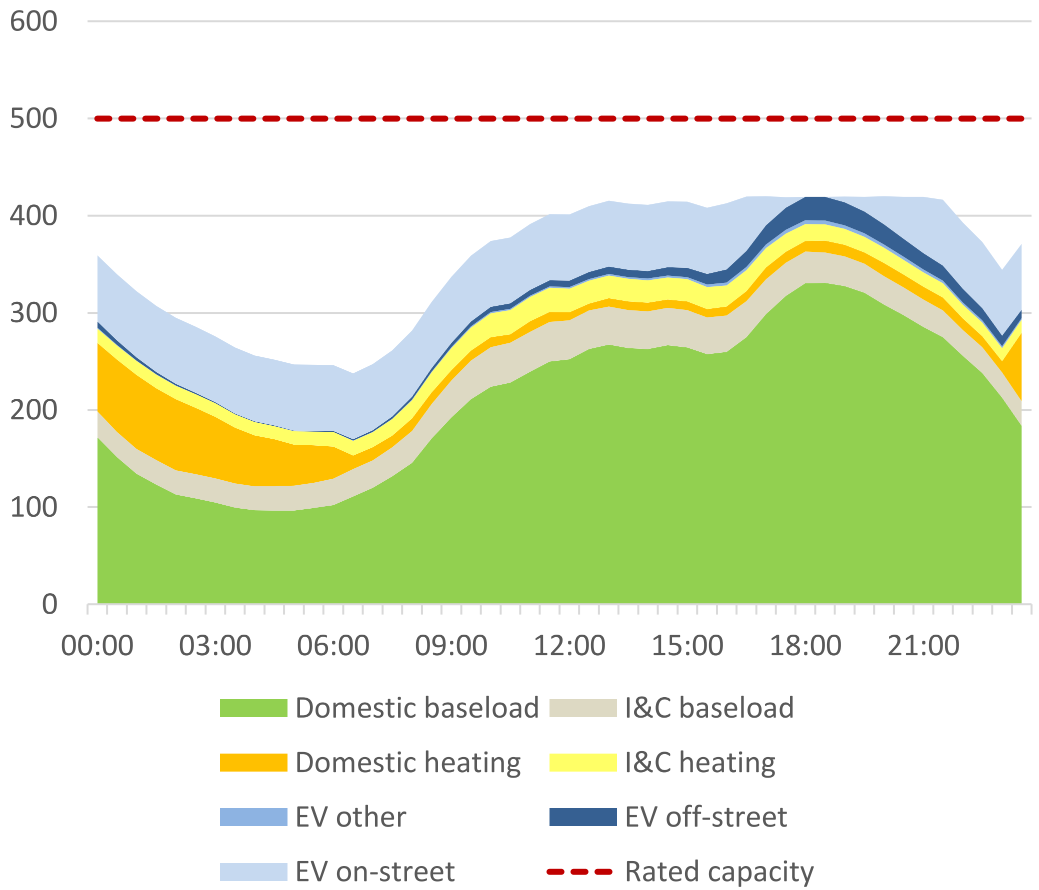
Substation locations

Constraints can be mitigated for the foreseeable future in locations where peak load growth is dominated by on-street charging and there is capacity to shift this demand

Task 3

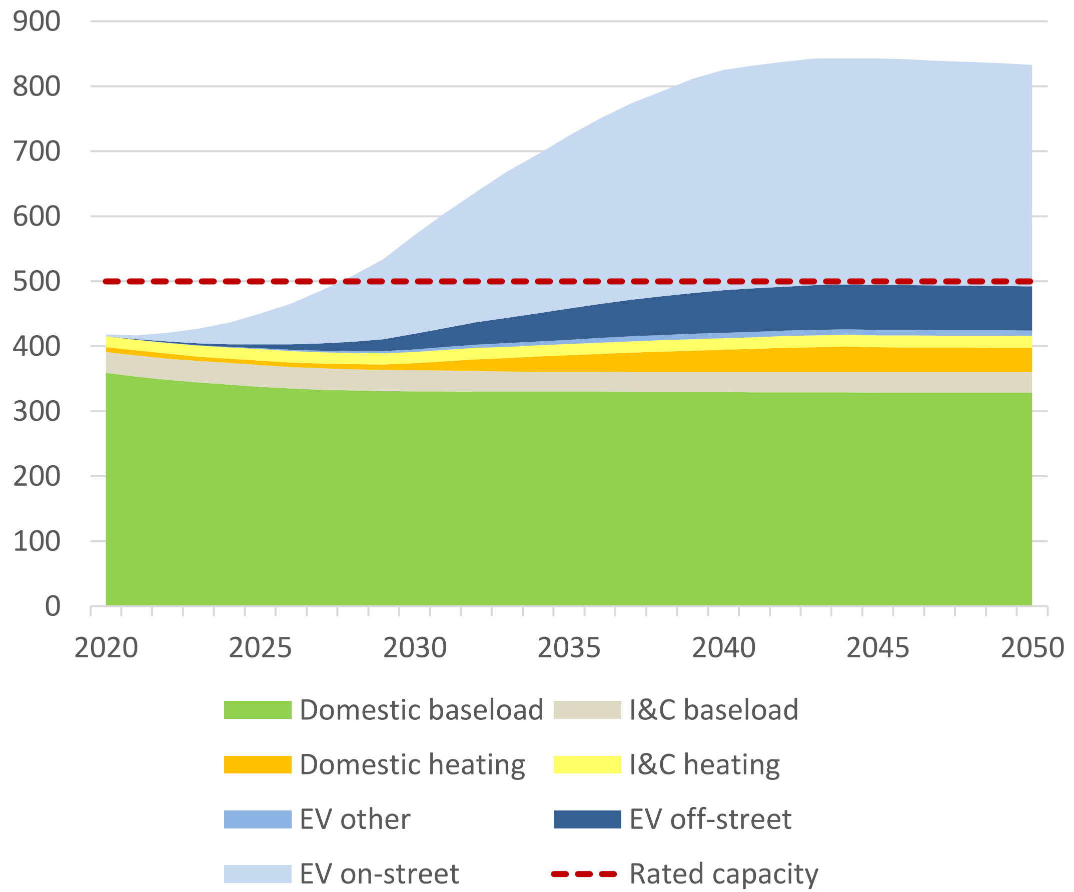
Profiles shown for London Central secondary substation

Peak day demand profile with smart on-street charging, 2030 (kW)



- Peak demand reduction of 152 kW achieved using smart on-street charging
- Smart on-street charging avoids network constraints

- Peak half hour winter load each year split by demand class (kW)



- As peak load growth is dominated by on-street charging, constraints can be managed by on-street smart charging into the future

Constraint mitigation from on-street smart charging is most feasible in urban locations with high levels of on-street charging demand

Task 3

Use case Substation archetype	Firm / rated capacity	Peak demand reduction in 2030	Key outputs	Potential for on-street smart charging
Brent secondary **Urban, mostly I&C**	500 kVA	9.2 kW	• Dominated by I&C loads • Very little potential for on-street smart charging up to 2030	
London Central secondary **Urban, mostly domestic**	500 kVA	152 kW	• Large peak in on-street charging in the evening • Spare capacity available overnight so demand can be shifted • Demand growth is dominated by on-street EV charging • On-street smart charging has potential for long-term constraint mitigation	
London Central primary **Urban, mostly domestic**	57.3 MVA	5.7 MW	• Overnight EV bus charging limits how much on-street charging demand can be shifted into overnight period • On and off-street charging flexibility could be capable of mitigating constraints up to 2035	
Brighton Central secondary **Urban, mostly domestic, night heating**	500 kVA	70 kW	• Night storage heating demand limits shifting of on-street demand into the overnight period	
Brighton Central primary **Urban, mostly domestic**	21.7 MVA	4.2 MW	• On-street smart charging can delay capacity breaches but underlying demand is high • On and off-street charging flexibility could be capable of mitigating constraints up to 2033	
Brighton Suburban secondary **Suburban, mostly domestic**	500 kVA	34 kW	• Off-street charging flexibility could help to mitigate constraints	
Rural secondary **Rural, mostly domestic**	50 kVA	9.3 kW	• High share of off-street parking means there is much more off-street than on-street charging demand	

I&C: industrial and commercial

Urban substations with high on-street demand and overnight capacity were found to be the sites with most opportunities to benefit from on-street flexibility

Shortlist filtering process

10 shortlisted substations

Can constraints be managed using smart charging?

No → **Substation is not considered further**

Yes ↓

Qualitative assessment of on-street smart charging potential:

- Parking suitability
- Charger power and availability
- Smart charging acceptance required
- Likely size of benefits

Most suitable substations are selected for commercial model analysis

Summary of findings

- Constraint management using smart charging was not possible on the Brent substations, as these were dominated by I&C loads
- The remaining urban substations (London Central and Brighton Central primaries and secondaries) consistently showed more potential for smart on-street charging benefits on qualitative assessment than the more rural substations
- As a result 4 substations were chosen for commercial model analysis:
  - London Central secondary
  - London Central primary
  - Brighton Central secondary
  - Brighton Central primary

Different commercial models are applicable to the substations being studied

Task 4

Commercial models considered

- **Timed connection**: CPOs can apply to only load the network at specific times. Analysis of charging event data has shown that demand can be shifted so this option is feasible, however if there is insufficient headroom or vehicles are only plugged in during peak times then not all demand can be shifted in this way
  - **Need to consider the minimum viable scale:** While CPOs can benefit from a timed connection on a single charger, network constraint mitigation is only feasible if many chargers are connected on a timed basis. This should be assessed in area planning, in line with the Charge Collective overall approach
- **Flexibility procurement**: UK Power Networks may agree to connect CPOs even if this could lead to a breach in firm/rated capacity and procure flexibility services to create sufficient headroom. Requires active management of the network by the DNO and paying for flexibility services when load would otherwise exceed capacity, but can be used in cases where demand cannot be completely shifted away from peak times

These results could be used to complete a cost-benefit analysis of managing constraints using commercial models considered compared to upgrading network capacity, or using price signals

Commercial models recommended for each substation

Use case Substation archetype	Level	Commercial model
London Central **Urban, mostly domestic**	Secondary	**Timed connection** due to high capacity in overnight period
London Central **Urban, mostly domestic**	Primary	Some capacity in overnight period (limited by EV bus charging) so a combination of **timed connection and flexibility procurement** may be suitable
Brighton Central **Urban, mostly domestic, night storage heating**	Secondary	Overnight storage heating means timed connections are not feasible so **flexibility procurement** would be required
Brighton Central **Urban, mostly domestic**	Primary	**Timed connection** due to high capacity in overnight period

Findings from this study can be used to inform further trials of on-street smart charging

Task 4

Recommendations from these findings

- **Assess ability to use on-street smart charging to shift load in the real world**
  - This could be done in the SmartSTEP trial[1]; even if shifting of load in the trial does not lead to mitigation of constraints this could prove its practical feasibility as a peak load reduction technique
  - Include assessment of the level of scale required/coordination of infrastructure needed to provide smart charging value
  - Further consumer research to understand what charge point tariff is required in order for consumers to participate
  - The financial incentive needed for CPOs/aggregators to shift on-street charging load could also be assessed
  - Cost and effort of alternative options (e.g. flexibility procurement, active network management) could be assessed to determine possible cost savings from on-street smart charging
- **Further study of the London Central and Brighton Central use cases**
  - If expected substation upgrade costs are available, the value of deferring network reinforcement could be assessed

This Charge Collective work, combined with the SmartSTEP trial[1] will enable the value of flexibility services to be assessed from all stakeholder perspectives – see matrix on the right

Key Themes	Charge Collective	SmartSTEP Trial	Post-SmartSTEP
Consumer Acceptance	Researched	Will be trialled	*Should SmartSTEP be successful from a consumer and commercial perspective, the focus of future work will be on the technical aspects for wider rollout and adopting industry standards*
Opportunity for Peak Load Shift	Modelled	Can be analysed based on real data	
Financial Value (for DNO, CPO, Consumer)	Commercial models explored	Can be analysed based on real data	
Technical Feasibility[2]	No work completed	Aspects will be trialled	More work needed

1 – SmartSTEP is a BEIS funded project, where 90 chargers deployed in Brent (as part of the STEP project) will be upgraded to a smart charging solution in Autumn 2021
2 – DNO communication to DCC

Executive Summary

Task 1: Evaluating Barriers and Opportunities

Literature review

Summary

Deep Dive – key projects and reports

Conclusions

Focus groups

Task 2: Analysis of Charging Behaviour + Substation Selection

Task 3: Assessment of Flexibility Opportunities

Task 4: Development of Use Cases and Commercial Models

Report Conclusions

Appendix

Summary of relevant trials – Trials with published literature on the results

Included in our literature review

Task 1

	Name	Organisation	Region	Study start/end	EVCP location	Brief description
1	Flexpower	• Amsterdam University of Applied Sciences • Elaad NL • Vattenfall	Amsterdam	March 2017 – March 2020	On-street	Project to test, improve and scale on-street smart public charging infrastructure. Charge stations limit power available at peak hours.
2	InterFLEX	• Enexis • TNO • ElaadNL	Eindhoven	January 2017 – January 2019	On-street, car park	As part of a larger study to investigate the technical and economic potential of local flexibilities, the Dutch demonstrator explored the use of public smart charging as flexibility on the distribution network. Charging infrastructure in a car park of a housing association.
3	**Demonstration of PEV Smart Charging and Supporting Grid Objectives project**	UCLA Smart Grid Energy Research Center	Los Angeles	2017 - 2018	Car park	Development and deployment of an EV charging system in Santa Monica. Demonstrated smart charging to optimise solar consumption, V2G and V2B in a car park.
4	smart solar charging Regio Utrecht	• Utrecht Council • LomboXnet • Last Mile Solution • ElaadNL	Utrecht	June 2015 - Ongoing	Public spaces, on-street	Project to develop and roll-out an AC bidirectional public charge point. A number of AC bidirectional public charge points have been rolled out in Utrecht, with the aim of balancing the grid and maximising solar consumption. Charge points can be used by all EVs, but flexibility services only available from fleet of shared EVs.

Summary of relevant trials – Ongoing trials with no published literature on the results

No literature to report on

Task 1

	Name	Organisation	Region	Study start/end	EVCP location	Brief description
5	GREEN CHARGE	European consortium		Spring 2019 - Ongoing	Car park	Three demonstration pilots around Europe, with the aim of developing a pre-bookable smart charging system. Oslo pilot trialling smart charging solution to balance local network, considering charging infrastructure in a housing association car park.
6	aVEnir	Enedis		July 2019 - Ongoing	Destination	Project led by French distribution network operator Enedis, aiming to test peak smoothing, smart charging and V2G from 250 chargers in malls and workplaces.
7	OSCD Orchestrating Smart Charging in mass Deployment	Austrian Institute Of Technology; Delft University of Technology; ElaadNL		2018 - Ongoing	Modelled	Project to research the idea that aggregated EVs can deliver required distribution network flexibility, though simulation of public, semi-public and private EV charging infrastructure, using real-life charging data.
8	**Smart charging in practice**	ElaadNl, Enexis, SunProjects, Stichting Limburg Elektrisch and Stichting Brabant		2018 - Ongoing	Unclear where the charge points are located	Project to demonstrate charging infrastructure in combination with local storage of solar energy. Fleet of shared EVs used for the demonstrator project.

Barriers and opportunities identified in literature

Key: Mostly opportunities identified — Balance of barriers and opportunities, solutions identified — Mostly barriers identified

Task 1

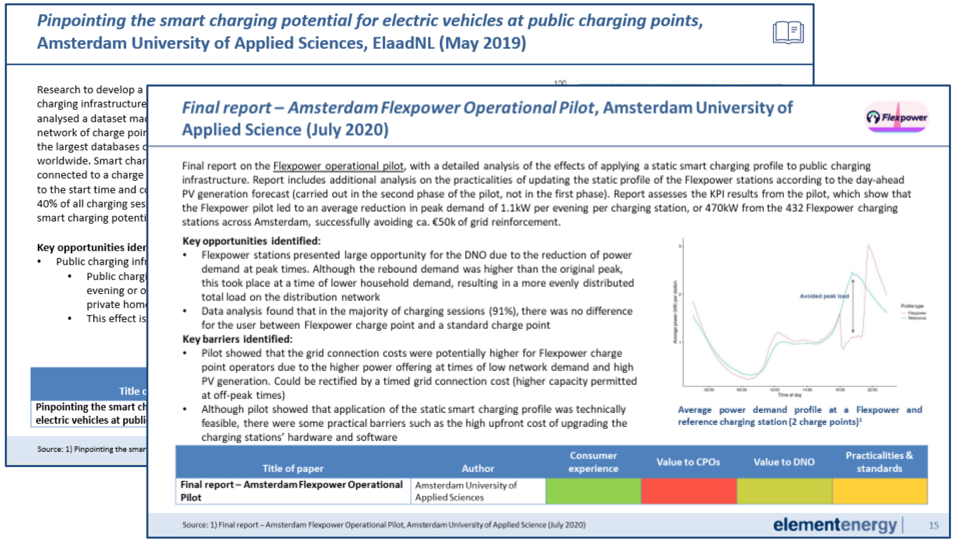
Title of paper	Trial	Author	Year	Region	Consumer experience	Value to CPO	Value to DNO	Practicalities & standards
SEEV4-city Flexpower 1: analysis report of the first phase of the Flexpower pilot	**1**	Amsterdam University of Applied Sciences	2019	Netherlands				
Behaviour research report Flexpower	**1**	Amsterdam University of Applied Sciences; ElaadNL	2020	Netherlands				
Final report - Amsterdam Flexpower Operational Pilot	**1**	Amsterdam University of Applied Sciences	2020	Netherlands				
D7.6 Lessons learned to draw business models in use cases NL#2 and NL#3	**2**	Enexis	2019	Netherlands				
D7.7 Raw demonstration results based on the KPI measurements of the Dutch demo	**2**	Enexis	2019	Netherlands				
Demonstrating Plug-in Electric Vehicles Smart Charging and Storage Supporting the Grid	**3**	UCLA Smart Grid Energy Research Centre (SMERC)	2018	California				
Smart Solar Charging: Bi-Directional AC Charging (V2G) in the Netherlands	**4**	ElaadNL	2017	Netherlands				
Maximising EV integration in LV-grids using the Universal Smart Energy Framework	**4**	Stedin	2017	Netherlands				
The Impact of Transition to Shared EVs on Grid Congestion and Management	**4**	Brinkel et al.	2020	Netherlands				
Infrastructures de recharge pour véhicule électrique (Charging infrastructure for electric vehicles)	No linked trials	CODA Stratégies	2019	France				
Note de position sur le smart charging et le V2X (Position paper on smart charging and V2X)		Colombus Consulting and Avere France	2020	France				
Pinpointing the smart charging potential for electric vehicles at public charging points		Amsterdam University of Applied Sciences, ElaadNL	2019	Netherlands				
Implementing Open Smart Charging		Element Energy for International ZEV Alliance	2019	Global				
Practical deployment of electric vehicle flexibility		Jedlix, USEF Foundation	2020	Netherlands				

A thorough literature review has been completed – key research is presented in detail, before concluding with a general overview of the barriers and opportunities found

Task 1

Deep dive – key projects and reports

- The next section of the report has key project summaries followed by detailed literature review slides
- Each slide includes a brief summary of the literature and key barriers and opportunities identified, and a summary of the barriers and opportunities



Barriers & Opportunities

Barriers and opportunities identified in literature have been reviewed and summarised across all four key themes:

Consumer Experience

Value to DNOs

Value to CPOs

Practicalities & Standards*

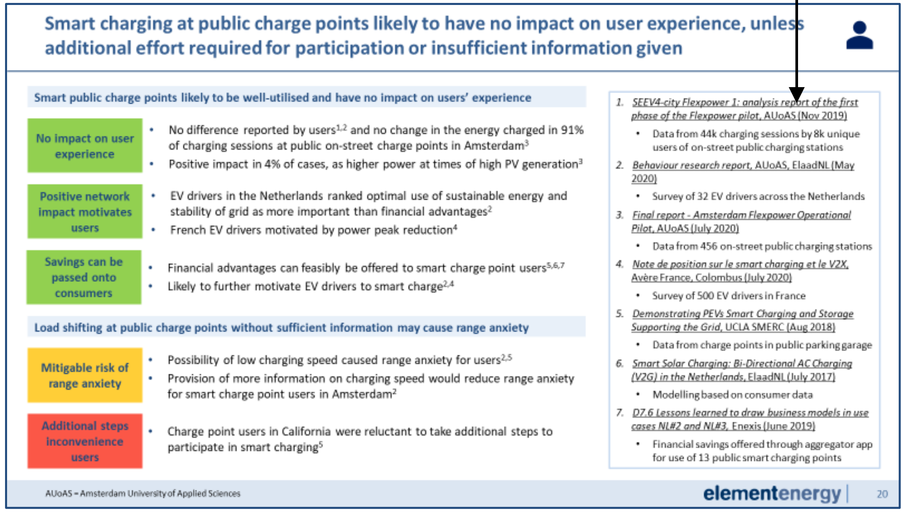
Links to dedicated slide

Key barriers and opportunities listed

Opportunity

Barrier with solution, opportunity with caveat

Barrier



Note: No UK specific trials or UK-focused literature was found, highlighting the need for the Charge Collective research and trial to understand regional differences

*Practicalities & Standards = Practical considerations for the implementation of the system e.g. technical feasibility or data exchange structure. Includes communication and technology standards

Executive Summary

Task 1: Evaluating Barriers and Opportunities

Literature review

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Deep Dive – key projects and reports

Key literature from projects

Other literature (not linked to projects)

Conclusions

Focus groups

Task 2: Analysis of Charging Behaviour + Substation Selection

Task 3: Assessment of Flexibility Opportunities

Task 4: Development of Use Cases and Commercial Models

Report Conclusions

Appendix

KEY PROJECT: Flexpower pilot, Amsterdam

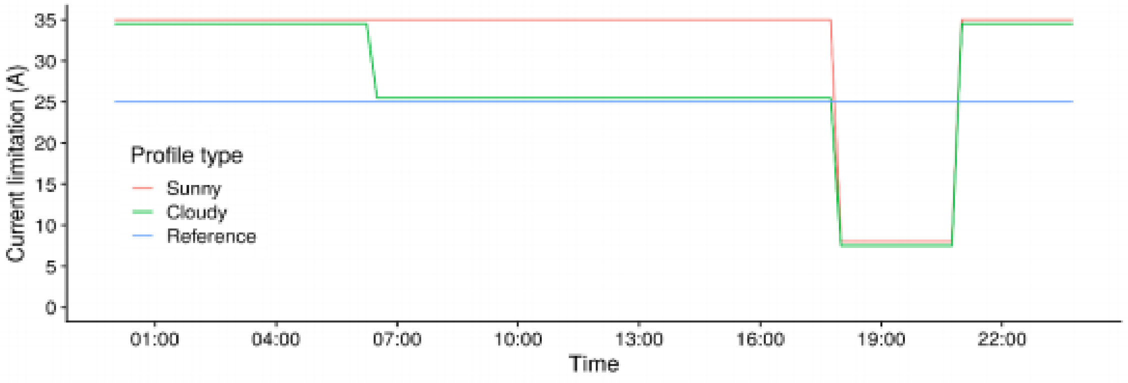
Task 1

Project summary

Project aim: Pilot aimed to increase its network of public charging stations, without overloading the distribution network. It also aimed to trial and roll out a smart public charging solution

- Vattenfall and the City of Amsterdam launched Flexpower in 2017 alongside the distribution network operator Liander, ElaadNL and the Amsterdam University of Applied Sciences
- Pilot rolled out a network of **public on-street charging stations** with a **static smart charging profile**
  - Power from charge points was limited between **7-8am** and **6-9pm (peak times)**
  - If local production of solar power forecast to be high, the power available from the charging stations outside of peak times was raised to maximise consumption
- First phase of the project was carried out between March 2017 and August 2018 with a small number of charge points
  - Phase 1 was used to demonstrate the concept of limited power at peak times
- Main phase, with **456 charging stations** operated by Vattenfall, ran between May 2019 and May 2020
  - Weather-dependent smart charging profile was introduced
- Flexpower stations with static smart charging profile still operational

Time-dependent current available from Flexpower charging stations under sunny and cloudy conditions compared to a regular public charging station in Amsterdam (Reference)[1]



Reviewed project reports

- *SEEV4-city Flexpower 1: analysis report of the first phase of the Flexpower pilot*, Amsterdam University of Applied Sciences (November 2019)
- *Behaviour research report Flexpower*, Amsterdam University of Applied Sciences, ElaadNL (May 2020)
- *Final report - Amsterdam Flexpower Operational Pilot*, Amsterdam University of Applied Sciences (July 2020)

SEEV4-city Flexpower 1: analysis report of the first phase of the Flexpower pilot, Amsterdam University of Applied Sciences (November 2019)

Task 1

Report published on the **findings from the first phase** of the Flexpower pilot (March 2017 – August 2018). First phase involved a static smart charging profile on 52 on-street public charging stations around Amsterdam. During Phase 1 of the trial, data was collected on ca. 44k unique charging sessions carried out by over 8000 users. Analysis of the data led researchers to conclude that the **static smart charging profile did not negatively impact EV users**, and that it was **possible to shift the EV charging peak** to later in the evening, improving the utilisation ratio of the low voltage electrical network and avoiding grid reinforcement investments.

Key opportunities identified:

- Trial found that the application of a static smart charging profile to the public charge points was technically feasible
- The smart charging profile has a positive impact on the experience of the charge point users as EVs could be charged faster during off-peak hours than at regular charging stations in Amsterdam (higher charging power permitted)
- Data suggested there was very little inconvenience to charge point users, as experience similar in both smart charging and non smart charging cases

Flexpower charging station[1]

Key barriers identified:

- A large peak in demand is seen once the restricted power is lifted, which could result in congestion issues. Report suggests that a smooth transition in the power profile might avoid creating a second, even higher, consumption peak
- No evidence that smart charging improves the charge point occupancy ratio (time spent charging vs plugged in, not charging), which reduces the value of the scheme to charge point operators

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
SEEV4-city Flexpower 1: analysis report of the first phase of the Flexpower pilot.	Amsterdam University of Applied Sciences				

Source: 1) Vehicle2Grid and Smart Charging: Learnings from 6 pilot cases, Amsterdam University of Applied Sciences (November 2019)

Behaviour research report, Flexpower, Amsterdam University of Applied Sciences, ElaadNL (May 2020)

Task 1

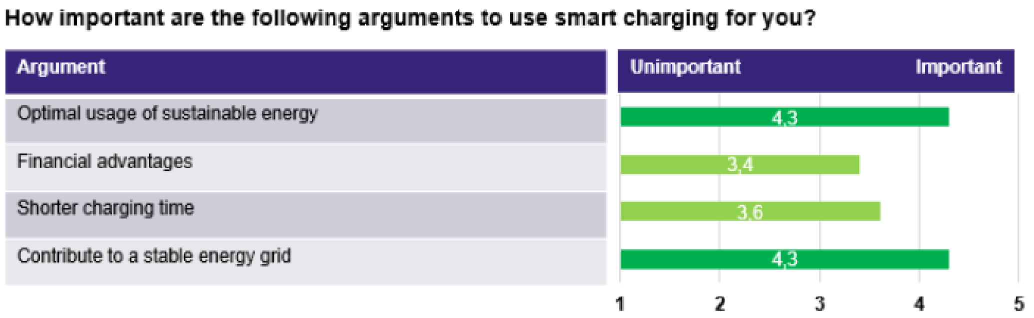
Report examining the results of surveys and interviews with EV drivers and BEV taxi drivers – interview with EV drivers that have used Flexpower stations, survey of taxi drivers that have used Flexpower stations, and survey with EV drivers across the Netherlands. Research aimed to examine awareness of the functioning of Flexpower charge points and users' motivations around using or avoiding the Flexpower stations. The sample size for both groups were small and some research was cancelled due to the pandemic. The report recommends further research could be undertaken. The research found that there is high awareness of Flexpower within the group of EV drivers, but lower awareness amongst taxi drivers. Report recommends that in the future, users of the charge point should be given information on the exact charging speed of the session to reduce inconvenience.

Key opportunities identified:

- Surveys showed that EV drivers were positively motivated to use the Flexpower charge points by the idea of contributing to network balancing
- Majority of respondents showed that smart charging experiences made no difference to the charging experience
- Majority of respondents ranked optimal use of sustainable energy and stability of the grid as more important than financial advantages

Key barriers identified:

- Respondents in all groups identified that lack of information around the charging speed of the session when plugging-in hindered their charging experience.
- This made taxi drivers and EV drivers concerned that their EV battery would not charge quickly enough, and motivated taxi drivers to avoid the Flexpower stations.



Results from survey with EV drivers across the Netherlands. Respondents asked to rate the importance of particular arguments to use smart charging[1].

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Behaviour research report Flexpower	Amsterdam University of Applied Sciences; ElaadNL				

Source: 1) *Behaviour research report*, Flexpower, Amsterdam University of Applied Sciences, ElaadNL (May 2020)

Final report – Amsterdam Flexpower Operational Pilot, Amsterdam University of Applied Science (July 2020)

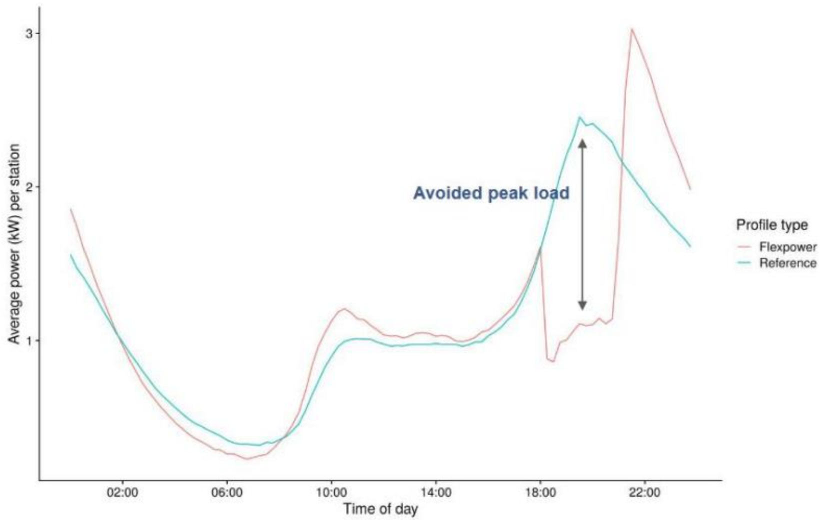
Final report on the Flexpower operational pilot, with a detailed analysis of the effects of applying a static smart charging profile to public charging infrastructure. Report includes additional analysis on the practicalities of updating the static profile of the Flexpower stations according to the day-ahead PV generation forecast (carried out in the second phase of the pilot, not in the first phase). Report assesses the KPI results from the pilot, which show that the Flexpower pilot led to an average reduction in peak demand of 1.1kW per evening per charging station, or 470kW from the 432 Flexpower charging stations across Amsterdam, successfully avoiding ca. €50k of grid reinforcement.

Key opportunities identified:

- Flexpower stations presented large opportunity for the DNO due to the reduction of power demand at peak times. Although the rebound demand was higher than the original peak, this took place at a time of lower household demand, resulting in a more evenly distributed total load on the distribution network
- Data analysis found that in the majority of charging sessions (91%), there was no difference for the user between Flexpower charge point and a standard charge point

Key barriers identified:

- Pilot showed that the grid connection costs were potentially higher for Flexpower charge point operators due to the higher power offering at times of low network demand and high PV generation. Could be rectified by a timed grid connection cost (higher capacity permitted at off-peak times)
- Although pilot showed that application of the static smart charging profile was technically feasible, there were some practical barriers such as the high upfront cost of upgrading the charging stations' hardware and software



Average power demand profile at a Flexpower and reference charging station (2 charge points)[1]

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Final report – Amsterdam Flexpower Operational Pilot	Amsterdam University of Applied Sciences				

KEY PROJECT: Interflex - Dutch Demonstrator, Eindhoven

Task 1

Interflex project and aims of the Dutch demonstrator

Interflex was a large study that investigated the interactions between stakeholders and the technical / economic potential of local flexibilities to relieve existing or prevent future grid constraints. The Dutch demonstrator was one of 6 across 5 European countries*.

Project aim: Dutch demonstrator aimed to explore how a DSO could use public smart charging alongside a stationary battery system to manage congestion on the distribution network.

Project stages/use cases:

1. NL#1: Deployment of a stationary battery for flexibility delivery
2. NL#2: Installation and deployment of smart charging EV stations to provide flexibility to the DSO
3. NL#3: demonstration of an integrated distribution network flexibility market with flexibility from public charging infrastructure or stationary battery

Reviewed project reports

- *D7.6 Lessons learned to draw business models in use cases NL#2 and NL#3, Interflex,* Enexis (June 2019)
- *D7.7 Raw demonstration results based on the KPI measurements of the Dutch demo, Interflex,* Enexis (December 2019)

Dutch demonstrator project summary

- The project was launched by Enexis (Dutch DSO), TNO (research and development organisation) and ElaadNL (knowledge centre on smart mobility) in January 2017
- The 13 smart charging points were located in the **car park of an apartment building** in the Strijp-S area of Eindhoven and additionally outside the building **on-street**
  - If not participating in the demonstrator, EVs could carry out ‘regular’ charging at the smart charging points
- The smart charging points could respond to flexibility requests from the local DSO via a **commercial aggregator** (Jedlix)
  - The aggregator could, upon receiving a flexibility request from the DSO, adapt the charging schedules on the EV charge points, thus providing the flexibility requested
  - Consumers signed up through the Jedlix app and were offered savings of 5 cents/kWh charged when participating in a flexibility request
- The final stage (NL#3) of the project involved testing the use of the **EV flexibility alongside stationary battery storage systems** that could also respond to the DSO’s flexibility request
  - Response to flexibility requests determined on the marketplace
- The Dutch demonstrator was completed in January 2019

Source: 1) *Flexible Charging of Electric Vehicles: Results of a large scale smart charging demonstration*, Amsterdam University of Applied Sciences (June 2020)

*Note: Dutch demonstrator only one of the 6 with relevance to smart public charging infrastructure. Others included investigation of microgrids, heating systems and stationary storage

D7.6 Lessons learned to draw business models in use cases NL#2 and NL#3, Interflex, Enexis (June 2019)

Report on the implementation of the second and third use cases of the Dutch demonstrator of the Interflex project. **Use case 2 (NL#2)** involves the implementation of the systems necessary to provide flexibility to the DSO (Enexis) through adjusting the charging profiles of EVs at public charging infrastructure. **The third use (NL#3)** case involves the demonstration of an integrated distribution system flexibility market where flexibility can be provided by two sources – a battery storage system or smart charging of EVs from public charging infrastructure. The Universal Smart Energy Framework (USEF) was used as the flexibility market framework. Preliminary results found that the number of EV charging sessions was not adequate to solve the congestion problem. Report suggests that investigation into incentives to increase engagement of EV drivers in using smart charging points is necessary.

Key opportunities identified:

- Demonstrator showed that it was technically feasible to establish smart public charging points and an integrated flexibility market with more than one flexibility source

Key barriers identified:

- Observed lower than expected usage of smart charging points. In particular, the report notes that the on-street charge points had noticeably lower usage than those within the car park, possibly due to high parking charges on-street
- Report notes a number of issues using open protocols such as difficulties communicating between different protocols, making it hard to match network congestion points with the relevant charge point
    - Time had to be spent expanding the communication protocols to ensure players could access necessary information flows
- High cost of hardware upgrades, including reinforcement to ensure sufficient capacity for the charge points
- Risk of high rebound peak which may result in another congestion that must be solved

Interflex Dutch demonstrator smart charging points within the apartment building car park[1]

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
D7.6 Lessons learned to draw business models in use cases NL#2 and NL#3, Interflex	Enexis				

D7.7 Raw demonstration results based on the KPI measurements of the Dutch demo, Interflex, Enexis (December 2019)

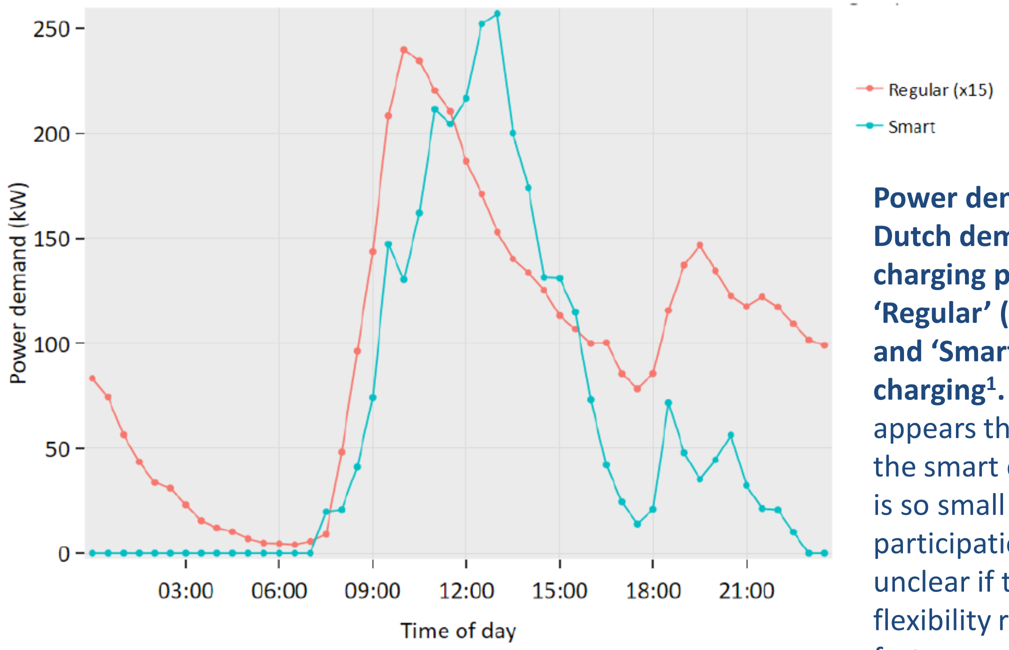
Report describes the raw demonstration results, KPI measurements and conclusions based on the Dutch demonstrator of the Interflex project. KPIs included the impact on the grid, the potential to shift EV demand and the local peak load reduction. A key conclusion from the Dutch demonstrators was that the flexibility offered by the public smart charging points was difficult to forecast, control, measure and obtain. This was largely due to the low participation of EV drivers.

Key opportunities identified:

- Demonstrator showed that it was technically feasible for a local flexibility market to provide a DSO with an alternative to grid reinforcement
- Price of flexibility from EV charging was low compared to the costs of flexibility from stationary storage or PV curtailment

Key barriers identified:

- Low EV driver participation in the demonstrator. Analysis of results showed that a substantial part of the EV drivers in a local area need to participate in smart charging for sufficient availability of flexibility from load shifting to solve local congestion problems
    - A result of the low participation was that EV flexibility is difficult to forecast, control, measure and obtain
- Although the local flexibility market is technically feasible, it is a complex system with a large number of stakeholders and dependency on forecasting reliability



Power demand at Interflex Dutch demonstrator smart charging points for 'Regular' (unmanaged) and 'Smart' (managed) charging[1]. Although it appears the load is shifted, the smart charging data set is so small (low participation) that it is unclear if this shift is due to flexibility requests or other factors

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
D7.7 Raw demonstration results based on the KPI measurements of the Dutch demo, Interflex	Enexis				

Source: 1) *D7.7 Raw demonstration results based on the KPI measurements of the Dutch demo*, Interflex, Enexis (December 2019)

KEY PROJECT: Demonstration of PEV Smart Charging and Supporting Grid Objectives project, California

SMERC UCLA Smart Grid Energy Research Center

Task 1

Project summary

- The Demonstration of PEV Smart Charging and Storage Supporting Grid Objectives project was conducted by the Smart Grid Energy Research Centre (SMERC) at UCLA in partnership with the California Energy Commission (CEC)
- Project was designed to develop and demonstrate charging infrastructure for smart charging, V2X and grid services
  - The infrastructure was demonstrated at public charge points in the City of Santa Monica
- Smart charging infrastructure was installed in the Santa Monica Civic Center Parking Structure
  - EV drivers could use the charge points for regular charging sessions, or register for the project to enable smart charging
  - Registered users were notified of demand response events during the charging sessions, at which point they could choose to switch off their charging session for 30 minutes and receive rewards.
  - Total power available per charging station was capped and split between the connected EVs (up to 4 at a time)
- Demonstration carried out from January 2017 – January 2018

Reviewed project reports

- *Demonstrating Plug-in Electric Vehicles Smart Charging and Storage Supporting the Grid*, UCLA Smart Grid Energy Research Center (SMERC) (August 2018)

UCLA SMERC Level 2 Smart Charger[1]: Chargers installed in the Santa Monica Civic Center Parking Structure allow one dedicated circuit to be shared among four EV charging sessions simultaneously.

Charger able to respond to demand response signals such as PV generation

Source: 1) *Demonstrating Plug-in Electric Vehicles Smart Charging and Storage Supporting the Grid*, UCLA Smart Grid Energy Research Center (SMERC) (August 2018); V2X = Where car is used as power supply, Vehicle to everything (including home and grid)

Report on the Demonstration of PEV Smart Charging and Storage Supporting Grid Objectives project. Report presents the development and deployment of a smart EV charging system in a car park in Santa Monica, California. Project involved the development of smart charging hardware and software (scheduling algorithm). Project found that smart charging stations at a public car park that attempted to maximise consumption of local PV production led to cost savings for the charge point operator, which could be passed onto the EV driver.

Key opportunities identified:

- Smart charging algorithm successfully caused the EV drivers to charge their vehicles with solar energy produced on-site, reducing the overall energy costs for charge point operators and drivers
- Smart charging software could also receive signals from the grid operator to start charging events and mitigate the impact of PV over-generation

Key barriers identified:

- A 4-hour parking limitation in the car park made the low charging power when multiple EVs connected to one charging station unacceptable to many drivers
- Drivers reluctant to log a user id and password into a kiosk when charging
- Lack of open protocols to communicate between the charge point, the car and the grid operator was challenging. Development would facilitate implementation
- Consumer behaviour prediction algorithms had to be adapted during the project as users were adapting their behaviour as they developed an understanding of the smart charging algorithm

Santa Monica Civic Center			
Month	**Energy Cost Without SC [$]**	**Energy Cost with SC [$]**	**Savings [$]**
January	583.04	522.68	60.36
February	620.85	553.41	67.44
March	668.71	594.18	74.53
April	451.58	410.43	41.15
May	566.35	505.83	60.52
June	975.43	623.12	352.31
July	1010.07	622.03	388.04
August	1040.65	647.94	392.71
September	1007.85	636.14	371.71
October	624.03	568.88	55.15
November	644.54	578.83	65.71
December	544.18	496.33	47.85

Energy cost savings at the Santa Monica Civic Center parking structure charge points in 2017[1]. Savings in cost of energy consumed passed onto CPO.

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Demonstrating Plug-in Electric Vehicles Smart Charging and Storage Supporting the Grid	UCLA Smart Grid Energy Research Center (SMERC)				

Source: 1) *Demonstrating Plug-in Electric Vehicles Smart Charging and Storage Supporting the Grid*, UCLA Smart Grid Energy Research Center (SMERC) (August 2018)

KEY PROJECT: Smart Solar Charging, Utrecht

Task 1

Project summary

Project aim: Develop an AC bidirectional public charge point to be rolled out in Utrecht, the Netherlands

- The Smart Solar Charging project was launched in 2015 by the Utrecht Sustainability Institute (project lead) in partnership with a consortium including LomboXnet (technical project leader), ElaadNL, Stedin and Last Mile Solutions
- An AC bidirectional public charge point was developed, and several were installed around Utrecht
  - Charge points aim to maximise consumption of local PV generation, and are capable of discharging EVs to balance the grid
  - Day-ahead load forecasting is used by the DSO (Stedin), who send out a flex request if necessary, which can be fulfilled by the bidirectional capability of the charge points
- The public charge points can be used for regular charging sessions by any members of the public, but the smart and bidirectional functionality of the charge points is only available to a shared fleet of AC bidirectional charging-enabled cars, provided by Renault ('We Drive Solar' project)

Reviewed project reports

- *Smart Solar Charging: Bi-Directional AC Charging (V2G) in the Netherlands*, ElaadNL (July 2017)
- *Maximising EV integration in LV-grids using the Universal Smart Energy Framework*, Dr. E.J. Coster, H.A. Fidder, M. Broekmans, Stedin (October 2017)
- *The Impact of Transition to Shared Electric Vehicles on Grid Congestion and Management*, Brinkel et al. (October 2020)

Vehicle from shared fleet at an AC bidirectional Smart Solar Charging project charge point in Utrecht[1]

Source: 1) Renault press release, [Fleet of 150 Renault ZOE for smart solar charging project](#) (2016)

Smart Solar Charging: Bi-Directional AC Charging (V2G) in the Netherlands, ElaadNL (July 2017)

Task 1

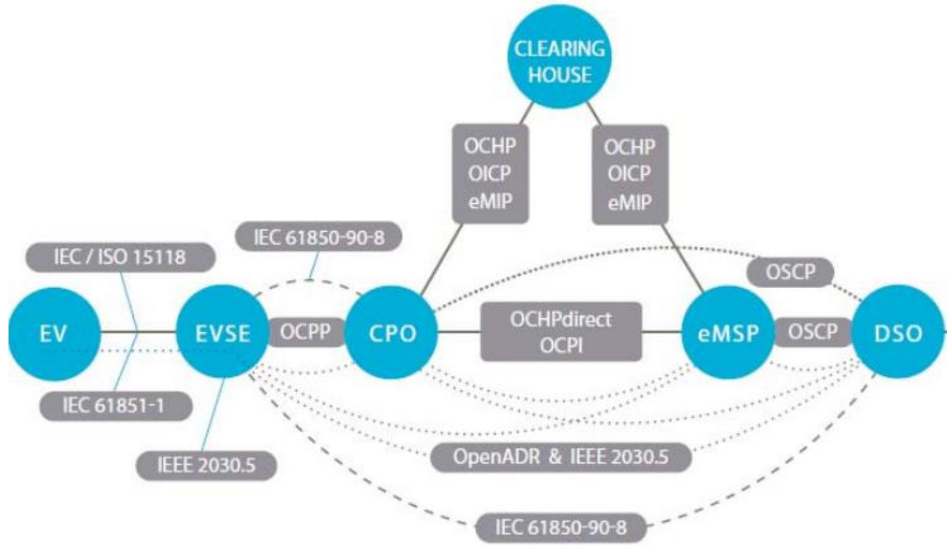
Report on the launch and initial results of the Smart Solar Charging project. Report summarises the design stage of the public AC bidirectional charging station, and the establishment of the framework (Universal Smart Energy Framework, USEF) and open protocols used to communicate between the DSO (Stedin), aggregator (Jedlix) and the charge points. Report indicates that a public network of the designed solar-controlled, AC bidirectional charge points have been installed in a number of residential areas in Utrecht and its surrounding cities alongside a shared fleet of AC bidirectional EVs, and that the system successfully responds to flex requests from the DSO. Report focused on the practicalities and standards that would be used for implementation, but does not include findings from actual implementation (e.g. value or consumer experience).

Key opportunities identified:

- Project successfully demonstrated that a system can be put in place by which public charge points can respond to flexibility service requests from the DSO, via an aggregator
  - Open standards/protocols can be used to implement an effective system

Key barriers identified:

- A number of regulatory barriers to the expansion of the service were identified:
  - Mostly involving barriers to the bidirectional charging of the shared EV fleet (e.g. double taxation on energy charged and discharged to the grid).
  - Also identified that the current energy taxation system in the Netherlands did not sufficiently incentivise the efficient use of locally generated renewable energy via smart charging



Overview of actors and related charging protocols in the Smart Solar Charging project pilot[1]

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Smart Solar Charging: Bi-Directional AC Charging (V2G) in the Netherlands	ElaadNL				

Source: 1) *Smart Solar Charging: Bi-Directional AC Charging (V2G) in the Netherlands*, ElaadNL (July 2017)

Maximising EV integration in LV grids using the Universal Smart Energy Framework, Stedin (October 2017)

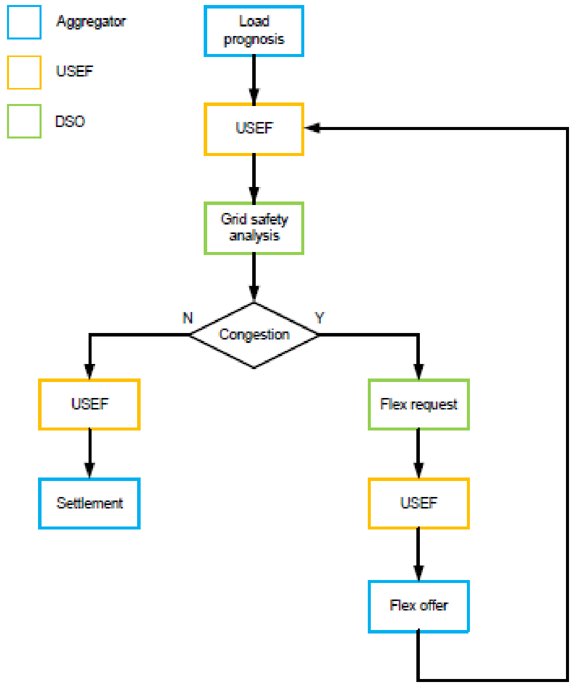
Report discussing the use of the Universal Smart Energy Framework (USEF) as a framework for the distribution system flexibility market in the Smart Solar Charging project in Utrecht. The paper gives an overview of USEF which describes stakeholder roles, interactions and communication protocols, and discusses preliminary results of its implementation in an LV grid. For the Smart Solar Charging project, the distribution network operator is Stedin. Results collected from the trial suggest that capacity management by USEF works well. However, the report suggests that further development of protocols to expand message exchange between DSO, aggregator and flexibility source (in this case a charge point) to multiple aggregators is required, and that reliable forecasting of demand from a small number of charge points was an issue.

Key opportunities identified:

- The distribution network flexibility market framework USEF works well as a framework to collect prosumers' flexibility and make it available to the distribution network operator

Key barriers identified:

- Pilot project found that despite the system involving only a single aggregator, it was difficult to successfully exchange messages (DSO – aggregator – charge point)
    - Report suggests that this will become a larger problem if the project expands to multiple aggregators
- Difficult to accurately forecast demand from charging stations when there was only a small number at feeder level. Report concluded that the system would work well when forecasting at MV feeder or MV/LV transformer level, where there is larger demand and thus more diversity



Flowchart of the general USEF process[1]

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Maximising EV integration in LV grids using the Universal Smart Energy Framework	Stedin				

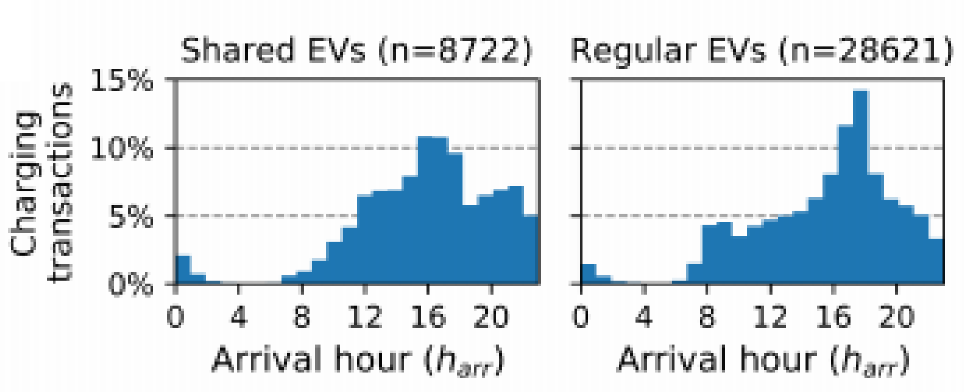
Source: 1) *Maximising EV integration in LV grids using the Universal Smart Energy Framework*, Stedin (October 2017)

The Impact of Transition to Shared Electric Vehicles on Grid Congestion and Management, Brinkel et al. (October 2020)

Study that aims to assess the charging behaviour of shared electric vehicles compared to regular EVs at on-street public charge points. It also assesses how the impact of shared EVs on the low-voltage electricity network compares to that of regular EVs, and to assess the opportunity for flexibility services from these two groups. Data was collected from 460 public charge points used by the 'We Drive Solar' shared EV fleet, and scenarios were developed with varying proportions of shared EVs. These scenarios were modelled and the overall charging profile as well as the potential for flexibility in each were assessed. Flexibility potential was calculated as the difference between the plugged-in time and the actual charging time for each charging session.

Key opportunities identified:

- On-street public charge points have the potential to provide flexibility services to the distribution network, with both high and low penetration of shared EVs in an area
- Shared EV service providers ensure over-capacity at all times. As a result, shared EVs on average spend longer plugged into the charge points but not charging than regular EVs, and so have a higher potential for flexibility services
    - Additionally, the departure times of shared EVs, and therefore the availability of flexibility, is easier to predict due to the service's reservation system



Distribution of charging sessions carried out by shared and regular EVs at charge points across Utrecht[1]

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
The Impact of Transition to Shared Electric Vehicles on Grid Congestion and Management	Brinkel et al.				

Executive Summary

Task 1: Evaluating Barriers and Opportunities

Literature review

Summary

Deep Dive – key projects and reports

Key literature from projects

Other literature (not linked to projects)

Conclusions

Focus groups

Task 2: Analysis of Charging Behaviour + Substation Selection

Task 3: Assessment of Flexibility Opportunities

Task 4: Development of Use Cases and Commercial Models

Report Conclusions

Appendix

Infrastructures de recharge pour véhicule électrique (Charging infrastructure for electric vehicles), CODA Stratégies (July 2019)

Study providing an overview of current charging infrastructure in France, and key actions for success of future EV and charging infrastructure deployment. Research included carrying out a survey with public and private operators of charge points. Survey discussed profitability of charge points, as well as the potential for flexibility services from charging infrastructure. The results of the surveys suggested that public charge points are not yet profitable, and are unlikely to be so in the near future. However, both public and private charge points operators indicated that there would be economic value to charging infrastructure flexibility, but cite the current lack of financial incentives and market barriers as key problems to developing this practice.

Key opportunities identified:

- Charge point operators believe that due to the high power drawn by charge points, there is economic value from flexibility services from charging infrastructure

Key barriers identified:

- Despite recognition that flexibility services from charging infrastructure could have economic value, 58% of survey respondents felt that it was unlikely that charge point operators would be able to make money from flexibility services in the future
    - Reasons cited included that the flexibility market in France would not allow this practice (regulatory barrier), and that there was currently a lack of financial incentive for charge point operators to develop the systems and infrastructure for future implementation

Report title page

Study was carried out by CODA Stratégies on behalf of the Ministère de l'Économie et des Finances and the Ministère de la Transition Écologique et Solidaire in France

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Infrastructures de recharge pour véhicule électrique	CODA Stratégies				

Note de position sur le smart charging et le V2X (Position paper on smart charging and V2X), Avère France, Colombus Consulting (July 2020)

Task 1

Position paper published by the 'Storage and Smart Grids' Task Force of Avère France. Paper describes the opportunities and barriers for smart charging and V2G in France. The report focuses on the economic feasibility of smart charging and V2G technologies, and the requirement for public policies that back them. The paper concludes that in France there are insufficient regulations and laws in place for the EV driver or the charge point operator to put smart charging or V2G practices into place. It calls for a number of measures including the simplification of the flexibility market mechanisms and changes to standard electricity network tariffs, to favour the roll-out of smart charging and V2G in the future.

Key opportunities identified:

- Smart charging and V2G offers opportunities for the distribution network to reduce reinforcement investments and operating costs
- Paper references a survey of 500 EV drivers in France, the results of which suggests that the vast majority of EV drivers would be willing to smart charge (home charge points) if motivated by reducing the power peak on the network

Key barriers identified:

- There are currently no fixed regulations around flexibility services provided by charging infrastructure to the distribution network in France
- French legislation is currently not favourable enough for charge point operators to implement public smart charging with tariffs that would attract users and compensate for potential inconveniences
- Lack of standardisation of data sharing between EVs, distribution networks and the relevant stakeholders (e.g. flexibility service providers) is a barrier to the integration of smart charging into the broader energy system

Report title page graphics[1]

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Note de position sur le smart charging et le V2X	Avère France, Colombus Consulting				

Source: 1) *Note de position sur le smart charging et le V2X*, Avère France, Colombus Consulting (July 2020)

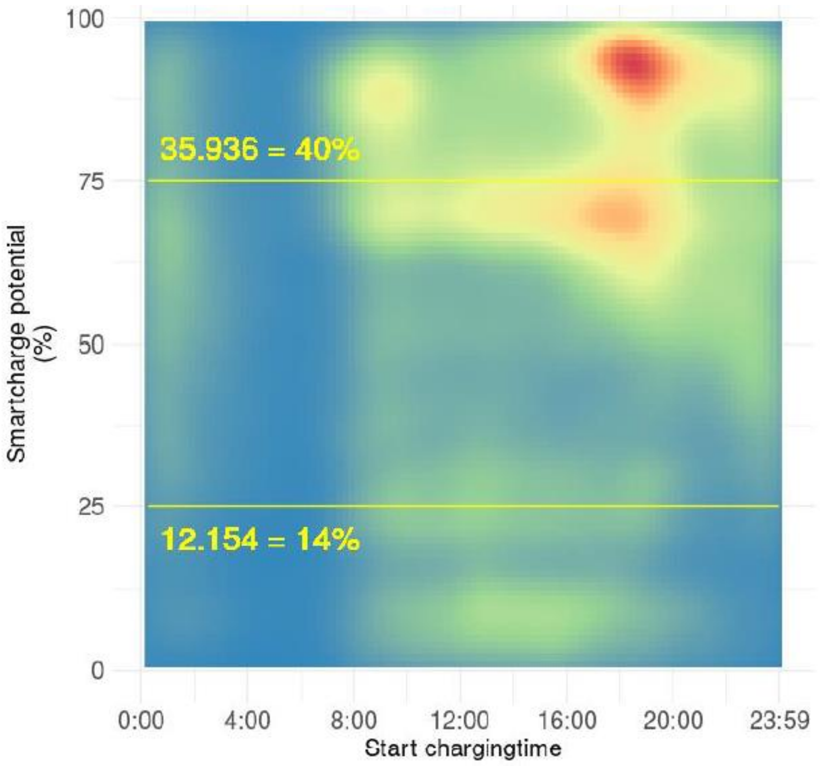
Pinpointing the smart charging potential for electric vehicles at public charging points, Amsterdam University of Applied Sciences, ElaadNL (May 2019)

Task 1

Research to develop a methodology to assess the potential for smart charging from public charging infrastructure and to suggest possible smart charging strategies. Researchers analysed a dataset made up of EVnetNL charging data (charging sessions from nationwide network of charge points) and G4/MRA-Electric data from urban areas. This dataset is one of the largest databases of charging sessions on public charging infrastructure available worldwide. Smart charging potential was defined as the proportion of time in which an EV is connected to a charge point, but is not charging. Charging sessions were categorised according to the start time and connection duration of charging sessions. The analysis found that ca. 40% of all charging sessions at public charging infrastructure start between 4-8pm and have a smart charging potential of 75% or higher.

Key opportunities identified:

- Public charging infrastructure has high potential for flexibility or smart charging potential
    - Public charging infrastructure observe long plug-in sessions that begin in the early evening or overnight, with low times of charging, similar to what is observed with private home chargers
    - Potential for flexibility is observed to increase with level of urbanity



Distribution of the smart charging potential for charging sessions of BEVs in Amsterdam[1]

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Pinpointing the smart charging potential for electric vehicles at public charging points	Amsterdam University of Applied Sciences, ElaadNL				

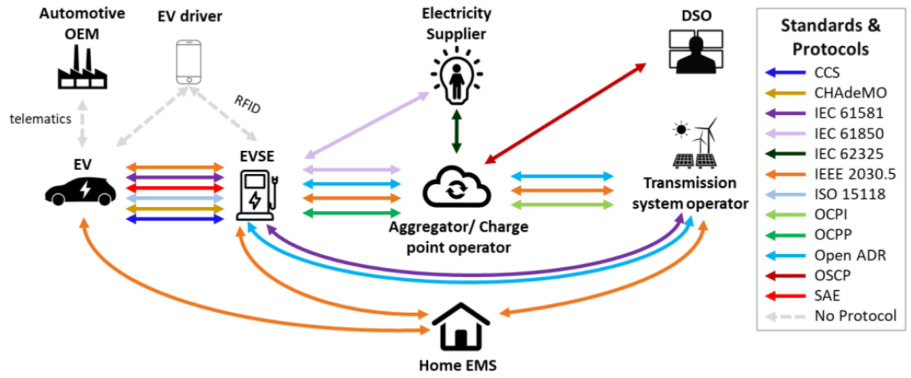
Implementing Open Smart Charging, Element Energy for International ZEV Alliance (October 2019)

Task 1

Paper outlining the potential benefits of smart charging and recommendations for policymakers to ensure that these benefits can be realised. The report explores the benefits of smart charging, and discusses the technical, consumer and institutional barriers to implementation of smart charging infrastructure. The report considers smart charging from different charge point locations, including off-street, destination, workplace and public smart charging infrastructure, and highlights the whole-system value of the widespread implementation of smart charging.

Key opportunities identified:

- Report notes that smart charging has the potential to postpone costly distribution network infrastructure upgrades
- Monetisation of smart charging savings at public charge points could improve the commercial case of charge point networks from the point of view of charge point operators
- Paid public EV charge points already have control and communication capabilities for billing customers – smart charging capability should not add significant further cost



Combinations of standards and protocols relating to EV smart charging[1]

Key barriers identified:

- Difficult for distribution network operators to precisely value and pass on savings from avoided network upgrades to charge point operators and EV drivers
- Need to ensure system optimises for the needs of both drivers and the energy system e.g. installations of a dense public charging network to ensure consumers are comfortable with smart charging
- Smart charging for congestion management in local distribution networks will need collection and sharing of high resolution data (including location data)

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Implementing Open Smart Charging	Element Energy for International ZEV Alliance				

Source: 1) *Implementing Open Smart Charging*, Element Energy for International ZEV Alliance (October 2019)

Practical deployment of electric vehicle flexibility, Jedlix, USEF Foundation (February 2020)

Task 1

Paper that summarises the roles and interactions of stakeholders in a distribution network flexibility market, as set out by the market framework USEF (Universal Smart Energy Framework). It discusses the current utilisation of EV flexibility based on the practical deployments of Jedlix, a Dutch EV flexibility aggregator. Jedlix is involved in a number of projects that offer DNOs constraint management through EV flexibility (Interflex and Smart Solar Charging). The paper describes the steps taken by Jedlix to offer flexibility services from public charge points to the DNO, in the context of the USEF framework, and the barriers faced trading flexibility in this way.

Key opportunities identified:

- Jedlix has shown that it is technically feasible to offer flexibility services to distribution networks from public EV charging infrastructure

Key barriers identified:

- Methodology is based on the expected day-ahead load profile. As the number of simultaneous charging sessions in a specific congestion area is typically low, the expected load and flexibility is hard to predict. Leads to undesirable deviations between the predicted and measured data

Jedlix uses the USEF framework to enable a direct flexibility trade with the DSO. The **steps for trading flexibility** in these projects are as follows:

1. Jedlix informs the DSO about the expected controllable EV load in a congestion area, by sending a day-ahead D-prognosis to the DSO
2. If there is a high risk of congestion, it can be mitigated by the aggregator and DSO trading flexibility
3. To realise the flexibility trade, Jedlix sends control signals to the CPOs involved (public CPs on-street or in a car park)
4. CPO validates whether the connected EV driver has a contract with Jedlix
5. If they do, CPO informs Jedlix and gives close to real-time data from the CP
6. Aggregated data of all controllable EV load in the congestion area is used by the DSO to validate Jedlix's product delivery

Title of paper	Author	Consumer experience	Value to CPOs	Value to DNO	Practicalities & standards
Practical deployment of electric vehicle flexibility	Jedlix, USEF Foundation				

CPO = Charge point operator, CP = Charge point

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Appendix

CONSUMER EXPERIENCE: Smart charging at public charge points likely to have no impact on user experience, unless additional effort required for participation or insufficient info given

Task 1

Smart public charge points likely to be well-utilised and have no impact on users' experience

No impact on user experience

- No difference reported by users[1,2] and no change in the energy charged in 91% of charging sessions at public on-street charge points in Amsterdam[3]
- Positive impact in 4% of cases, as higher power at times of high PV generation[3]

Positive network impact motivates users

- EV drivers in the Netherlands ranked optimal use of sustainable energy and stability of grid as more important than financial advantages[2]
- French EV drivers motivated by power peak reduction[4]

Savings can be passed onto consumers

- Financial advantages can feasibly be offered to smart charge point users[5,6,7]
- Likely to further motivate EV drivers to smart charge[2,4]

Load shifting at public charge points without sufficient information may cause range anxiety

Mitigable risk of range anxiety

- Possibility of low overall charging speed (across session as a result of reduced power during smart charging periods) caused range anxiety for users[2,5]
- Provision of more information on charging speed would reduce range anxiety for smart charge point users in Amsterdam[2]

Additional steps inconvenience users

- Charge point users in California were reluctant to take additional steps to participate in smart charging[5]

1. *SEEV4-city Flexpower 1: analysis report of the first phase of the Flexpower pilot*, AUoAS (Nov 2019)
   - Data from 44k charging sessions by 8k unique users of on-street public charging stations
2. *Behaviour research report*, AUoAS, ElaadNL (May 2020)
   - Survey of 32 EV drivers across the Netherlands
3. *Final report - Amsterdam Flexpower Operational Pilot*, AUoAS (July 2020)
   - Data from 456 on-street public charging stations
4. *Note de position SUR LE SMART CHARGING ET LE V2X*, Avère France, Colombus (July 2020)
   - Survey of 500 EV drivers in France
5. *Demonstrating PEVs Smart Charging and Storage Supporting the Grid*, UCLA SMERC (Aug 2018)
   - Data from charge points in public car park
6. *Smart Solar Charging: Bi-Directional AC Charging (V2G) in the Netherlands*, ElaadNL (July 2017)
   - Modelling based on consumer data
7. *D7.6 Lessons learned to draw business models in use cases NL#2 and NL#3*, Enexis (June 2019)
   - Financial savings offered through aggregator app for use of 13 public smart charging points

AUoAS = Amsterdam University of Applied Sciences

VALUE TO CPO: There is limited evidence of the value of public smart charging infrastructure to charge point operators

Task 1

- Value to charge point operators (CPOs) was generally not a key focus in the reviewed literature around public smart charging

Smart charging may improve the commercial case of charge point networks, but exact value is unclear

Possibility of additional economic value	• Monetisation of savings from smart charging or flexibility services may improve commercial case of public charge point networks[2,3] • Maximising consumption of local renewables observed to lower energy costs[1]
Unnecessary to pass savings onto consumers	• Charge point users may not need financial incentives to smart charge[4,5], improving the business case for CPOs
Limited modelling of economic value	• Value to CPOs not directly modelled or analysed in the reviewed literature • French CPOs believe there is a lack of financial incentive to develop these practices[3] (note: practice restricted in France)
No improvement in CP occupancy	• No improvement in occupancy ratio demonstrated[6,7], which reduces the value of smart charging to CPOs
Potential increase in costs	• Smart charging observed to increase connection costs for CPOs[7] • Only the case when charging power increased to reduce renewable curtailment

1. *Demonstrating PEVs Smart Charging and Storage Supporting the Grid*, UCLA SMERC (Aug 2018)
    - Data from charge points in public car park
2. *Implementing Open Smart Charging*, Element Energy for International ZEV Alliance (Oct 2019)
3. *Infrastructures de recharge pour véhicule électrique*, CODA Stratégies (July 2019)
    - Survey of 52 French CPOs
4. *Note de position SUR LE SMART CHARGING ET LE V2X*, Avère France, Colombus (July 2020)
    - Survey of 500 EV drivers in France
5. *Behaviour research report*, AUoAS, ElaadNL (May 2020)
    - Survey of 32 EV drivers across the Netherlands
6. *SEEV4-city Flexpower 1: analysis report of the first phase of the Flexpower pilot*, AUoAS (Nov 2019)
    - Data from 44k charging sessions by 8k unique users of on-street public charging stations
7. *Final report - Amsterdam Flexpower Operational Pilot*, AUoAS (July 2020)
    - Data from 456 on-street public charging stations

CPO = Charge point operator, AUoAS = Amsterdam University of Applied Sciences

VALUE TO DNO: Smart public charging can reduce peak demand on the distribution network and minimize local renewable curtailment

Task 1

With sufficient participation, smart public charging can offer flexibility services to a DNO

Flexibility services feasible	• Achievable[1,2,3,4] including integration with local flexibility market[4,5,6,7] • Smart charge points in Amsterdam reduce peak demand by 1.1kW, leading to total peak reduction of 470kW and €50k network saving
High suitability of public charging behaviour	• Long plug-in sessions with low charging time over times of peak demand observed, particularly in urban residential areas[8] • 40% of Dutch public charging sessions have a smart charging potential* of 75% or higher[8]
Low-cost flexibility	• Cost of EV flexibility observed to be lower than that of stationary storage or renewables curtailment[4]
High participation necessary	• Substantial proportion of EV drivers in a local area must participate for sufficient flexibility availability[4] • Low participation issue for forecasting and obtaining EV flexibility[4,7]

Rebound demand peak observed not to negatively impact the distribution network

Congestion from rebound peak unlikely	• Load shifting risks a rebound peak in demand[2,3,5] • Demonstrated that additional network congestion not caused and overall network demand smoothed[3]

1. *Demonstrating PEVs Smart Charging and Storage Supporting the Grid*, UCLA SMERC (Aug 2018)
   - Data from charge points in public car park
2. *SEEV4-city Flexpower 1: analysis report of the first phase of the Flexpower pilot*, AUoAS (Nov 2019)
   - Data from 44k charging sessions by 8k unique users of on-street public charging stations
3. *Final report - Flexpower Operational Pilot*, AUoAS (Jul 2020)
   - Data from 456 on-street public charging stations
4. *D7.7 Raw demonstration results based on the KPI measurements of the Dutch demo*, Enexis (Dec 2019)
   - Data from pilot of 13 public smart charge points, operated through a flexibility market
5. *D7.6 Lessons learned to draw business models in use cases NL#2 and NL#3*, Enexis (June 2019)
   - Data from pilot of 13 public smart charge points, operated through a flexibility market
6. *Maximising EV integration in LV grids using the Universal Smart Energy Framework*, Stedin (Oct 2017)
   - Data from smart public charge points, operated through a distribution network flexibility market
7. *Practical deployment of electric vehicle flexibility*, Jedlix, USEF Foundation (February 2020)
8. *Pinpointing the smart charging potential for EVs at public charging points*, AUoAS, ElaadNL (May 2019)
   - Analysis of data from charging sessions across the Netherlands, 2015-2016
9. *Implementing Open Smart Charging*, IZEVA (Oct 2019)

DNO = Distribution Network Operator, AUoAS = Amsterdam University of Applied Sciences

*In this study, smart charging potential defined as the proportion of time in which an EV is connected to a charge point, but is not charging

PRACTICALITIES & STANDARDS: The system required for charge points to offer flexibility services is complex and requires high volume of data sharing

Task 1

Flexibility services from public charging infrastructure requires a complex communication system

Technically feasible

- Demonstrated the application of a static smart charging profile[1,2]
- Shown integration with a local flexibility market via an aggregator[3,4,5], alongside multiple flexibility sources[3,4]

Complex system with many stakeholders

- High volume of data must be shared between many stakeholders[4,5,6,7]
- Open communication protocols exist, but may have to be expanded, or issues communicating between different protocols[4]

Market framework demonstrated

- Market frameworks created and demonstrated, e.g. USEF[4,5,8] sets out stakeholder rules, interactions and necessary communication protocols
- Observed difficulty exchanging messages despite single aggregator[5]

Accurate load forecasting difficult

- Complex system heavily reliant on accurate load forecasting[4,5,8]
- Difficulty forecasting observed, particularly with low participation[4] or with small numbers of charge points at LV feeder level[5]

Public charge points are suitable for flexibility services

High suitability of public charge points

- Public charging behaviour is suitable for participation in flexibility services[8]
- Paid public EV charge points already have control and communication capabilities – smart charging capability should not add significant cost[6]
- In some cases, hardware or software upgrades may be required[1]

1. SEEV4-city Flexpower 1: analysis report of the first phase of the Flexpower pilot, AUoAS (Nov 2019)
   - Data from 44k charging sessions by 8k unique users of on-street public charging stations
2. Final report - Flexpower Operational Pilot, AUoAS (Jul 2020)
   - Data from 456 on-street public charging stations
3. D7.7 Raw demonstration results based on the KPI measurements of the Dutch demo, Enexis (Dec 2019)
   - Data from pilot of 13 public smart charge points, operated through a flexibility market
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   - Data from smart public charge points, operated through a distribution network flexibility market
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7. Note de position SUR LE SMART CHARGING ET LE V2X, Avère France, Colombus (July 2020)
8. Practical deployment of electric vehicle flexibility, Jedlix, USEF Foundation (February 2020)
9. Pinpointing the smart charging potential for EVs at public charging points, AUoAS, ElaadNL (May 2019)
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Objectives and Methodology

Task 1

Focus Group Objectives

Charge Collective consumer focus groups aimed to **assess consumer attitudes to flexible public charging** (on-street or in public, residential car parks)

— Results will help understand the feasibility of public smart charging: identifying main barriers and opportunities

Key research questions:

1. Would consumers be **willing** to use a smart public charge point?
2. If so, what **conditions** would be necessary?
3. What would **motivate consumers** to smart charge at public charge points?
4. What would **concern consumers** about public smart charging?
5. What concerns do consumers have around **information sharing and billing**?

Methodology

- Participants were recruited through a **consumer sign-up survey,** sent out to residents by local authorities, and through a community group
- We aimed to carry out 3 focus groups, with 4 – 6 participants in each
- The focus groups were carried out over **Zoom**, and consisted of:
  1. Explanation of UK Power Networks' role in the energy system
  2. Explanation of smart charging concept
  3. Discussion of research questions
- We asked a combination of prompted and unprompted questions:
  - 8 questions in total (5 prompted, 3 unprompted) (see appendix)
  - Participants asked to respond to unprompted questions through **general discussion** or writing on the **Zoom whiteboard**
  - Responses to prompted questions collected through a **Zoom poll**
- Questions were developed based on discussions with UK Power Networks and **findings of the literature review** (first part of Task 1)

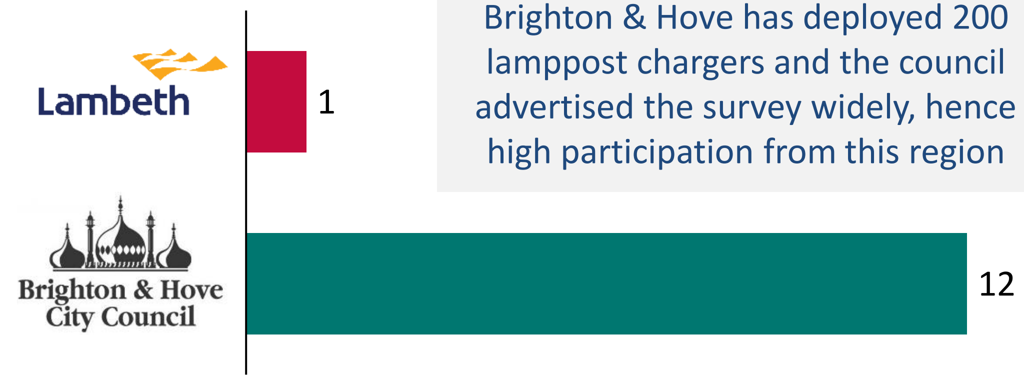
The majority of participants in the Charge Collective focus groups were: from Brighton, drive an EV and charge their vehicle at a public on-street charge point

Task 1

Profile of Focus Group Participants

- **Three consumer focus groups** were carried out as part of the project with a total of **13 participants** from **Lambeth** or **Brighton & Hove**
- The majority of participants (10) drive a **BEV**, while 2 own a PHEV and one participant does not yet have an EV
- Most participants (8) use **public on-street charge points** for the majority of their EV charging
- No participants use a charge point in a residential car park*
- Given the status of the market, participants can all be considered ‘early adopters’ so might not be representative of all future EV drivers

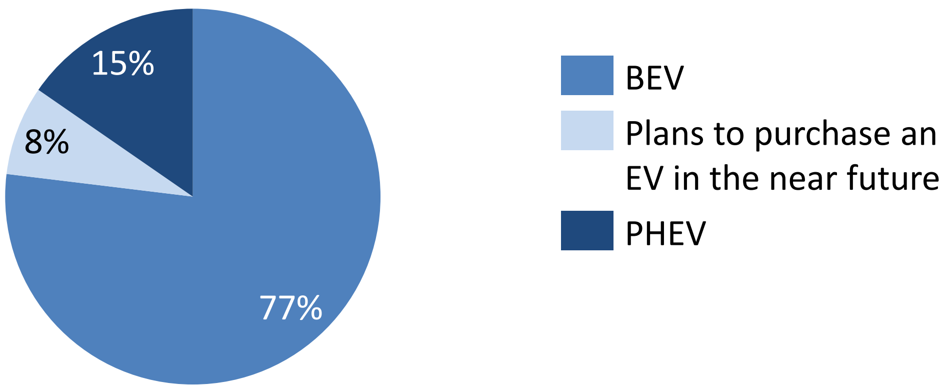
Local Authority



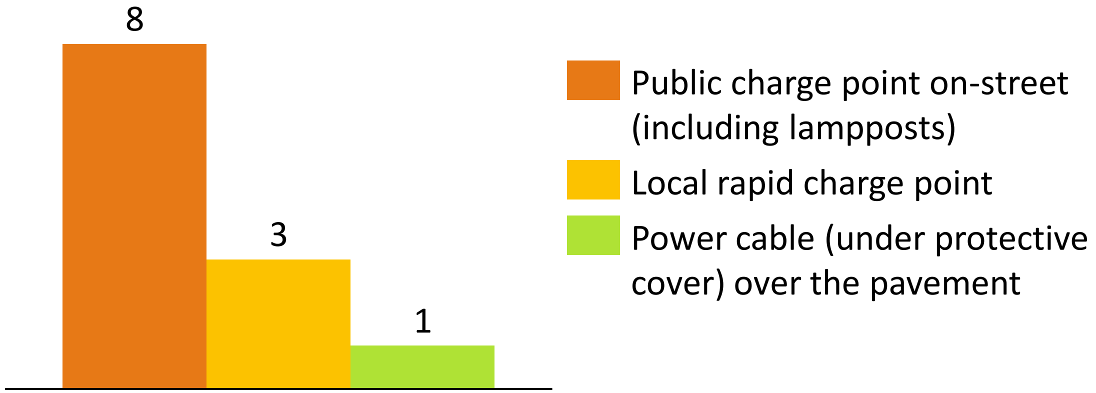
Brighton & Hove has deployed 200 lamppost chargers and the council advertised the survey widely, hence high participation from this region

Note, lamppost chargers in Brighton & Hove are not provided by Ubitricity

EV Ownership



Main EV charging method



*The scope of UK Power Networks’ research includes public, residential car parks. We anticipate concerns to be similar to those of on-street charge point users.

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Appendix

Results from polls and general discussion during the focus groups have been grouped into key themes for presentation of findings

Task 1

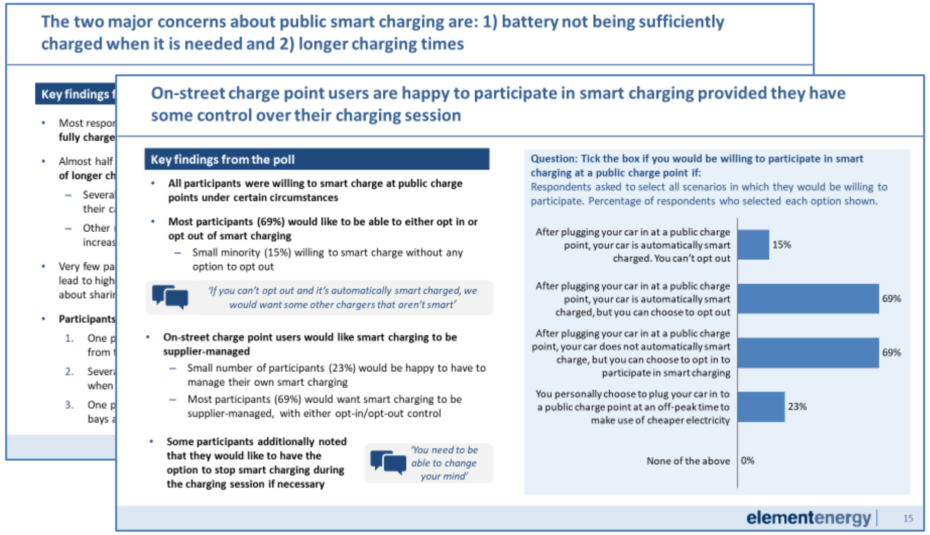
Results summary – quantitative and qualitative results

- Findings from the focus groups have been grouped into 5 key themes:
  - ❖ **Concerns around existing (not smart) charging infrastructure (hardware and software)**
  - ❖ **Concerns about public smart charging**
  - ❖ **Information access and data sharing while smart charging**
  - ❖ **Participation requirements for smart charging**
  - ❖ **Motivations to smart charge**
- Some results are quantitative from responses to Zoom polls, other qualitative results are from general discussion and open questions
- Results relating to existing, non-smart public charging infrastructure are presented first, before moving onto results specific to smart charging

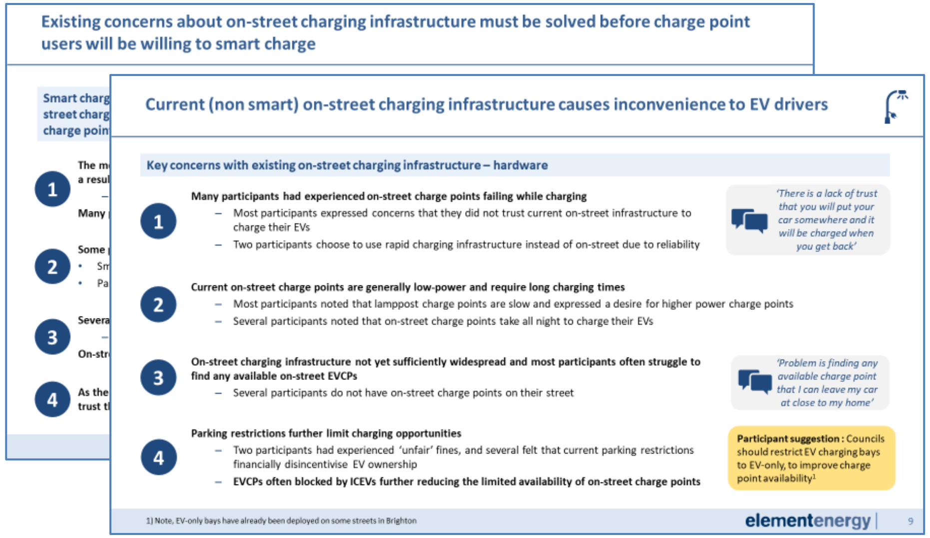
Key quotes from participants

Participant suggestion : points or suggestions we found important, but not strictly related to public smart charging

Poll results and relevant discussion points



Summary of key points from polls and general discussion



Current (non smart) on-street charging infrastructure causes inconvenience to EV drivers

Task 1

Key concerns with existing on-street charging infrastructure – hardware

1

Many participants had experienced on-street charge points failing while charging

- Most participants expressed concerns that they did not trust current on-street infrastructure to charge their EVs
- Two participants choose to use rapid charging infrastructure in car parks instead of on-street chargers due to reliability

'There is a lack of trust that you will put your car somewhere and it will be charged when you get back'

2

Current on-street charge points are generally low-power and require long charging times

- Most participants noted that lamppost charge points are slow and expressed a desire for higher power charge points
- Several participants noted that on-street charge points take all night to charge their EVs

3

On-street charging infrastructure not yet sufficiently widespread and most participants often struggle to find any available on-street EVCPs

- Several participants do not have on-street charge points on their street

'Problem is finding any available charge point that I can leave my car at close to my home'

4

Parking restrictions further limit charging opportunities

- Two participants had experienced 'unfair' fines, and several felt that current parking restrictions financially disincentivise EV ownership
- Residents may be subject to time restrictions when parking outside of their permit areas
- **EVCPs often blocked by ICEVs** further reducing the **limited availability** of on-street charge points

Participant suggestion : Councils should restrict EV charging bays to EV-only, to improve charge point availability*

Note: lamppost chargers in Brighton & Hove are not provided by Ubitricity;
*EV-only bays have already been deployed on some streets in Brighton

Current (non smart) charge point payment and management platforms are not trusted

Task 1

Key concerns with existing on-street charging infrastructure – software

1

Charge point users struggle to get charge point network apps to work

- Many participants had experienced the app crashing or failing during a session
- Particular issues noted around starting and ending the charging session
- A number of participants noted that they would **prefer to use contactless or a swipe card** to pay for their charging

'Stressful wondering if [the app] is going to work this time'

'You have to know how to cheat the app to get the charge started'

2

There is a lack of clarity around the price of charging at on-street charge points

- Many participants felt it was difficult to anticipate how much a charging session would cost
- Several participants noted that certain apps currently do not tell users final cost of their charging session
- Some participants felt concerned by the **inconsistent tariff structures**
  - Several participants had experienced 'hidden fees' or unclear payment mechanisms

'You just plug it in [to a charge point] and write a blank cheque'

3

Interoperability is a key concern to on-street charge point users

- Many participants expressed frustration at the large number of different apps and platforms from different public charging networks, which complicates charge point use

'There's lots of different companies and they've all got different apps – I find that annoying'

Note: lamppost chargers in Brighton & Hove are not provided by Ubitricity

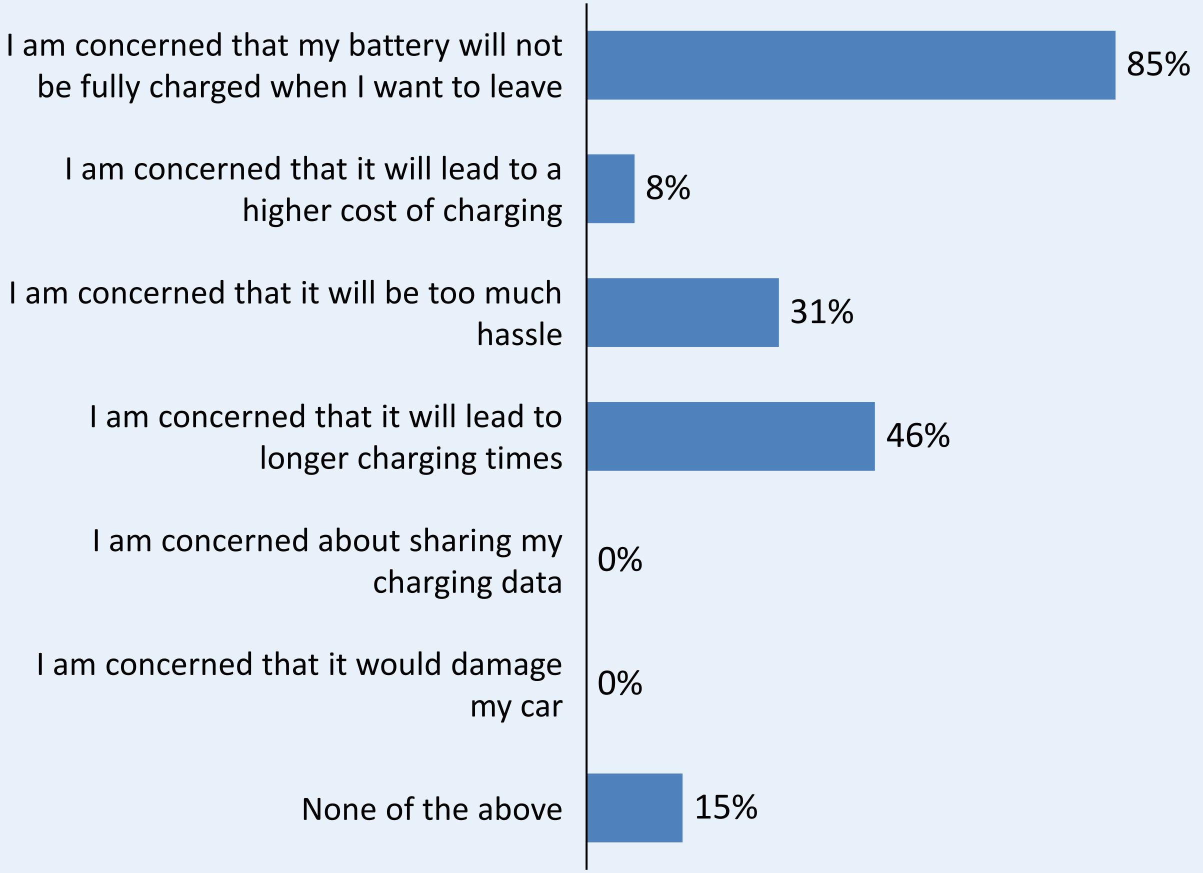
The two major concerns about public smart charging are: 1) battery not being sufficiently charged when it is needed and 2) longer charging times

Task 1

Key findings from the poll

- Most respondents (85%) felt concerned that **their car would not be fully charged** by smart on-street charge points when they needed it (e.g. due to longer charging times or delayed start times)
- Almost half of respondents (46%) were concerned by the **possibility of longer charging times**
  - Several respondents noted that this is tied into concerns about their car not being fully charged when it is needed
  - Other respondents felt this raised other concerns, such as the increased vandalism risk and reduced charge point availability
- Very few participants (8%) felt worried that smart charging might lead to higher charging costs, and no respondents were concerned about sharing data or smart charging damaging their car
- **Participants noted some other concerns** not listed in the poll:
  1. One participant was concerned by the possibility of hassle from the charge point network apps not working
  2. Several participants expressed concerns about vandalism when plugged in on-street, particularly overnight
  3. One participant had concerns of competition for charging bays at times of cheap, off-peak electricity

Question: What about smart charging would potentially be a concern to you?
Respondents asked to select all options they viewed as a potential concern.
Percentage of respondents who selected each option shown* **(n = 13)**



* Note: sum of responses higher than 100%, as respondents could select multiple options.

Existing concerns about on-street charging infrastructure must be solved before charge point users will be willing to smart charge

Task 1

Key concerns with smart charging

Smart charging exacerbates existing concerns around on-street charging. If on-street charging infrastructure were plentiful, reliable and of sufficient power, charge point users would be happy to smart charge

'I would be willing to use any charge point that would get my car charged by the time I need it… I don't think I would be more or less inclined to use it because it is smart.'

1 **The most common smart charging concern, not having sufficient charge when needed, is largely a result of the low power of current on-street charge points**

- Most participants felt would not be able to fully charge an EV and smart charge overnight
- **Many participants thought that higher power charge points would be needed** e.g. 11kW+

'My experience is, on a lamppost charge point, it's not enough to charge my car completely overnight'

2 **Some participants noted that their concern of longer charging times was related to a worry that this would lead to even lower availability**

- Smart charging would lead to EVs blocking charge points as well as ICEVs
- Participants felt that on-street charge points are not yet widespread enough to mitigate this concern

3 **Several participants were concerned by the longer or overnight charging times as this would increase the risk of vandalism**

- Two participants thought that this concern would be reduced if they were able to charge overnight on their own street
- **On-street charge points must be more widespread for all charge point users to feel comfortable charging overnight**

4 **As the charging management interface is already considered unreliable, there is a lack of trust that the apps would be able to handle the increased complexity of smart charging**

'There're horrible interface problems just getting the point to charge at all, let alone giving it complicated information'

When smart charging, on-street charge point users would like access to a lot of information about their charging session

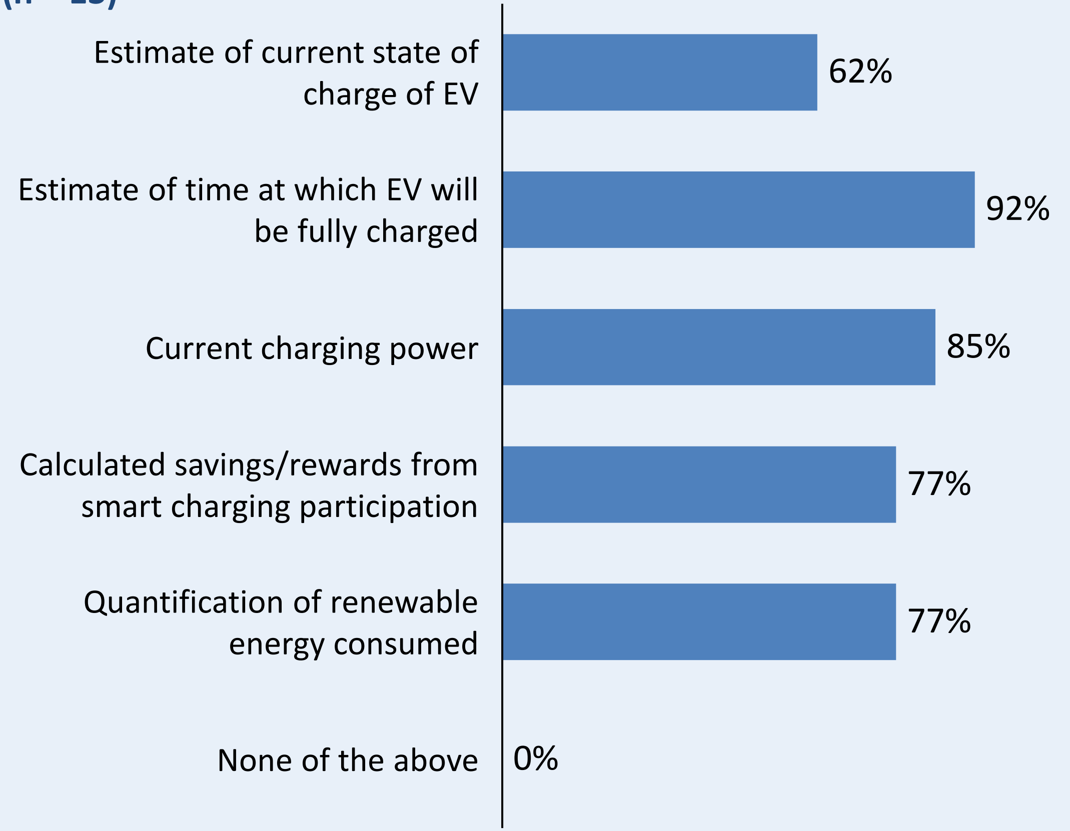
Task 1

Key findings from the poll

- **Most respondents want access to practical information**, such as the time at which their EV will be fully charged (selected by 92% of respondents) and the power at which their EV is currently charging (selected by 85%)
  - Most respondents noted that they view this information as essential
- The majority of respondents would also like to see information about their charging session such as **rewards or quantification of renewable energy**
  - Many respondents felt that although this information would be interesting, they **did not consider it essential**
- Several participants commented that in addition to the options shown in the poll, they would like to ensure they had access to the **total cost at the end of a charging session**
  - They noted that some charge point apps do not clearly present this
- One participant felt that if they have to share their data with the charge point operator, they should be able to access it

Participant suggestion to prevent bay-hogging: EV drivers should be able to request to use charge points via the app (form a virtual queue), to encourage charge point users to move when they have enough charge

Question: What information would you like to have access to while participating in public smart charging?
Respondents asked to select all options they would like to have access to. Percentage of respondents who selected each option shown*
(n = 13)



* Note: sum of responses higher than 100%, as respondents could select multiple options.

On-street charge point users are happy to participate in smart charging provided they have some control over their charging session

Task 1

Key findings from the poll

- **All participants were willing to smart charge at public charge points under certain circumstances**
- **Most participants (69%) would like to be able to either opt in or opt out of smart charging**
  - Small minority (15%) willing to smart charge without any option to opt out*

'If you can't opt out and it's automatically smart charged, we would want some other chargers that aren't smart'

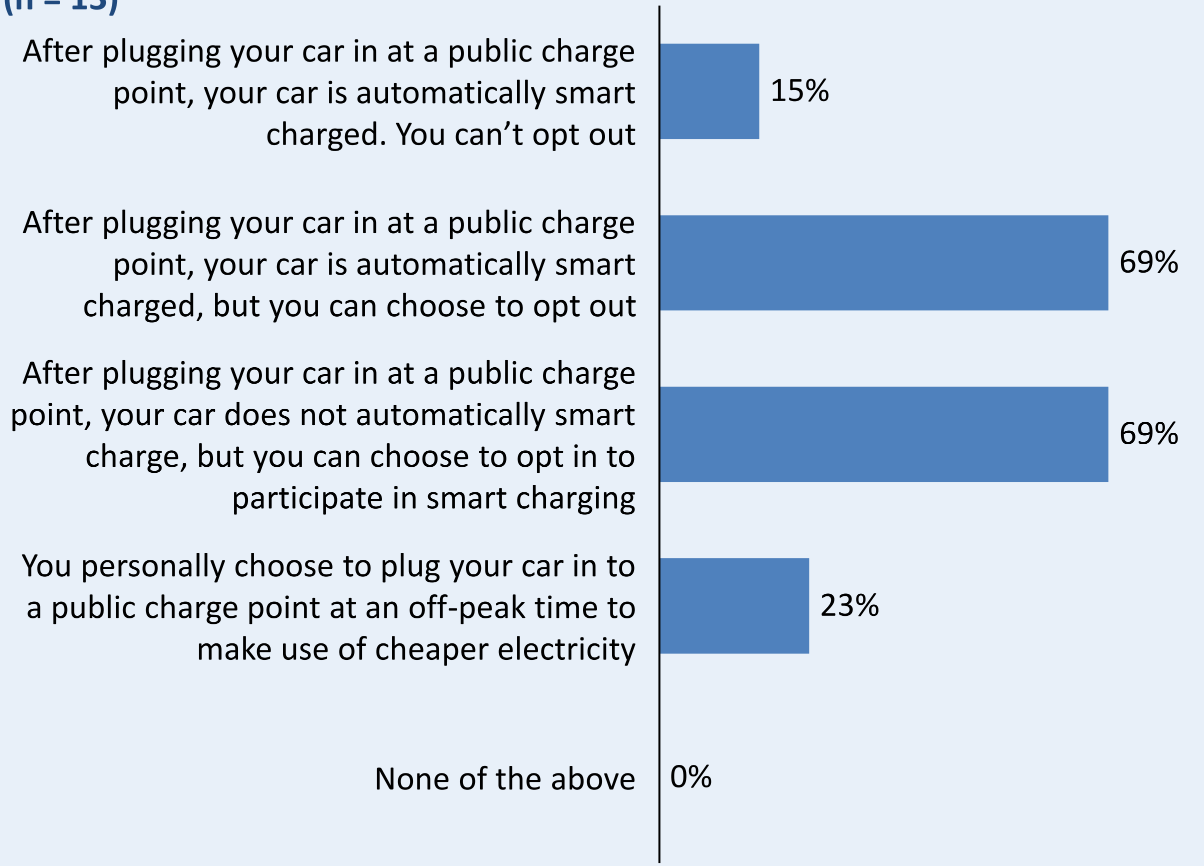
- **On-street charge point users would like smart charging to be supplier-managed**
  - Small number of participants (23%) would be happy to have to manage their own smart charging
  - Most participants (69%) would want smart charging to be supplier-managed, with either opt-in/opt-out control
- **Some participants additionally noted that they would like to have the option to stop smart charging during the charging session if necessary**

'You need to be able to change your mind'

Question: Tick the box if you would be willing to participate in smart charging at a public charge point if:

Respondents asked to select all scenarios in which they would be willing to participate. Percentage of respondents who selected each option shown †

(n = 13)



*Unclear if respondents fully understood option of smart charging without the option to opt out. These respondents later stated they would only smart charge if they could opt out, in contradiction to their poll response; † Note: sum of responses higher than 100%, as respondents could select multiple options.

A minimum percentage charge guarantee is necessary for users to participate in smart charging

Task 1

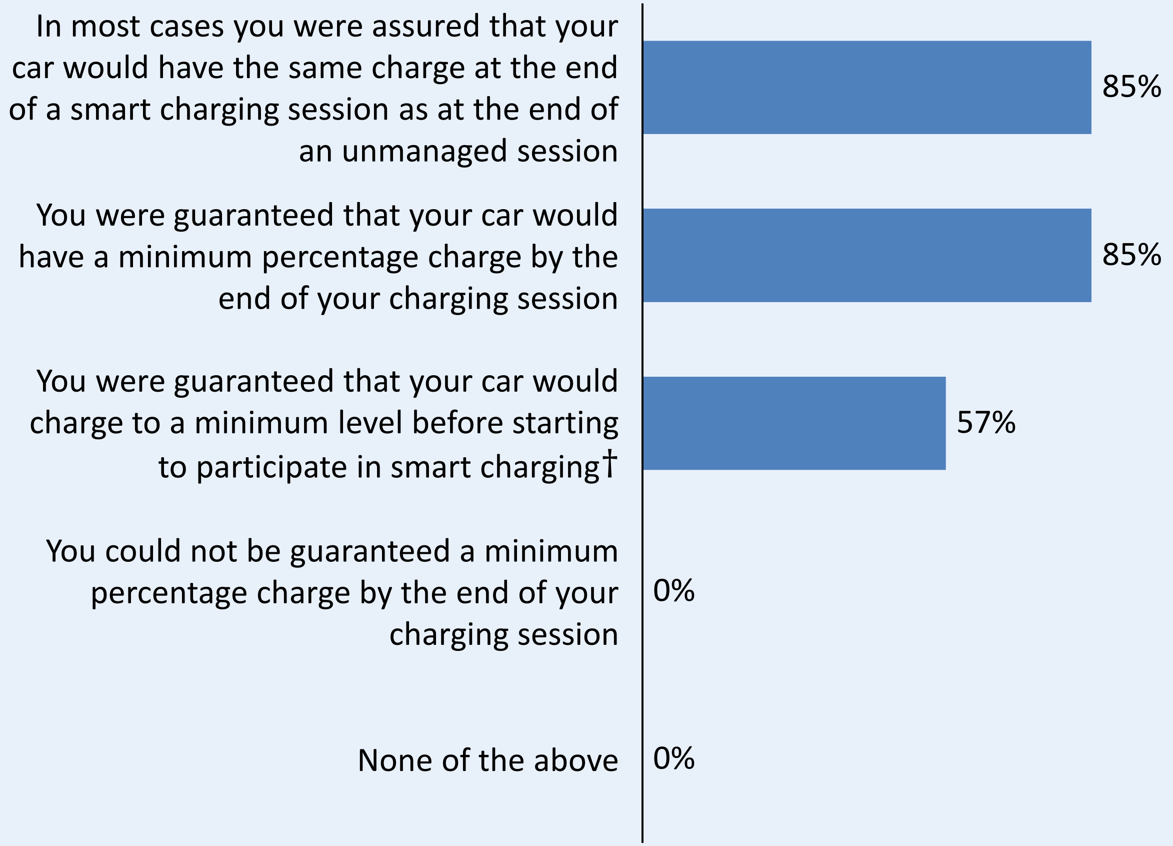
Key findings from the poll

- **On-street charge point users are willing to smart charge provided they can be guaranteed a minimum percentage charge**
  - None of the participants would be willing to use a smart public charge point if there were no guarantee
- Participants would **prefer to be guaranteed a minimum percentage charge by the end of the charging session,** as opposed to **the car charging to a minimum level before starting to smart charge**† (85% vs. 57%)
- When asked **what percentage charge** participants would like to have guaranteed, there was a wide range in percentages specified: **30%** to **100%**
  - Participants commented that the level of charge depends not only on the person, but also factors such as:
    1. **Charging infrastructure availability**
    2. **Mileage** – which depends on **time of year** and **other factors** (e.g. lockdown restrictions)
  - Some participants concerned that current EV drivers have insufficient knowledge to set a minimum charge level
  - Many participants would like to be able to set the minimum percentage charge on a **session-by-session basis**

Question: Tick the box if you would be willing to participate in smart charging at a public charge point if:

Respondents asked to select all scenarios in which they would be willing to participate. Percentage of respondents who selected each option shown*

(n = 13)



*Note: sum of responses higher than 100%, as respondents could select multiple options.

† This option was presented to only 7 focus group participants. For this option, n = 7. For all other options n = 13.

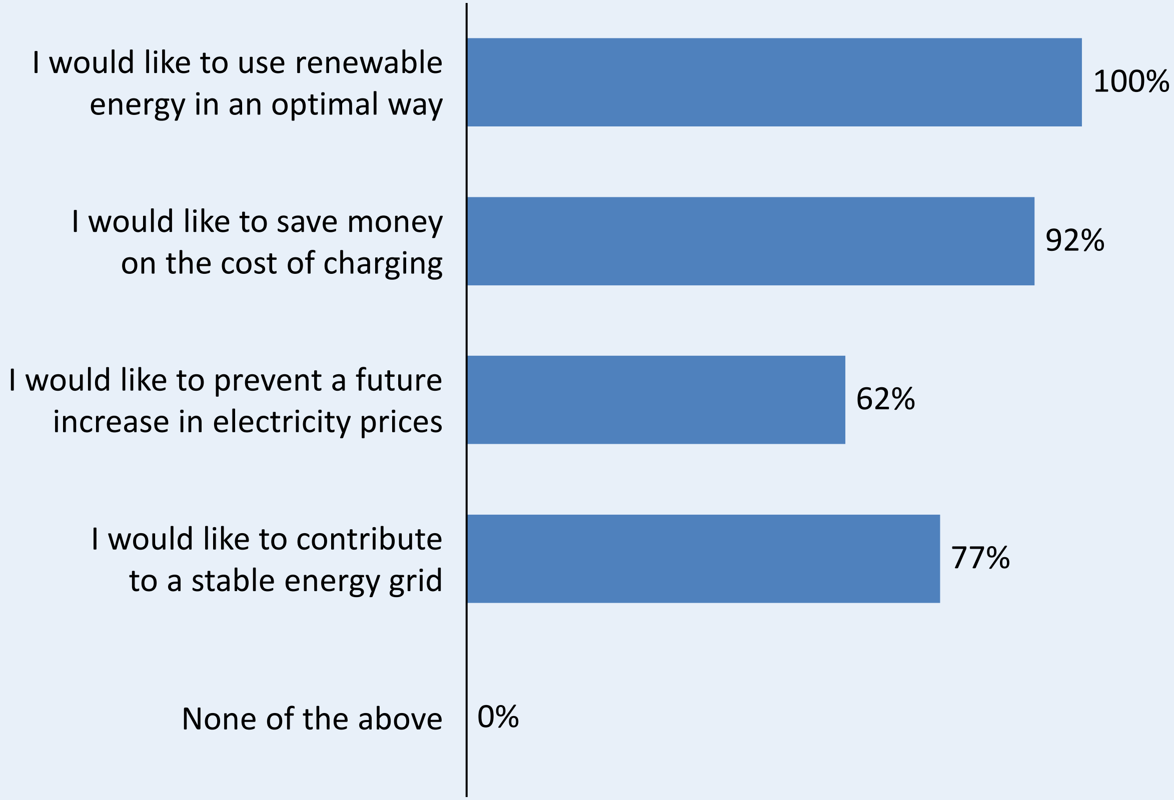
The main motivators for public smart charging are: 1) optimal use of renewable energy and 2) short-term cost savings

Task 1

Key findings from the poll

- **Most of the advantages of smart charging presented in the poll are viewed as important**
- **All participants felt that using renewable energy in an optimal way was important to them**
  - The optimal use of renewable energy was more widely considered important than contribution to a stable energy grid, although most participants (77%) were additionally motivated by this advantage
  - Participants are early EV adopters and might be more likely to be motivated by 'green' advantages than the general population[1]
- **The vast majority of respondents (92%) consider the possibility of saving money on the cost of charging as important**
- **Other advantages identified by participants** were:
  1. Specifically reducing the requirement for fossil fuels on the electricity system at peak times
  2. One participant noted that since smart charging would require longer charging times, local authorities would likely relax restrictions on on-street charge point plug-in times

Question: Which of the following advantages of smart charging are important to you?
Respondents asked to select all options they viewed as important.
Percentage of respondents who selected each option shown* **(n = 13)**



1) Element Energy consumer segmentation work for DfT (2,000 GB car buyers, 2015) showed early adopter segments had stronger 'green attitudes' than the other consumer segments
* Note: sum of responses higher than 100%, as respondents could select multiple options.

The optimal use of renewable energy is more important to on-street charge point users than cost savings

Task 1

Key advantages of smart charging

1

Optimal use of renewable energy is a major motivator to smart charge at public charge points

- When asked which of the advantages of smart charging were **most important** to the focus group participants, optimal use of renewable energy was the **most frequent response**

'It's not just a token thing of using renewable energy, but also about attacking the use of fossil fuels in the system'

2

Charge point users consider optimal use of renewable energy as more important than cost savings

- Although most participants (92%) consider cost savings as an important advantage of smart charging, optimal use of renewable energy was cited as the **most important advantage** more frequently
- This is consistent with results from Amsterdam's Flexpower trial, which found that Dutch EV drivers ranked optimal use of sustainable energy as more important than financial advantages[1]

'Electric cars are so much cheaper to run than old combustion engine cars that for cost to be the driving factor it would have to be a big differential'

3

Short term savings in charging cost was considered more important than reducing the cost of electricity in the long-term and for everyone

- No participant viewed the long-term reduction of electricity cost as the most important advantage

'Preventing a future increase in electricity price is not a primary motivator for me … saving money on the cost of charging is more important'

Source: 1) Behaviour research report, Flexpower, Amsterdam University of Applied Sciences, ElaadNL (May 2020)

Discussions of Focus Groups results with stakeholders: 1) Brighton & Hove City Council

Brighton & Hove City Council

Task 1

Focus group findings were presented to Brighton & Hove City Council and key issues with current infrastructure discussed

1

The Council confirmed that infrastructure reliability had been an issue but infrastructure is new and has been recently improved

- Key problem was a **complicated process to start charging sessions,** which has now been streamlined
- **Reliability** of the lamppost chargers has been improved

2

Some cases of vandalism of on-street charge points have been observed but vandalism of plugged-in cars is not common

- Charge point vandalism has impacted user experience

3

Issue of charge points blocked by ICEVs is recognised and the council plans to introduce more EV-only parking bays[1], prioritising those with high requests from residents

- However, **parking is at a premium** in Brighton, making it difficult to introduce EV-only bays: waiting lists for parking permits, and petitions against introduction of charge points in certain streets
- Council pointed out that **parking zone restrictions do not apply to existing EV-only bays**, which was not clear to focus group participants

4

Council is making efforts to future-proof their public charging infrastructure

- On-street charging infrastructure is **capable of load balancing** (able to respond to DSO signal) but functionality not used yet
- Looking at **pre-booking** for rapid charge points, particularly to accommodate **volume of visitors** to Brighton at certain times of the year

Note: 1- Currently 22 lamppost chargers are in EV-only bays in Brighton and council aims to have 50 by May 2021

Discussions of Focus Groups results with stakeholders: 2) Trojan Energy

TROJAN ENERGY LTD

Task 1

Focus group findings were discussed with Trojan Energy (operator of novel on-street charge point technology) to gain their perspective on the feasibility of the users' preferences around smart charging

Trojan Energy can share collected data with customers via the cloud

1. Trojan Energy recognised the need to ensure **billing information is accessible to all customers**
   - Plan to look into providing invoices by different means, including SMS, email and via a smartphone app

2. Noted that it is **difficult for a CPO to provide an estimate of current state of charge (SoC)**
   - Older EV models do not communicate their SoC with charge points (unless rapid)

3. **EV models additionally vary their charging behaviour, making it difficult to estimate required charging time**
   - Trojan Energy will collect data on different models' charging behaviours and smart charging needs, to provide better estimates

- Trojan Energy plan to give users control over smart charging

4. **Trojan Energy plan to offer charge point users the ability to opt out of smart charging sessions**
   - Offer an '**override button**' to opt out of smart
   - Could additionally ask users to confirm they are happy to smart charge at the start of charging sessions

5. **Offering a guaranteed minimum state of charge would be difficult** due to older EV models not communicating SoC and the varied charging behaviours of EV models
   - Trojan Energy are considering this as a **future functionality**

Note: Trojan Energy are considering introducing a virtual queue system for their charge points, as suggested in a focus group

Discussions of Focus Groups results with stakeholders: 3) On-street charge point provider

Task 1

Issues currently faced by focus group participants were discussed with an on-street charge point provider, in addition to the feasibility of the participants' conditions for using smart public charge points (information, level of control and assurance)

Charge point provider believe current concerns will disappear over time

1 **They do not see the degree of infrastructure unreliability reported during the focus groups**

- Charge points very rarely fail during charging sessions, but there are issues with users **unable to start sessions**

2 Actual cases of **vandalism of cars are rare**, but vandalism of the charge point itself common

3 **Charge point provider recognises the importance of clear, accessible billing from CPOs to customers**

- **Tariff clearly advertised** to users, but agree population likely to **struggle to translate** this to total price

4 **Believe that frustration with low power of on-street chargers will lessen as infrastructure becomes more commonplace**

- Lampposts offer a **low cost, convenient option** for residential charging, different from rapid offering

- User development costs too high to justify smart charging

5 **Provider believes that customers will need a high degree of control over their smart public charging sessions**

- Lamppost charge points marketed as an alternative to private home charge points: consequently users require a similar level of control as they would have with private charge points

6 However, **development time required by providers to offer charge point users the desired functionality is not yet logistically feasible**

- For a CPO, the **challenge in smart charging** is ensuring a **positive user experience** (information and control needs)
- Requires **high development time and costs**

7 **Smart public charging will become necessary**

- However, **utilisation not high enough yet** to justify the **development costs**

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If public charging infrastructure were plentiful and reliable, users would be willing to smart charge provided they have sufficient information and control

Task 1

This research has provided useful insight into consumer attitudes towards smart charging at public, on-street charge points* but also an understanding of the barriers faced by current users of on-street charging infrastructure, which is still in its infancy.

Existing on-street charging infrastructure is not trusted

- Many participants had experienced failure of on-street charge points while charging
- Most participants were frustrated by the low-power and limited availability of on-street charge points

Smart charging exacerbates existing concerns

- Biggest concern about public smart charging was the possibility of the battery not being charged when needed and long charging times, and most participants linked this to the low power and availability of on-street charge points
- If on-street charging infrastructure was plentiful, reliable and of sufficient power, users would be happy to smart charge

On-street EVCP users want information about their smart charging session

- Most participants viewed practical information about their smart charging session, such as charging time, as essential
- Other information, such as quantification of renewable energy consumed, were viewed as interesting but not essential

Level of control and assurance needed at smart public EVCPs

- All participants were willing to smart charge at public charge points, as long as they could either opt in or opt out
- A guarantee of a minimum percentage charge from smart charging session was a prerequisite for participating in public smart charging

Public EVCP users motivated by renewable energy

- Participants consider optimal use of renewable energy as a more important advantage of smart charging than cost savings
- Short term cost savings were considered more important than long-term electricity price reduction

* The scope of UK Power Networks' research includes public, residential car parks. We anticipate concerns to be similar to those of on-street charge point users. Two participants discussed their use of local, destination car parks. It was negatively viewed, largely due to parking restrictions.

Focus groups gave new insight into consumer attitudes towards smart charging, but some key results are in alignment with other consumer findings

Task 1

Public smart charging trials have similar consumer findings

1

Largest concerns amongst focus group participants related to the low charging power of on-street charge points

- **Amsterdam's Flexpower trial** found that smart public charge point users were concerned that their EV battery would charge too slowly[1]
  - Taxi drivers avoided the smart public charge points, due to concerns of long charging times[1]
- During a **smart charging trial in a public car park in California**, the power was increased as EV drivers felt the speed was unacceptable[2]

2

Focus group participants consider optimal use of renewable energy more important than cost savings

- Consistent with findings from **Amsterdam's Flexpower trial**, in which EV drivers ranked optimal use of sustainable energy as more important than financial advantages, although drivers were also somewhat motivated by cost savings/rewards[1]

3

Results of UK Power Networks' previous research[3] [on non-EV owners] around on-street charging needs aligned with focus group findings about the importance of environmental impact to consumers and concerns around on-street charging security[3]. UK Power Networks' research also suggests that future EV drivers will be willing to be **flexible** with their charging behaviour[3].

Government consultation aligns with focus group findings

The UK government has launched a consultation on public EV charging, with a focus on improving the consumer experience across four areas[4], which align with the focus group findings:

1. **Making it easy to pay:** Convergence towards fewer apps, and a payment option not reliant on use of a smart phone
2. **Opening up charge point data**
3. **Using a single payment metric (p/kWh):** Standardisation of payment structures to p/kWh, to aid user understanding
4. **Ensuring a reliable network:** Charge point operators should ensure a minimum 99% reliability across the infrastructure

Our focus group key concerns around existing on-street charge points included the **inconsistent and unclear pricing structures**, the **lack of interoperability** and **infrastructure unreliability**

Source: 1) Behaviour research report, Flexpower, ElaadNL (2020); 2) Demonstrating Plug-in Electric Vehicles Smart Charging and Storage Supporting the Grid, UCLA Smart Grid Energy Research Center (2018); 3) UKPN EV on-street charging trial, UKPN (2021); 4) Open consultation: the consumer experience at public chargepoints, DfT, OZEV (2021)

Network Operators should plan for flexibility services from public chargers, but CPO engagement is required to ensure infrastructure is sufficient to allow for smart charging

Task 1

- Smart charging is deemed **advantageous** by public charge point users, and it is likely that smart public charge points will be used by customers
- UK Power Networks can plan for **flexibility services from public charging infrastructure** reducing the impact of charging on the grid in the future

- **Existing problems with on-street charging infrastructure** must be solved before consumers will be willing to smart charge
- **Communication with charge point operators** is needed from the DNOs to ensure that charging infrastructure is sufficient and reliable, and payment systems are clear before smart charging is integrated

- On-street charge point users want charging infrastructure capable of **sufficient power** for charging in reasonable timeframes (generally 11kW or above)
- This will in turn put **additional pressure on the network** and increase the need for smart charging from public charging infrastructure

DNO = Distribution Network Operator

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Data on charge point location and utilisation were used to inform this project

Task 2

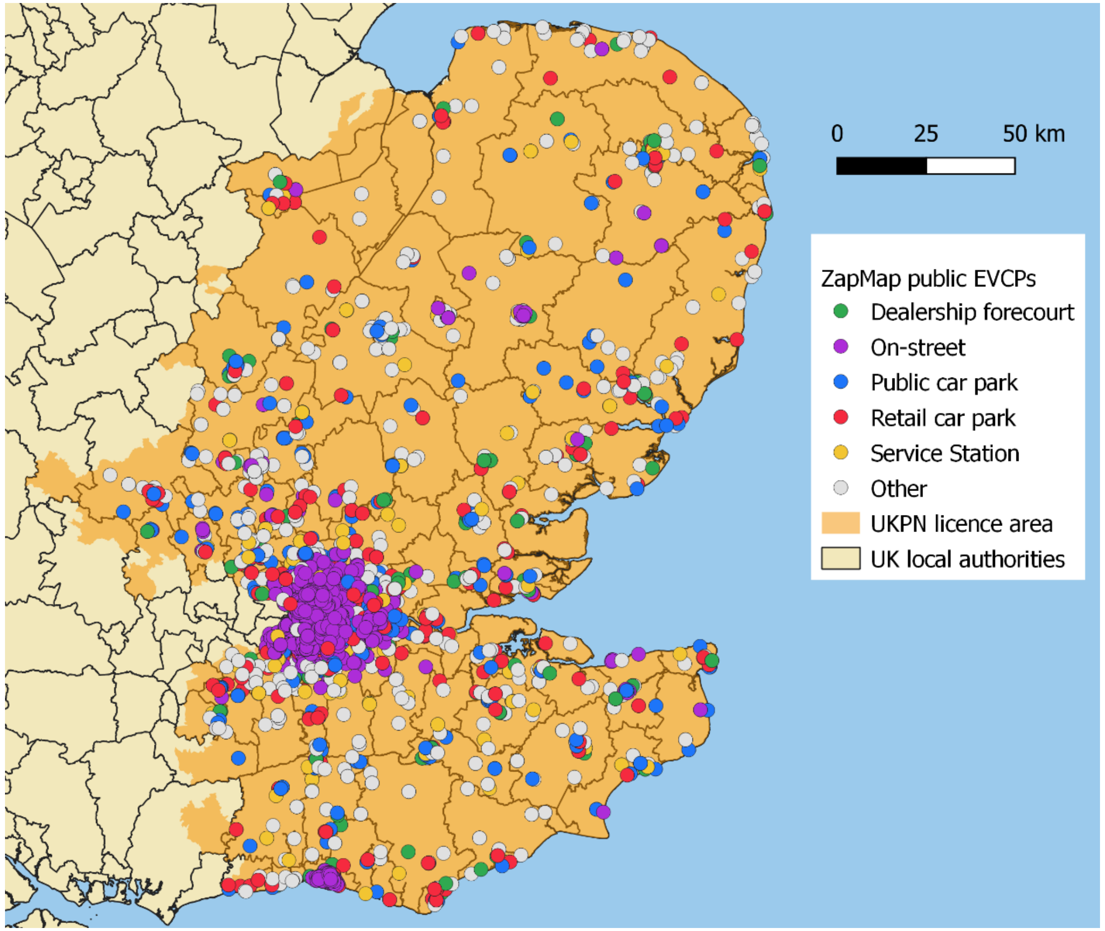
Data sources available for the project

- **ZapMap data – locations of public EVCPs in the UK Power Networks licence area**
  - Gives locations of all public charge points listed by ZapMap up to September 2020
  - The location type of each point is provided. The top 5 location types are on-street, public car park, retail car park, dealership forecourt, service station
  - For each location, the number of slow, fast, and rapid connectors are provided

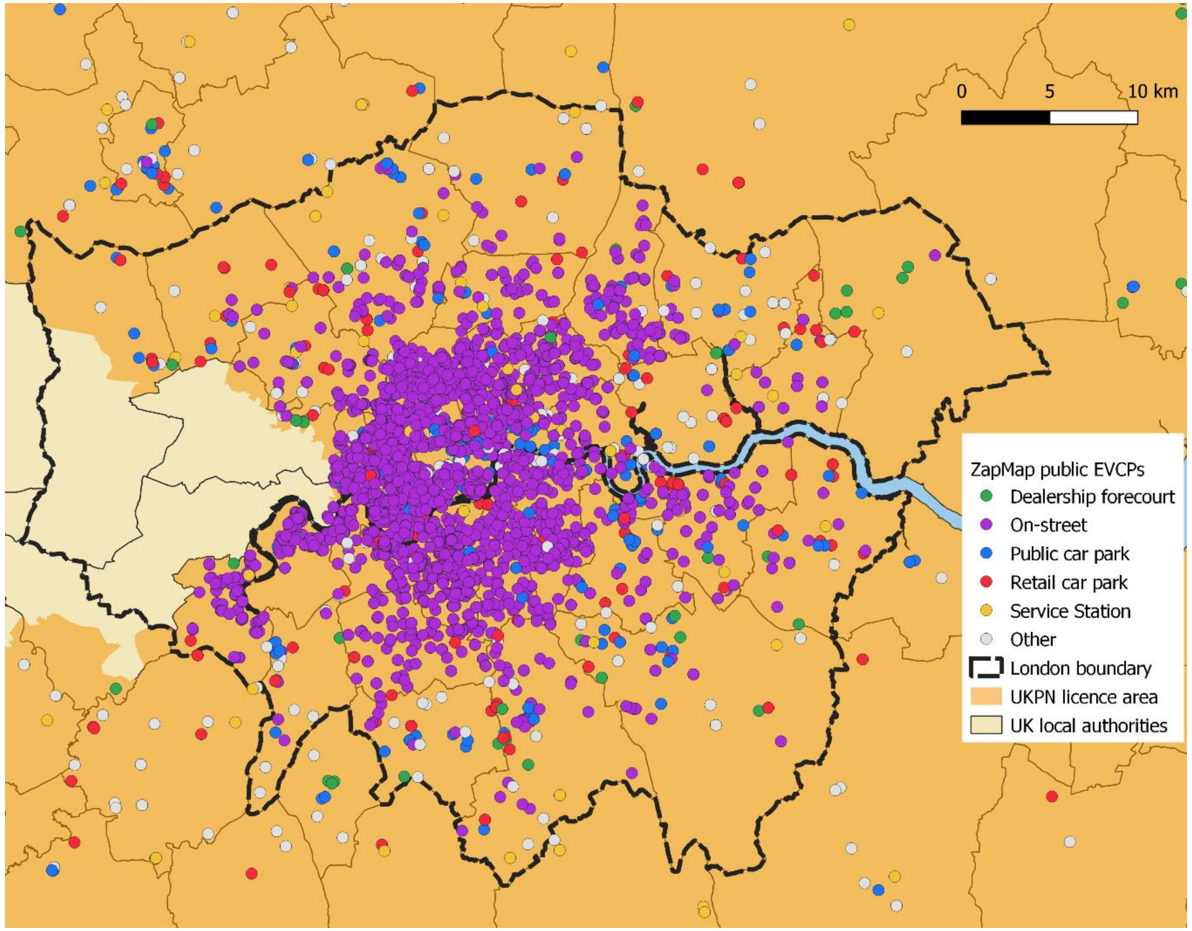
- **Large public on-street charge point dataset**
  - Data has been provided for March 2019 – March 2021
  - Full charging event data was provided for a random sample of the operator's charge points – 100,000 events from 1,300 charge points

ZapMap data shows there is a high concentration of on-street chargers in London and in Brighton, other charger types are more evenly distributed across the UK Power Networks licence area

Public EVCPs across UK Power Networks licence area



Zoom on London



4,900 EVCPs in London, 200 in Brighton and Hove LA, 6,800 in UK Power Networks licence area

ZapMap public EVCP locations were provided by UK Power Networks, 2020 data

LA: Local Authority

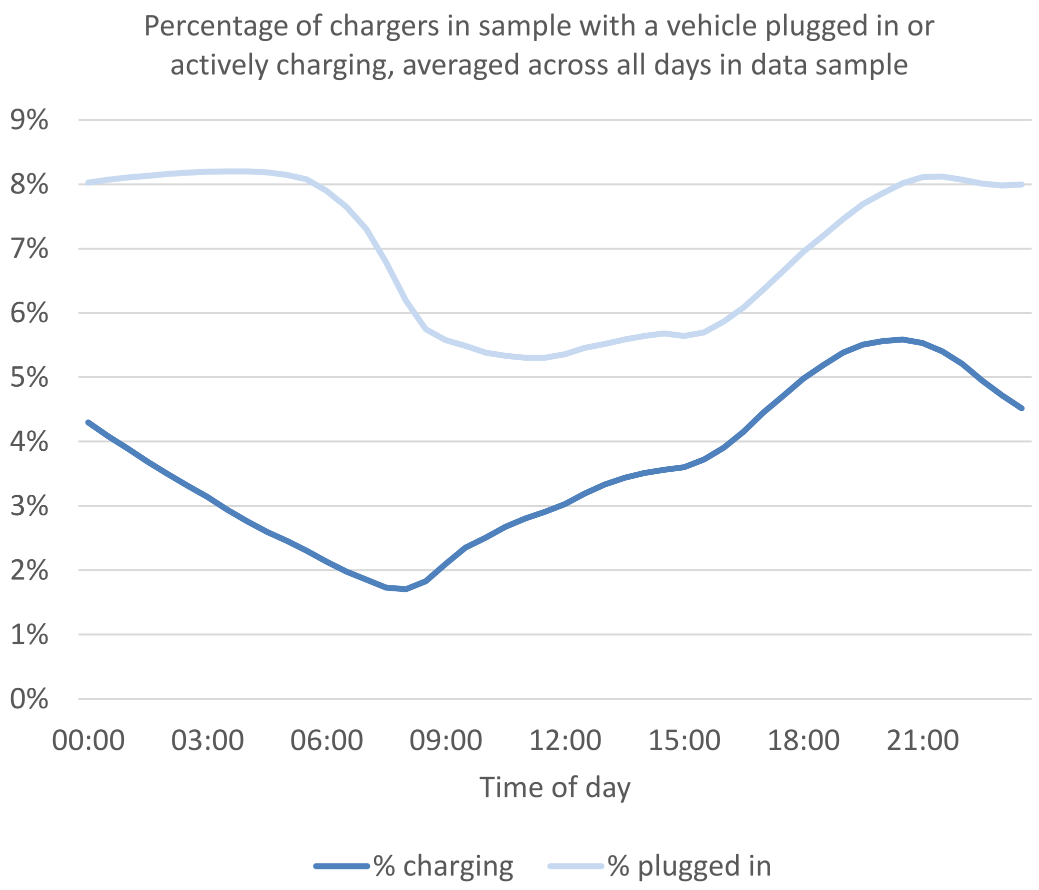
Data provided by a large charge point operator was used to create EV plug-in and charging profiles

Task 2

Data context

- Full charging event data was provided for a sample of 1,300 chargers and 100,000 charge events
- Data is from March 2019-March 2021 – COVID has not had a large impact on the data from March 2020 onwards, as shown in the following slides
- Charging event data has been further analysed to gain additional insights, such as the distribution of charging event duration and how this varies throughout the day

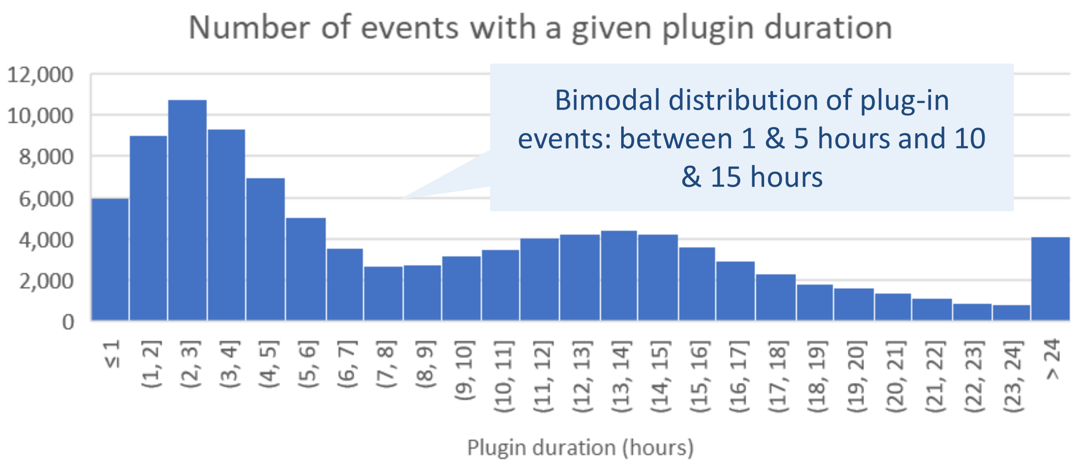
Plug-in and charging profiles

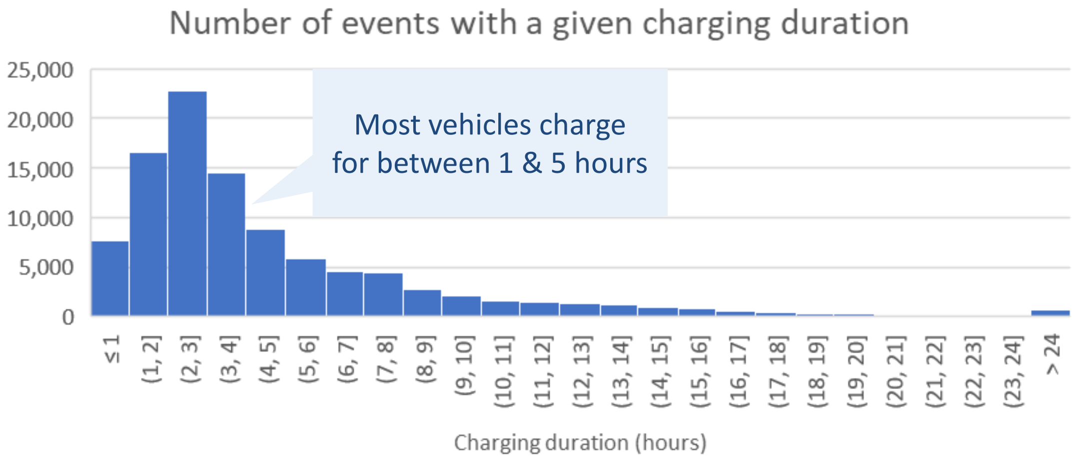


Bimodal distribution of plug-in event durations – shorter events are predominantly during the day, longer events predominantly overnight

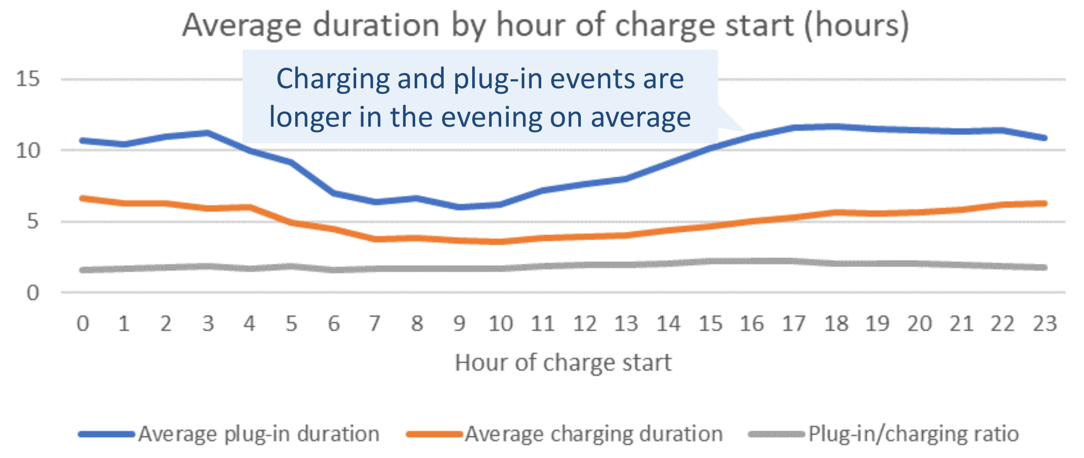
Task 2

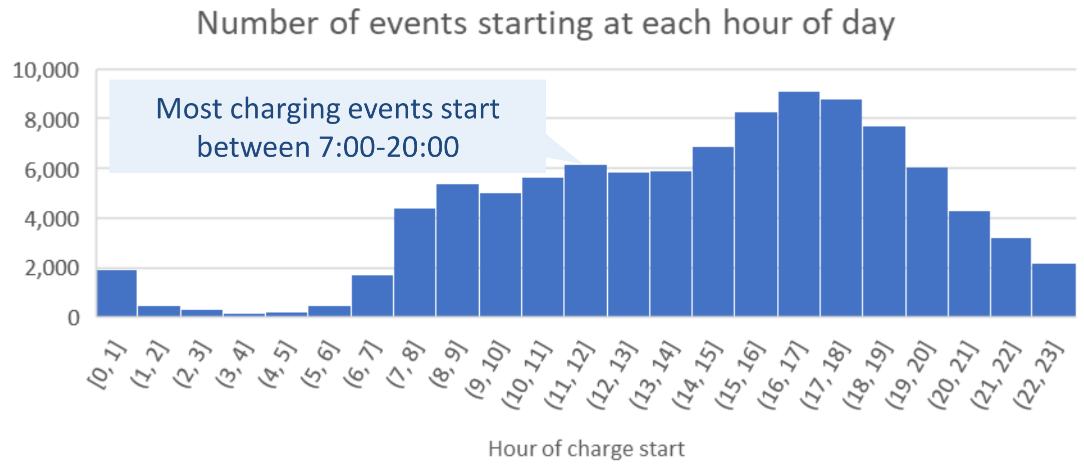
Distribution of plug-in and charging durations





Variation by hour of day

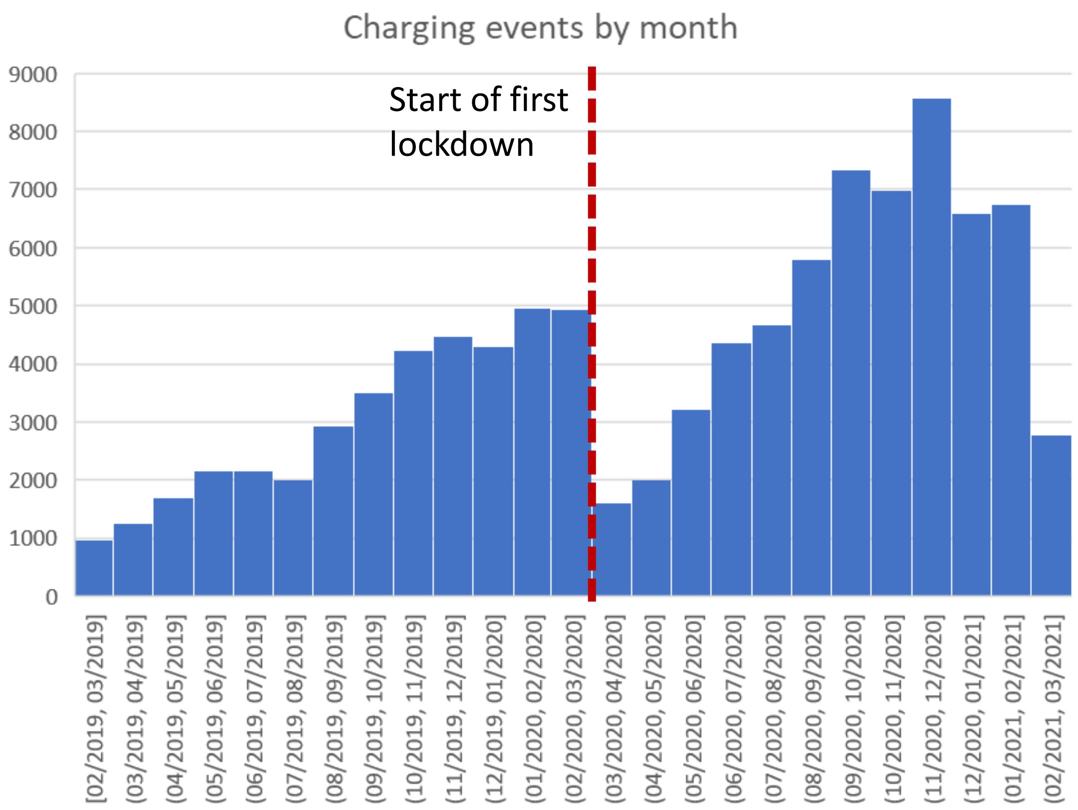




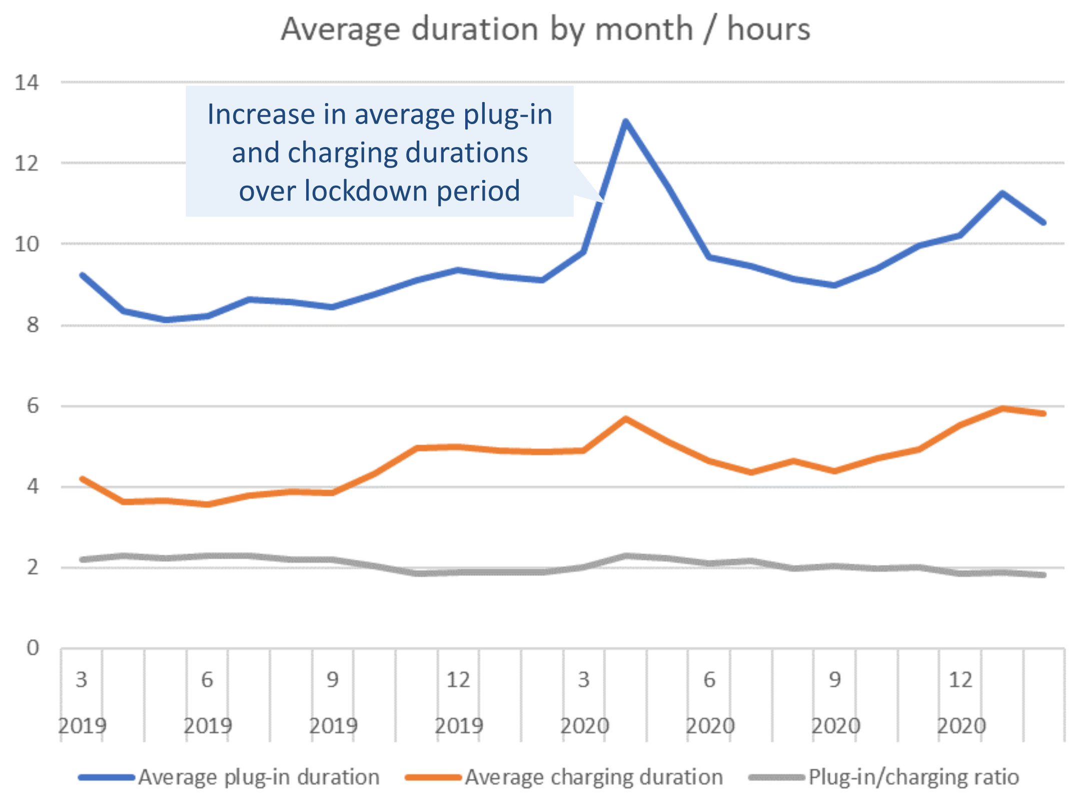
Impact of COVID-19: while behaviour did change from March-May 2020 and Jan 2021, very few events in these periods means effect on the overall average charging & plug-in profiles is negligible

Task 2

Effect on number of events



Effect on event duration



Note the effect of longer charging and plug-in durations over the lockdown period has been identified as negligible for two reasons:

1. The increase in charging and plug-in duration only lasts for 3 months (10% of the total time period we have data for)
2. Within the affected months the number of charging events is about half what it was in previous months

This means that overall only about 5% of charging events show this increase in charging and plug-in duration

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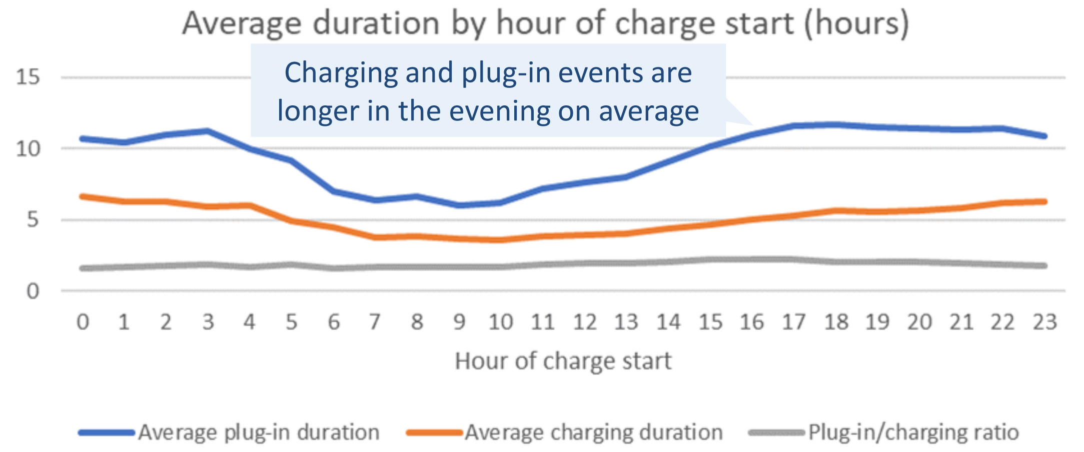
Analysis of charging event data suggests there is potential for demand to be shifted using on-street smart charging

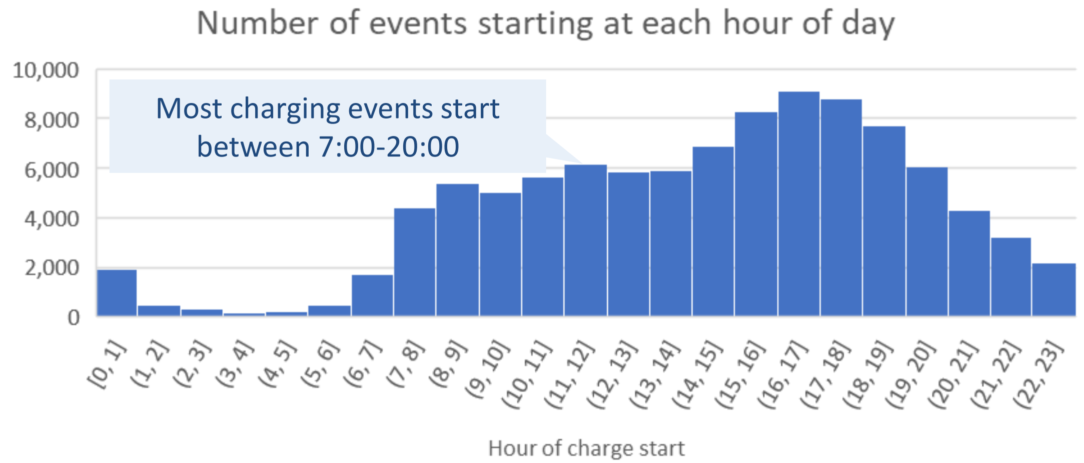
Task 2

Findings from charge event data

- A sample of charging event data containing 100,000 events from 1,300 charge points from a large charge point operator has been provided for March 2019 – March 2021
- The most common times for charging events to start are between 14:00 and 20:00
- Events starting between these times have an average plug-in duration of > 10 hours, while charging duration is ~5 hours
- Underlying domestic load on the distribution network tends to peak in the early evening – therefore there is potential to shift on-street charging demand away from this time and into the overnight period to minimise peak demand
- On-street charging demand shifting may not be as effective on assets dominated by industrial and commercial loads, which tend to peak during the day

Variation in charging behaviour by hour of day





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Study substations have been selected based on several criteria of interest

Task 2

Substation selection

A selection of **5 secondary substation use cases** have been identified:

- One of the Brent substations from the STEP trial
- A central London substation from a densely populated, low off-street parking access area
- A central Brighton substation
- A suburban Brighton substation
- A rural substation with (comparatively) low off-street parking access

>> For each of these, the upstream primary substation has been analysed when firm capacity is approached or breached on this primary. If not, another primary substation from the same use case which breaches constraints has been chosen

Selection criteria

- High level of on-street residential charging demand – this depends on several factors, including:
  - High EV car and van uptake (taxis and PHVs do more en-route charging so less important)
  - Low levels of off-street parking access
  - High numbers of domestic customers
- A diverse set of substations, from cities, towns, and more rural areas
  - Brighton and London have been included to give two different city contexts
- Expected to approach or breach firm or rated capacity in the next 10 years

SFS: Strategic Forecasting System, a load forecasting model currently being developed by Element Energy for UK Power Networks

10 substations have been selected for analysis

Task 2

Substation names

- Substations have been selected based on data and forecasts from preliminary strategic forecasting system (SFS) model – development of SFS is ongoing so projections will continue to be refined
- Note the upstream primary for the rural secondary substation has not been considered as it does not breach constraints – an alternative rural primary substation was selected

Brent
- Secondary: Kensal Rise
- Primary: Kimberley Road

Central London
- Secondary: St Barnabas Road E17
- Primary: Waterloo Road

Central Brighton
- Secondary: Powis Grove
- Primary: South Hove

Suburban Brighton
- Secondary: Bevendean Road
- Primary: Queens Park

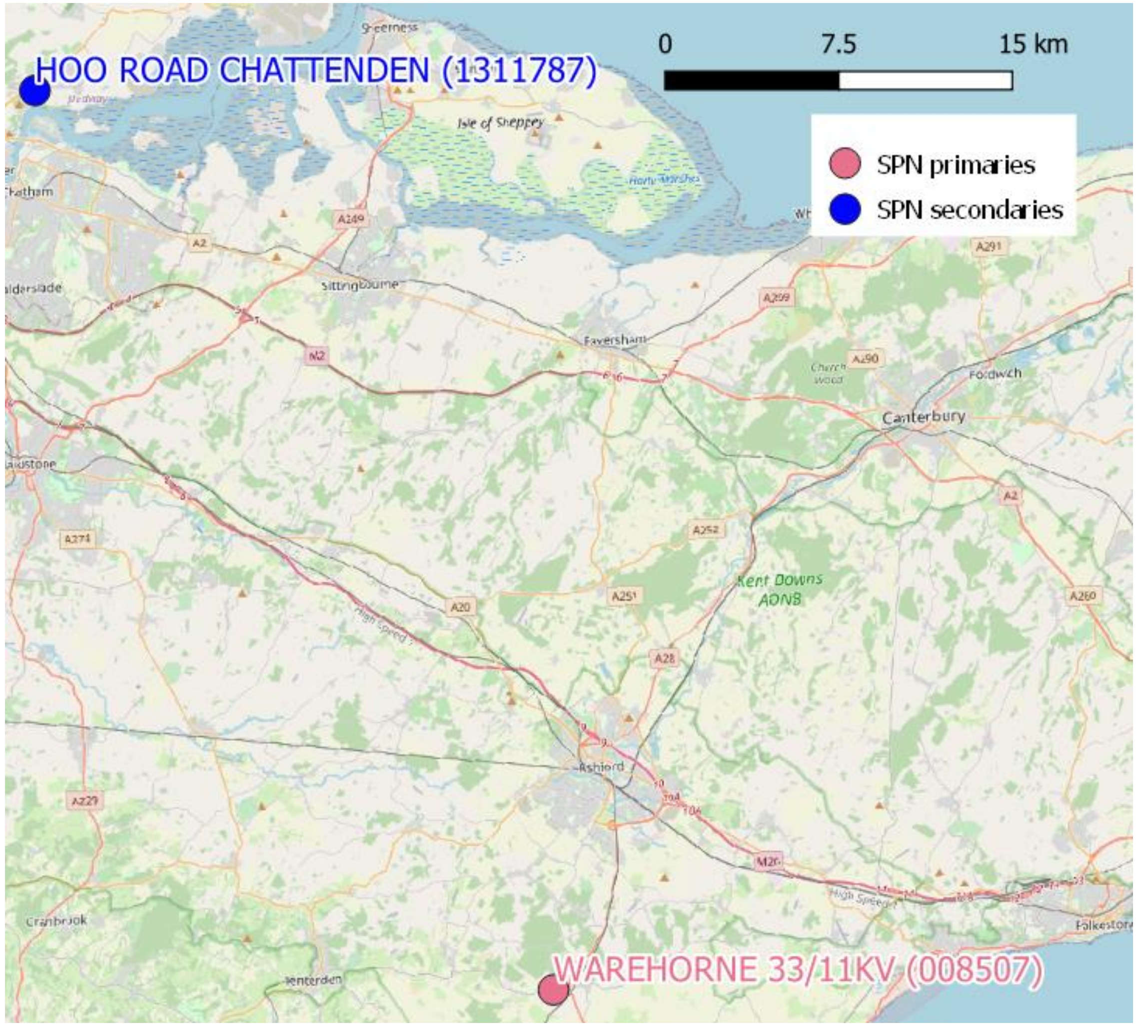
Rural
- Secondary: Hoo Road Chattenden
- Primary: Warehorne

Substation locations

Substation locations are shown in more detail in maps below

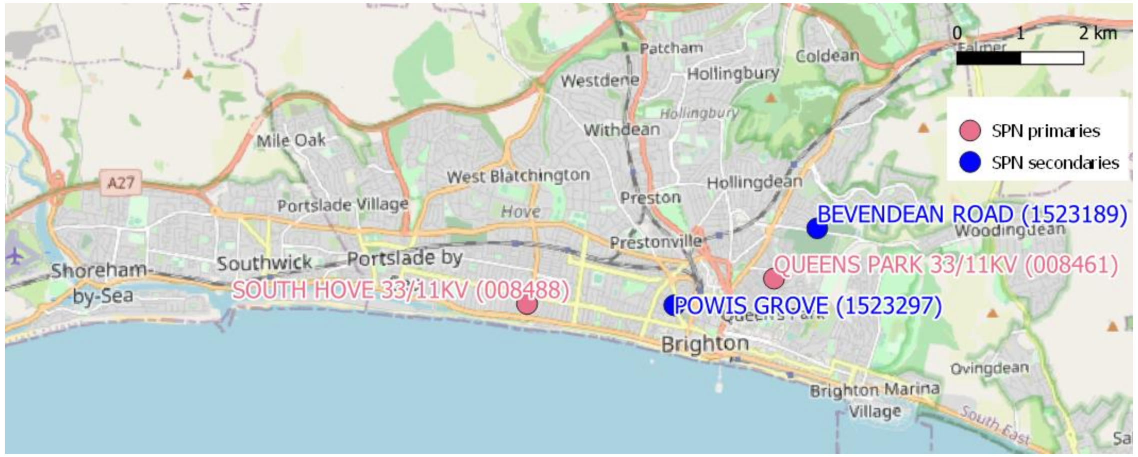
Task 2

Zoom on south-east (rural substations)

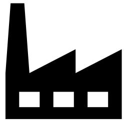
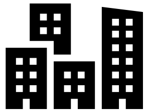
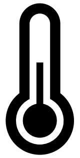
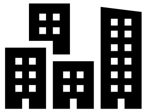
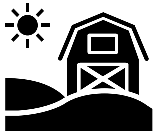


Zoom on London

Zoom on Brighton



The selected substations have been categorised into archetypes

Use case	Level	Archetype	Characteristics	
Brent	Secondary	Urban, mostly I&C	• Densely populated • High I&C base load • High on-street parking	
Brent	Primary			
London Central	Secondary	Urban, mostly domestic	• Densely populated • High domestic base load • High on-street parking	
London Central	Primary			
Brighton Central	Secondary	Urban, mostly domestic, night heating	• Densely populated • High domestic base load • High on-street parking • High overnight heating demand	
Brighton Central	Primary	Urban, mostly domestic	• Densely populated • High domestic base load • High on-street parking	
Brighton Suburban	Secondary	Suburban, mostly domestic	• Intermediate population density • High domestic base load • Intermediate on-street parking	
Brighton Suburban	Primary			
Rural	Secondary	Rural, mostly domestic	• Sparsely populated • High domestic base load • Low on-street parking	
Rural	Primary			

Demand projections have been considered for the four DFES scenarios

Task 2

Scenario World	Steady Progression	System Transformation	Consumer Transformation	Leading the Way
Net-Zero by 2050?	No	Yes	Yes	Yes
Core Demand				
Energy efficiency	Low	Medium	High	High
Building stock growth	Medium	Medium	Medium	Medium
Low-Carbon Transport				
Cars and vans: decarbonisation	Low	Medium	Medium	High
Heavy duty vehicles: decarbonisation	Baseline	Hydrogen world	High electricity	Fast rollout
Decarbonised Heating				
Heating technologies	Low electrification	Medium electrification with conversion to H_2	High electrification	Early high electrification
District heat uptake	Low	Medium	High	High
District heat supply	Baseline	Decarbonised gas	High electrification	Decentralised
Battery Storage				
Domestic battery storage	Low	Medium	High	Medium
I&C behind-the-meter battery storage	Medium	Low	High	Medium
Co-located battery storage	Low	Medium	Medium	High
Flexibility				
Flexibility	Low	Medium	High	High

Each of the DFES scenarios has been analysed – results from the maximum peak demand scenario for each substation have been documented in this report

Flexibility includes user-controlled EV smart charging, which is applied to vehicles parking off-street at home only. The impact of externally managed smart charging (i.e. not managed by the EV driver) and V2G charging is not modelled in this study.

The selected secondary substations are likely to breach constraints before 2030

Task 2

Secondary

Peak demand and constraints expected at selected secondary substations

Use case	Substation name	Scenario*	Peak demand in 2030 / kW	Rated capacity / kVA	On-street residential peak demand in 2030 / kW	Year constraints first expected
Brent	Kensal Rise	System Transformation	598	500	7.47	2021
London Central	St Barnabas Road E17	System Transformation	541	500	135	2028
Brighton Central	Powis Grove	System Transformation	546	500	96.6	2021
Brighton Suburban	Bevendean Road	System Transformation	521	500	19.2	2021
Rural	Hoo Road Chattenden	System Transformation	56	50	5.36	2028

**For each substation the DFES scenario which leads to maximum peak load has been analysed*

Source: analysis of SFS results

In most cases the upstream primary substation from each of the selected secondary substations breach constraints by 2030

Task 2

Primary

Peak demand and constraints expected at selected primary substations

Use case	Substation name	Scenario*	Peak demand in 2030 / MW	Rated capacity / MVA	On-street residential peak demand in 2030 / MW	Year constraints first expected
Brent	Kimberley Road	Consumer Transformation	52.8	43.7	5.30	2021
London Central	Waterloo Road	Consumer Transformation	59.5	57.3	6.10	2030
Brighton Central	South Hove	Consumer Transformation	22.9	21.7	2.69	2030
Brighton Suburban	Queens Park	Leading The Way	20.7	22.5	2.61	2031
Rural	Warehorne**	System Transformation	4.40	3.59	0.212	2027

**For each substation the DFES scenario which leads to maximum peak load has been analysed*

***Note the upstream primary from the selected rural secondary substation has not been studied as it does not breach constraints. A different rural primary substation which breaches constraints has been selected instead*

Source: analysis of SFS results

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Study substations have been selected based on several criteria of interest, including high predicted on-street charging demand and capacity breaches in next 10 years

Task 2

Substation selection process

- A pair of primary and secondary substations from 5 distinct use cases were analysed in this study
- Use cases were selected to be representative of the range of EV uptake, reliance on on-street charging, and network loading conditions across UKPN's network. Factors captured included:
  - Level of off-street parking access
  - Rural, suburban, and urban locations
  - Substations with predominantly domestic and predominantly I&C customers
- Within each of these use cases, substations were selected which are expected to show constraints in the next 10 years and with high levels of on-street residential charging demand

Substation locations

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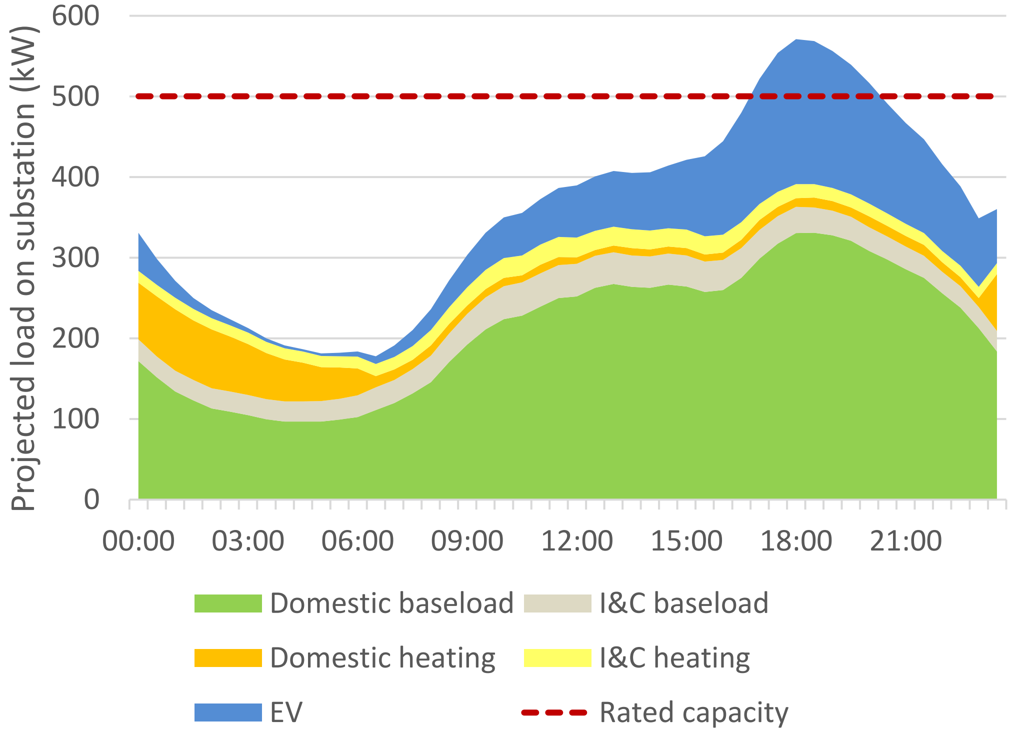
Demand can be broken down by class for each of the substations

Task 3

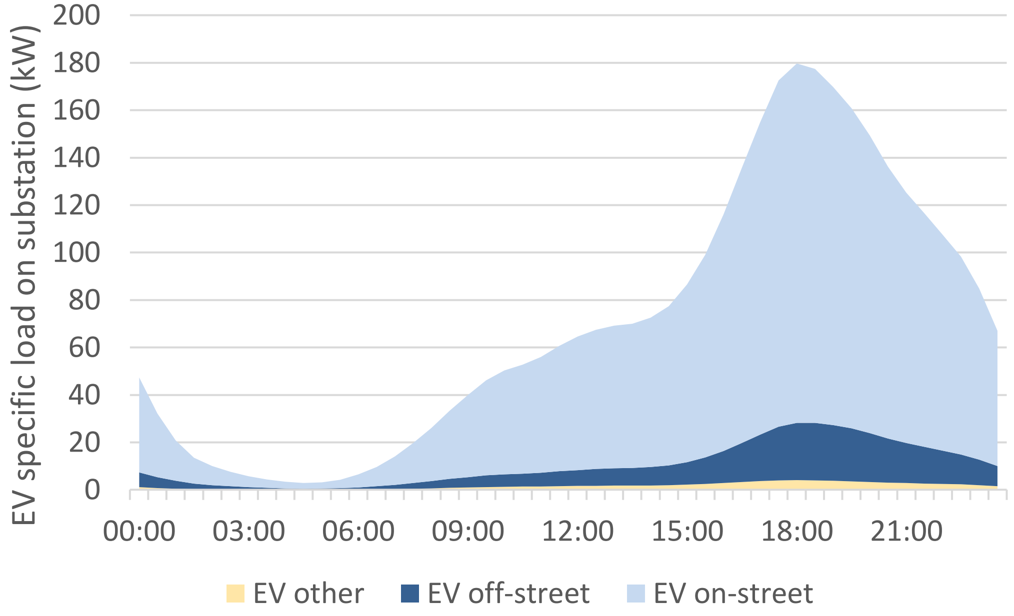
Breakdown of peak day load profile by demand class (kW)

- Outputted directly from SFS
- Load profile shown below is for the London Central secondary substation in a peak winter day in 2030 for the System Transformation DFES scenario



Breakdown of EV specific demand (kW)

- EV demand is split into residential on-street, residential off-street and other EV demand, based on projected annual EV demand in each of these classes in the SFS
- On-street EV demand is assumed to be flexible throughout the day – maximum power at a given half-hour is constrained based on below plug-in profile



I&C = Industrial & Commercial

Plug-in profiles are used to inform how on-street charging demand could be shifted to lower overall peak load on a given substation

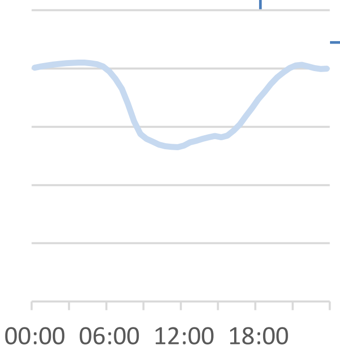
Task 3

Schematic demand shifting process

Plug-in profiles (produced in task 2)

Normalise to 1 kWh / day

Normalised plug-in profiles



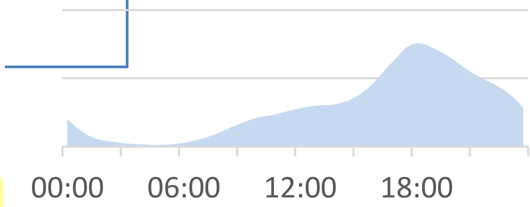
Scale by peak daily on-street charging demand

Maximum achievable charging demand at each half-hour

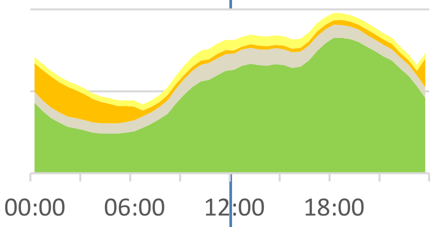
Peak day unmanaged EV load profile at a given substation

Scale by share of annual EV demand that is on-street

Peak day unmanaged on-street EV load profile at a given substation

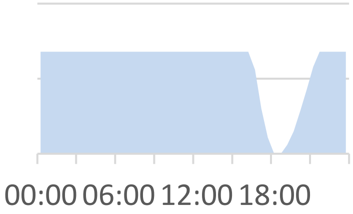


Peak day underlying load profile at a given substation



Distribute on-street charging demand throughout the day without exceeding maximum at any half-hour and while minimising total peak demand

On-street smart charging profile for the substation



Legend

- CPO data
- Data from SFS
- Intermediate calculation
- Output

CPO = Charge Point Operator

The following slides show the network impact of shifting on-street residential charging load

Minimum number of 7 kW EV charge points needed to meet peak smart on-street charging demand[2]. This provides context on how expected on-street charging demand translates to charger deployment and highlights areas where timed connections could be used to mitigate constraints

Task 3

For each of the selected substations, the following have been assessed:

1

Peak daily load profiles with **unmanaged** on-street residential charging in 2030

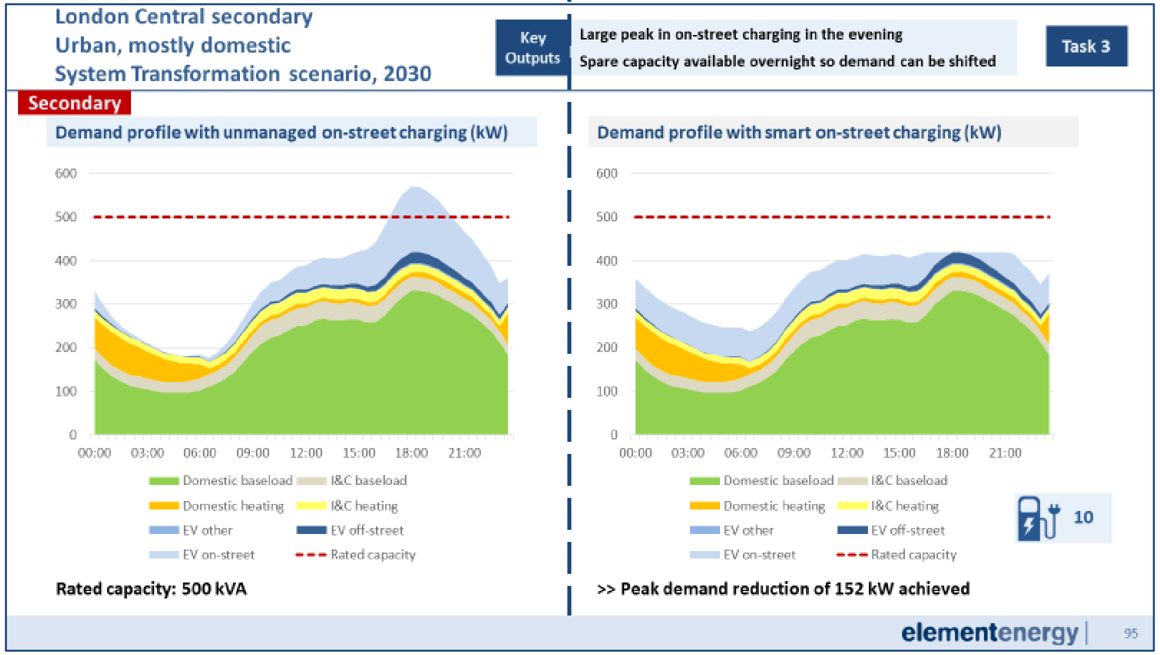
2

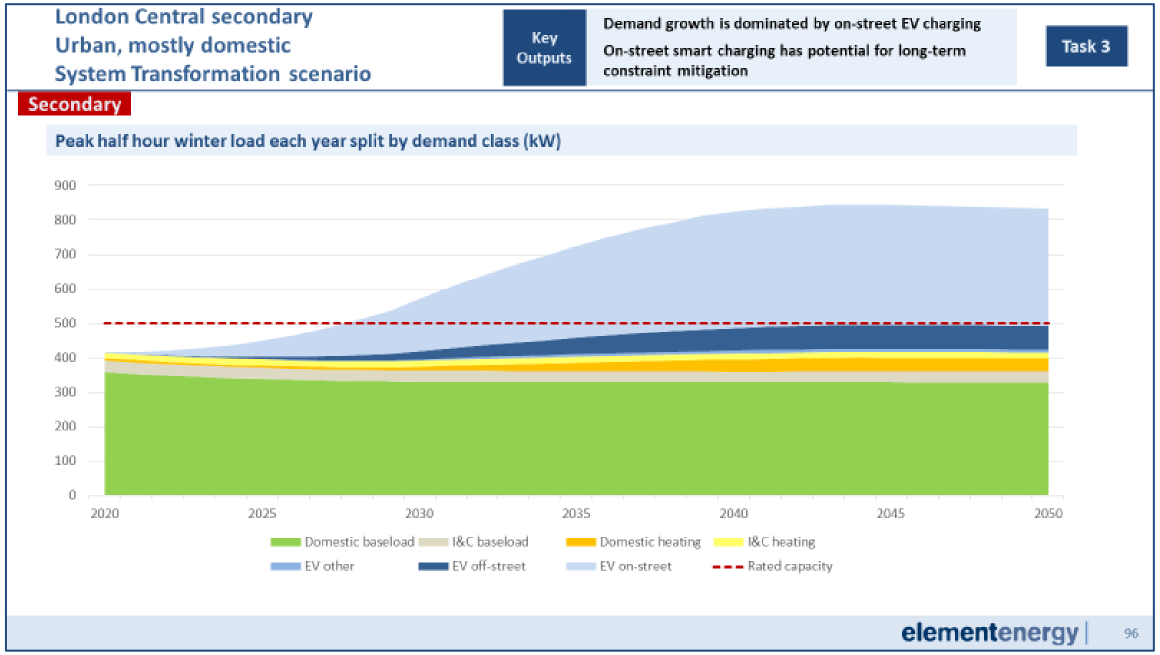
Peak daily load profiles with **smart** on-street residential charging in 2030[1]

3

Annual peak load split by demand class each year until 2050

Profiles which show interesting results are presented in the following slides. All other slides are in the Appendix





These profiles allow us to determine if on-street flexibility can be used to shift load away from peak and mitigate constraints

In cases where on-street flexibility can be used to shift load (assessed using profiles 1 and 2), this shows how long constraints can be mitigated

1: Load is redistributed by the process shown in the previous slide

2: Calculated using peak smart on-street charging demand, assuming 100% utilisation of charge points

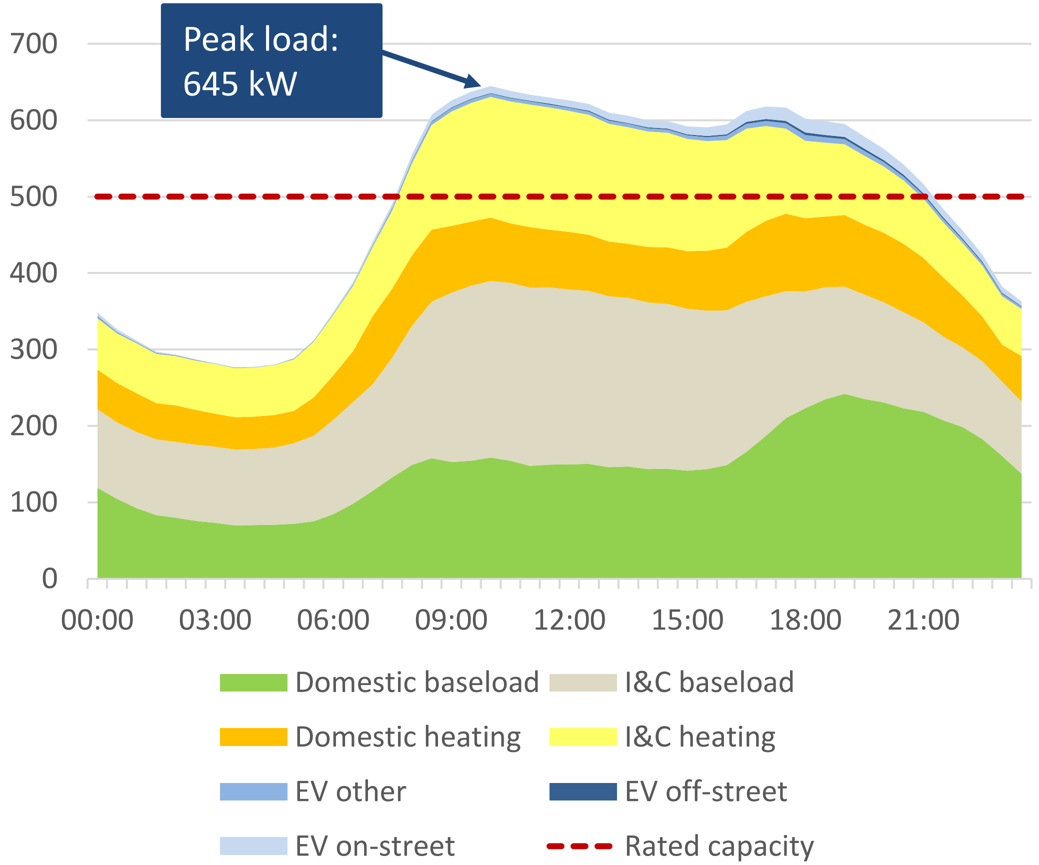
Brent secondary
Urban, mostly I&C
System Transformation scenario, 2030

Key Outputs	
	Dominated by I&C loads
	Very little potential for on-street smart charging in 2030

Task 3

Secondary

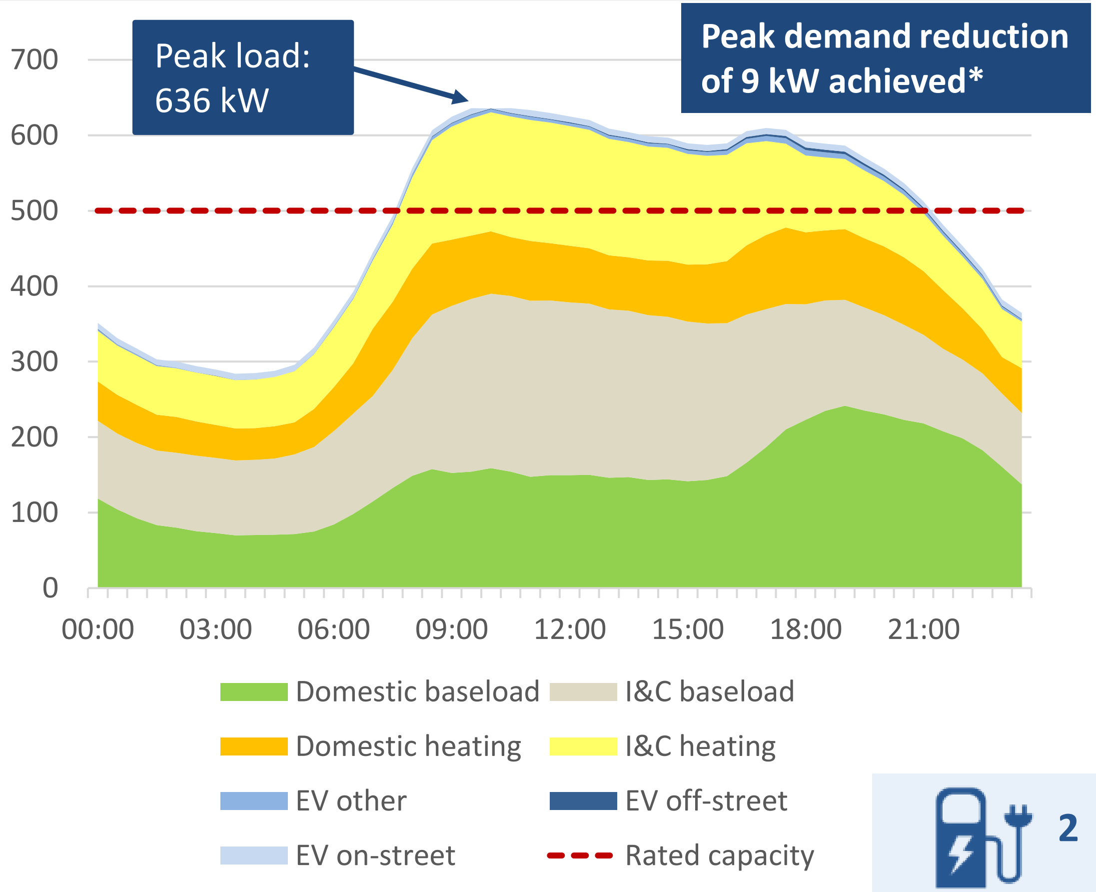
Demand profile with unmanaged on-street charging (kW)



Maximum unmanaged on-street charging demand: 18 kW

Rated capacity: 500 kVA

Demand profile with smart on-street charging (kW)



Maximum smart on-street charging demand: 8 kW

2

This represents the **minimum** number of 7 kW chargers needed

*Note this does not necessarily equate to the reduction in maximum on-street charging demand as the overall peak in load on the substation may not occur at the same time as maximum on-street charging demand

elementenergy
an ERM Group company

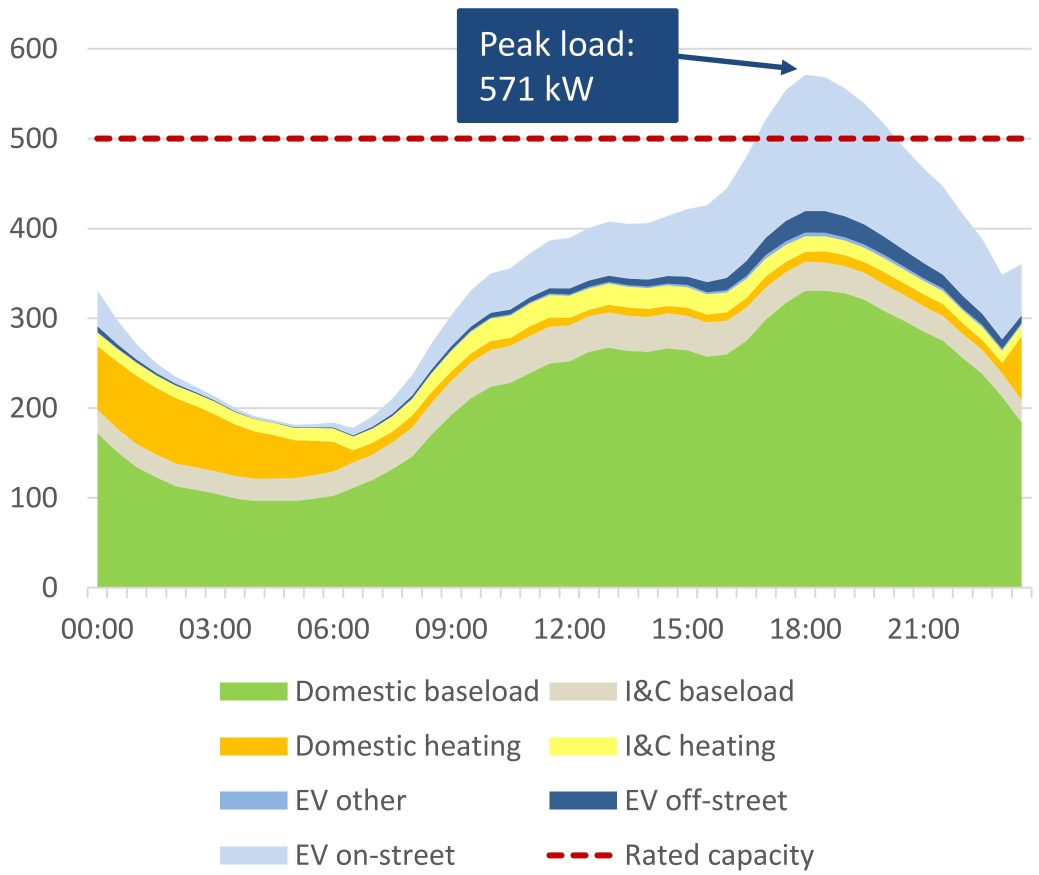
London Central secondary
Urban, mostly domestic
System Transformation scenario, 2030

Key Outputs	
	Large peak in on-street charging in the evening
	Spare capacity available overnight so demand can be shifted

Task 3

Secondary

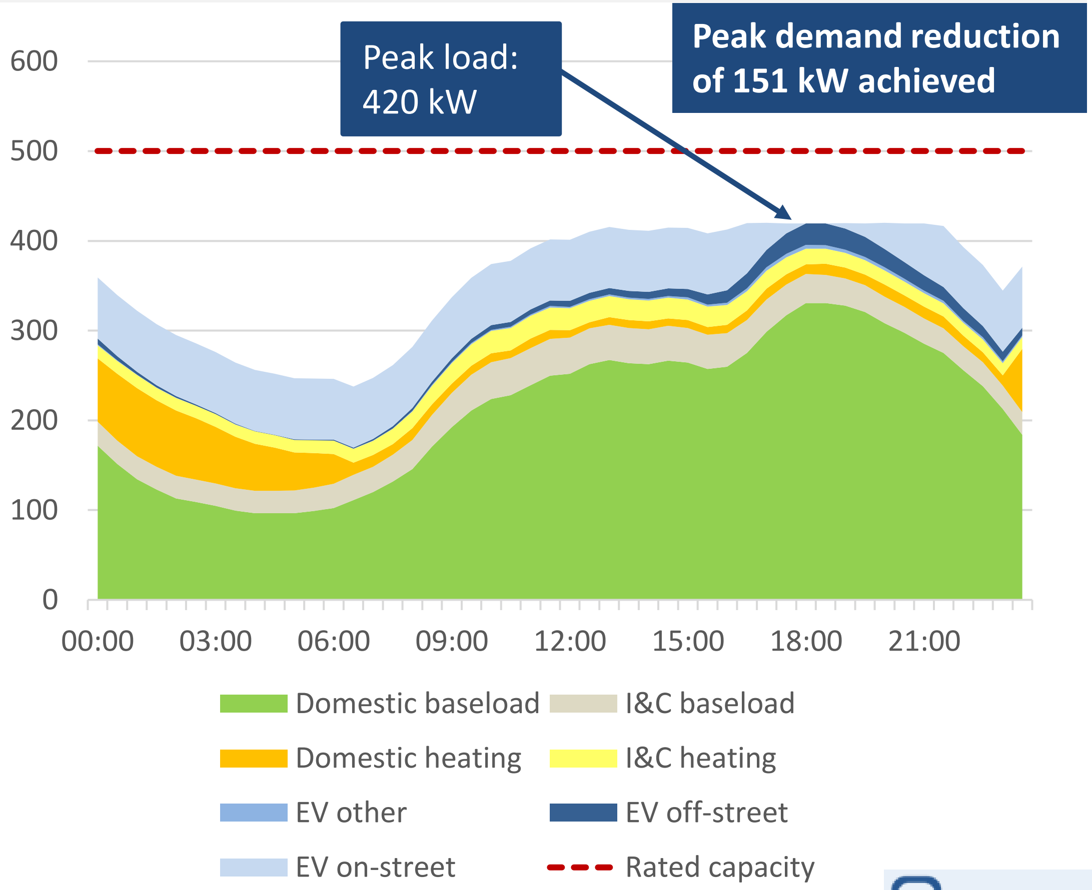
Demand profile with unmanaged on-street charging (kW)



Maximum unmanaged on-street charging demand: 152 kW

Rated capacity: 500 kVA

Demand profile with smart on-street charging (kW)



Maximum smart on-street charging demand: 68 kW → 10

London Central secondary
Urban, mostly domestic
System Transformation scenario

Key Outputs

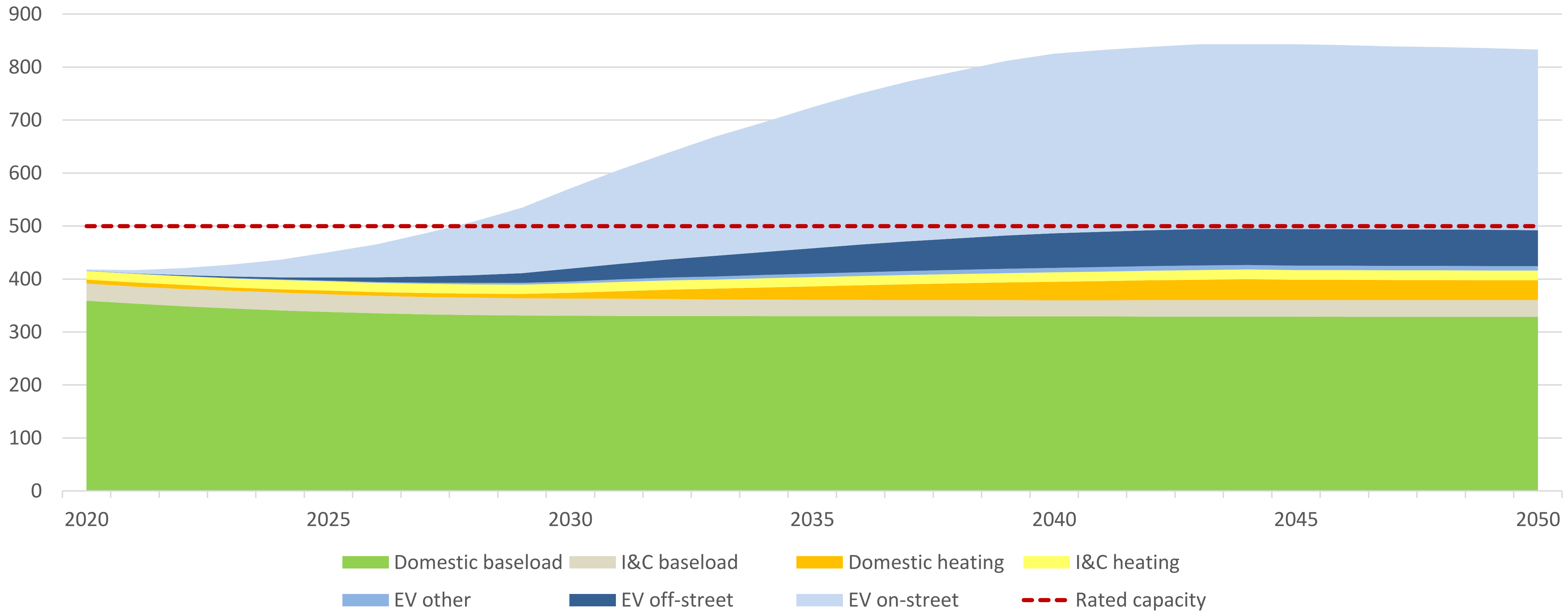
Demand growth is dominated by on-street EV charging

On-street smart charging has potential for long-term constraint mitigation

Task 3

Secondary

Peak half hour winter load each year split by demand class (kW)



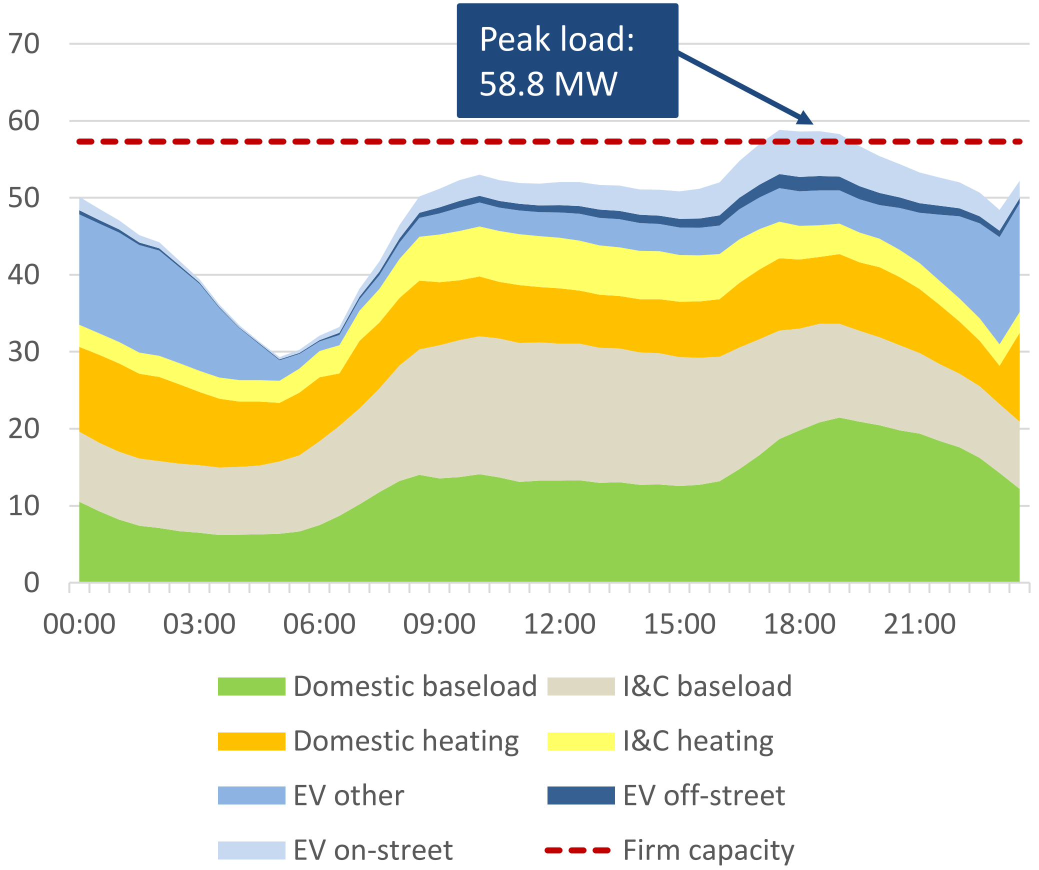
London Central primary
Urban, mostly domestic
Consumer Transformation scenario, 2030

Key Outputs — **Peak demand can be reduced but overnight EV bus charging limits how much on-street charging demand can be shifted into overnight period***

Task 3

Primary

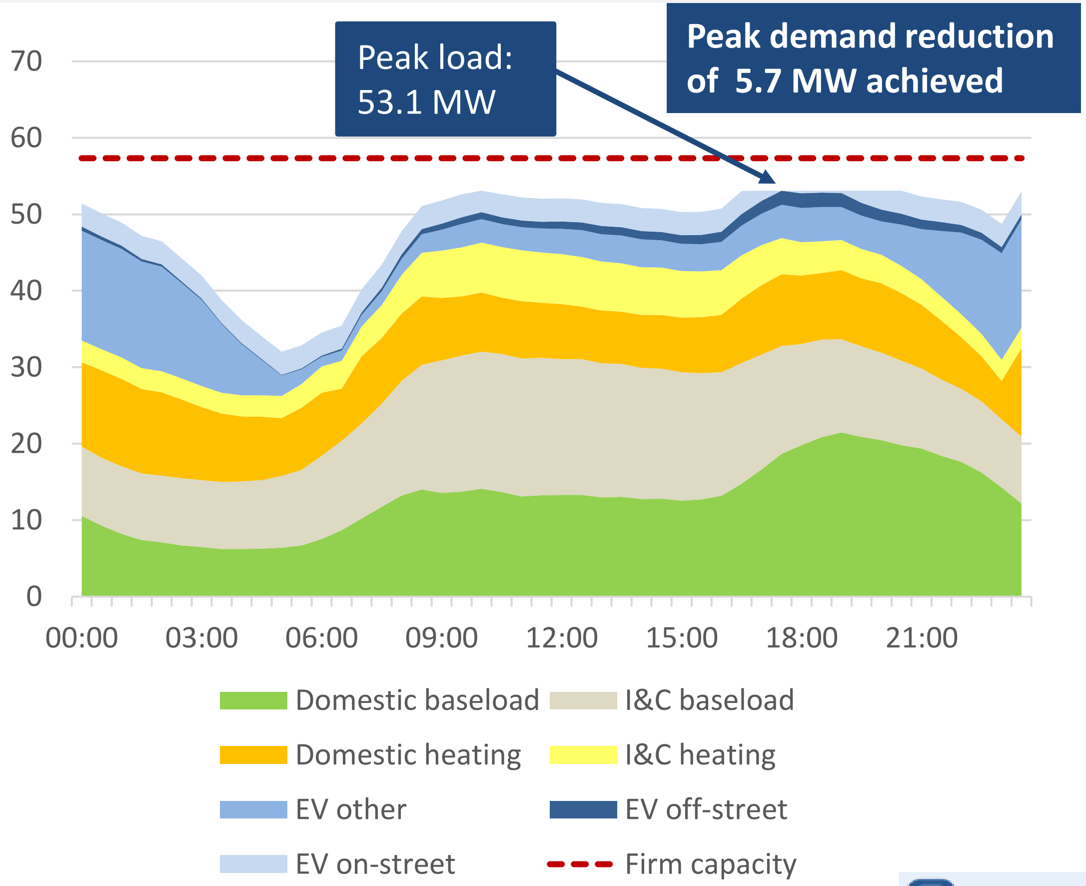
Demand profile with unmanaged on-street charging (MW)



Maximum unmanaged on-street charging demand: 5.9 MW

Firm capacity: 57.3 MVA

Demand profile with smart on-street charging (MW)



Maximum smart on-street charging demand: 3.0 MW → 430

*On this substation, the "EV other" category shown in demand profiles consists predominantly of EV bus charging

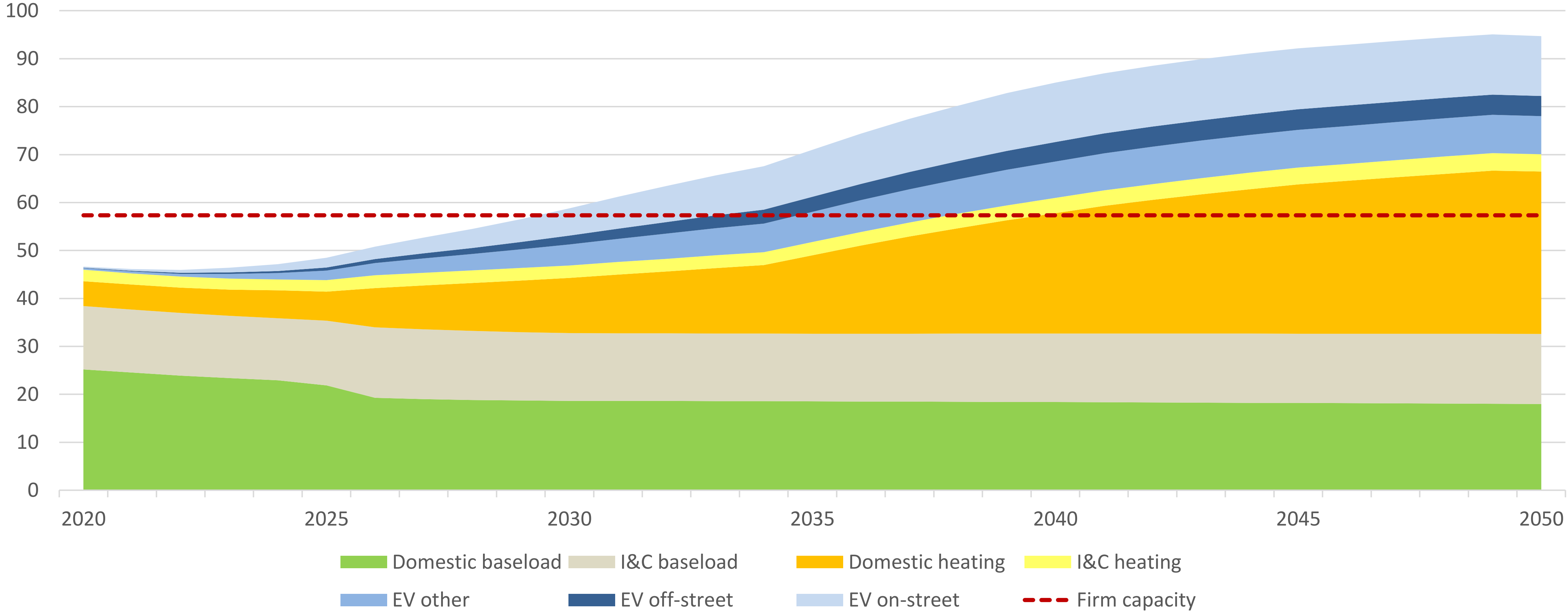
London Central primary
Urban, mostly domestic
Consumer Transformation scenario

Key Outputs On and off-street charging flexibility could be capable of mitigating constraints up to 2035

Task 3

Primary

Peak half hour winter load each year split by demand class (MW)



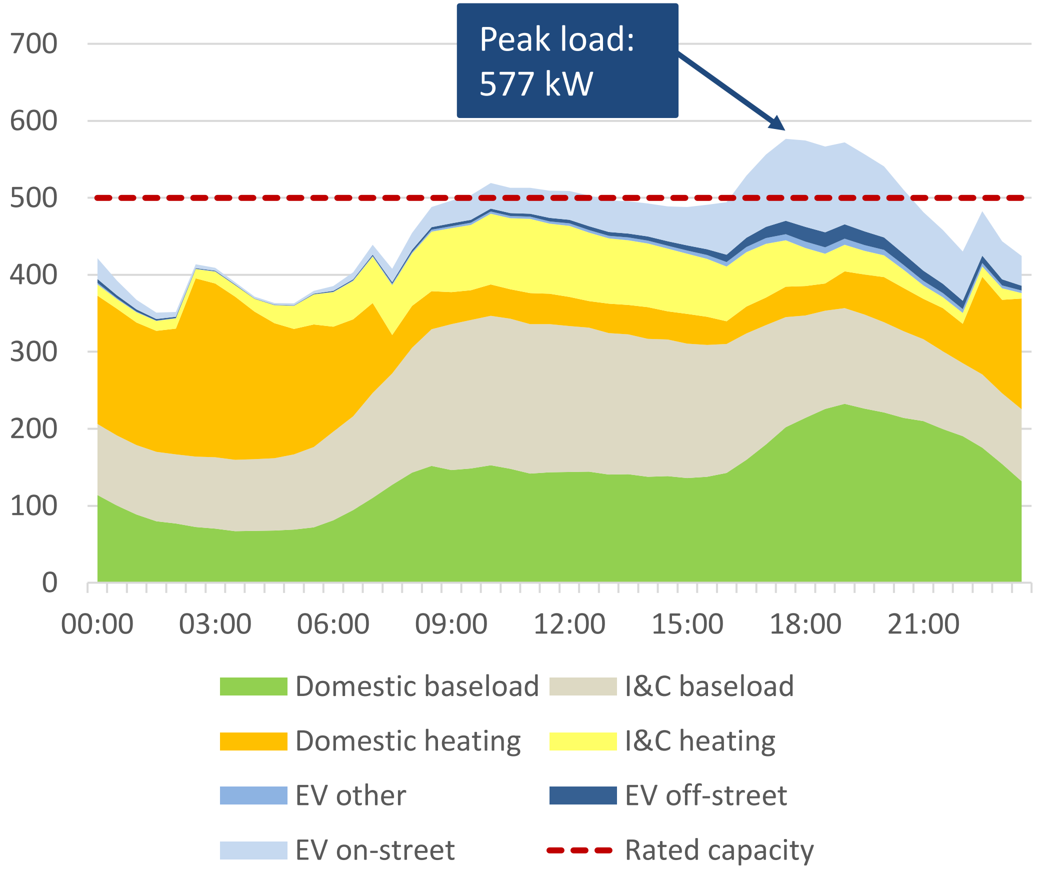
Brighton Central secondary
Urban, mostly domestic, night heating
System Transformation scenario, 2030

Key Outputs Peak demand can be reduced but night storage heating demand limits shifting of on-street demand into the overnight period

Task 3

Secondary

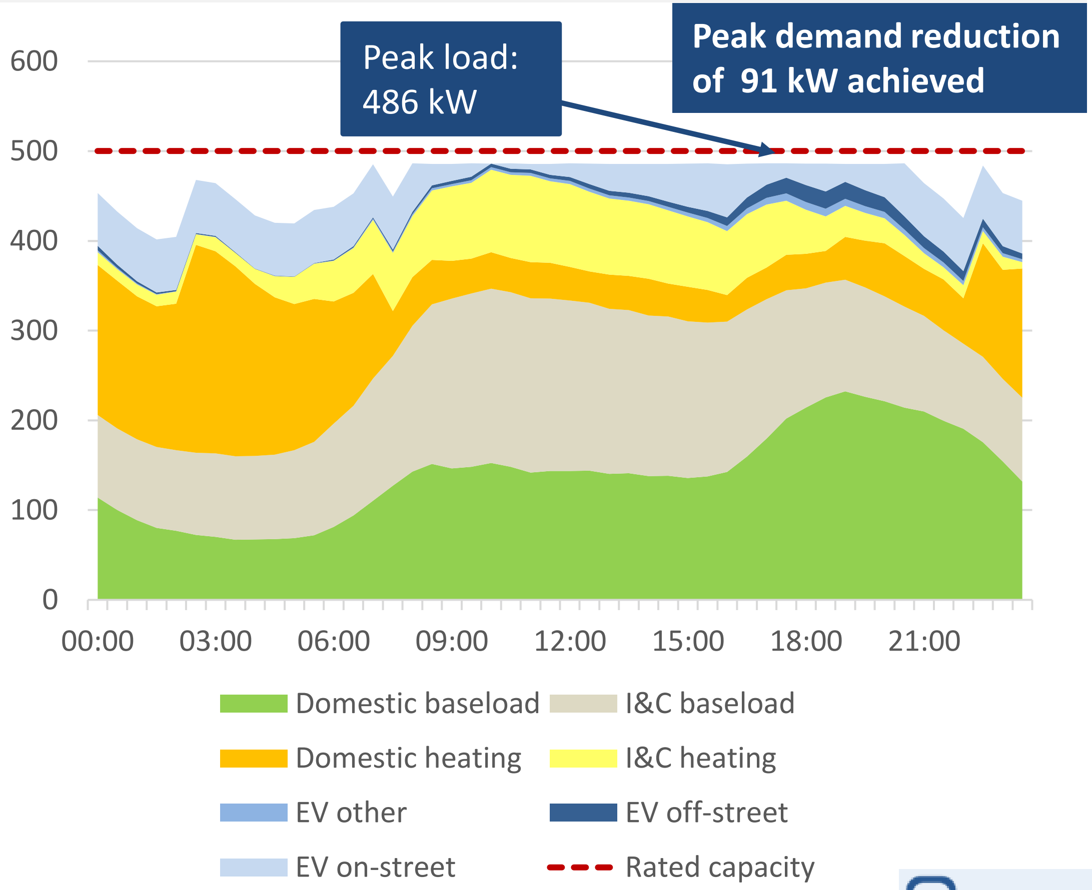
Demand profile with unmanaged on-street charging (kW)



Maximum unmanaged on-street charging demand: 112 kW

Rated capacity: 500 kVA

Demand profile with smart on-street charging (kW)



Maximum smart on-street charging demand: 59 kW → 9

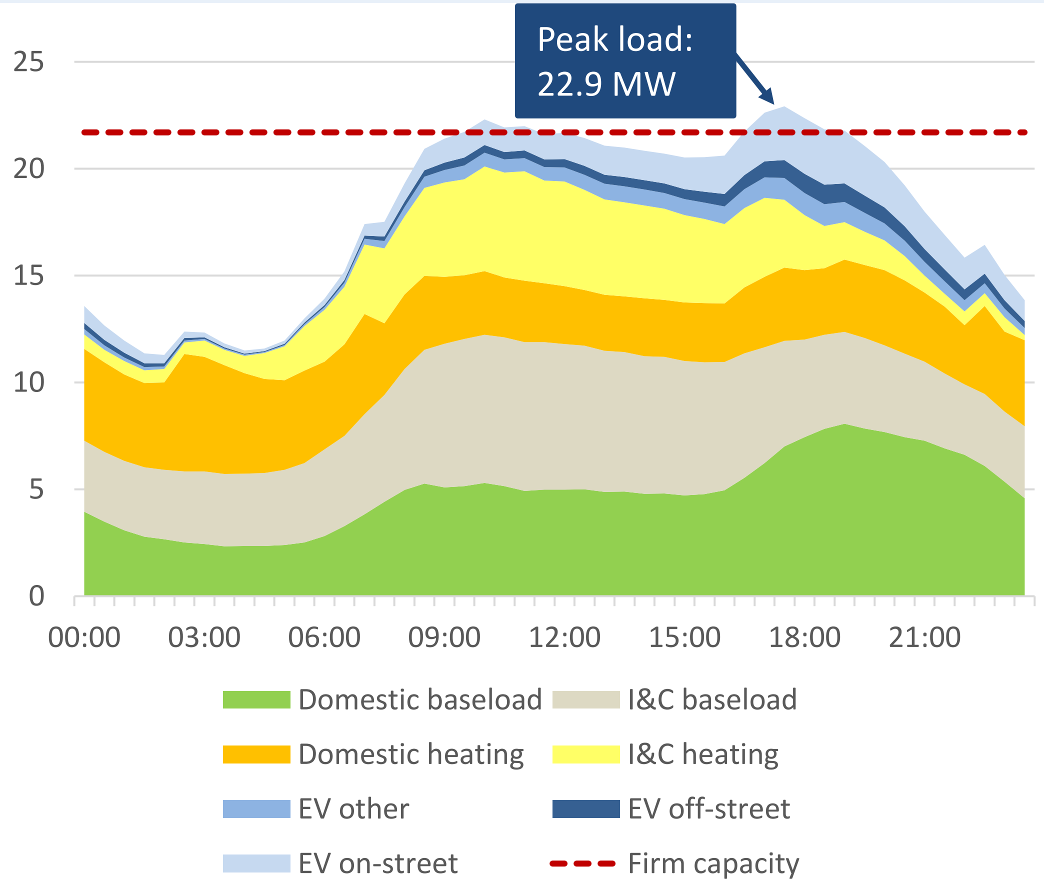
Brighton Central primary
Urban, mostly domestic
Consumer Transformation scenario, 2030

Key Outputs On-street smart charging can delay capacity breaches but underlying demand is high

Task 3

Primary

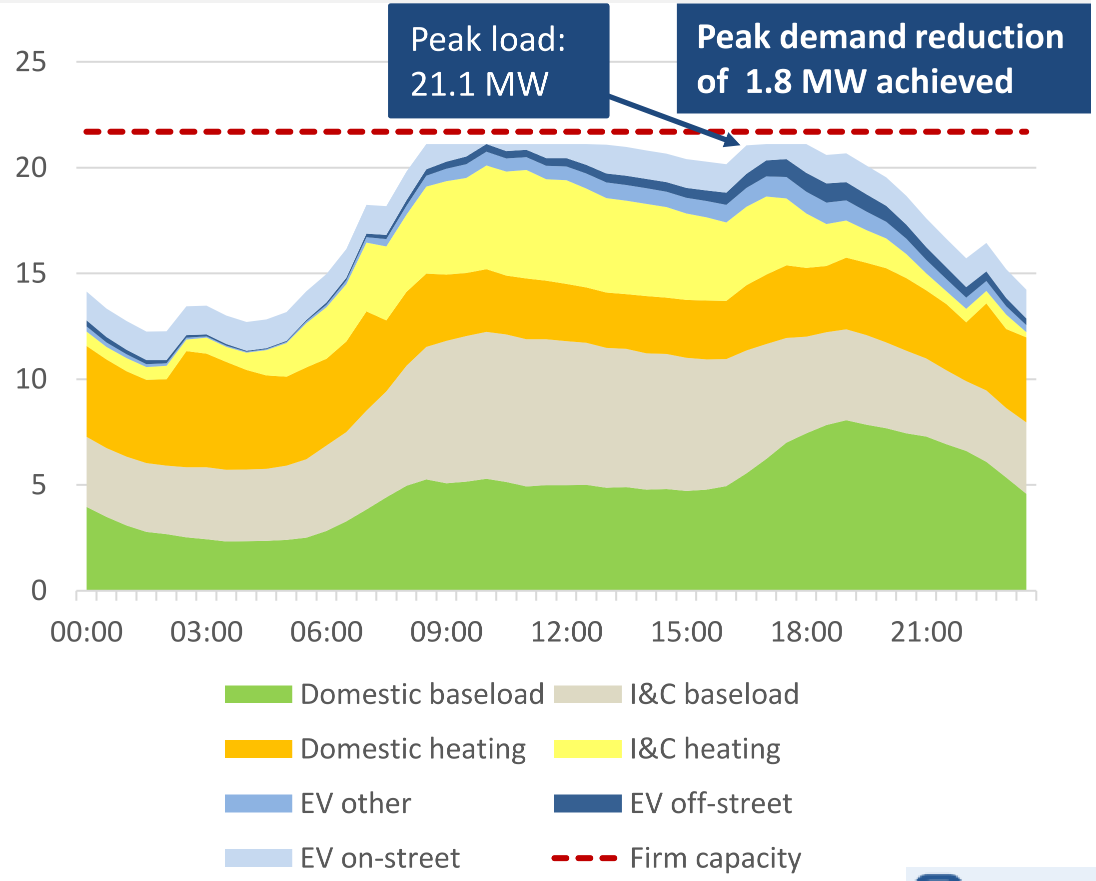
Demand profile with unmanaged on-street charging (MW)



Maximum unmanaged on-street charging demand: 2.6 MW

Firm capacity: 21.7 MVA

Demand profile with smart on-street charging (MW)



Maximum smart on-street charging demand: 1.4 MW

200

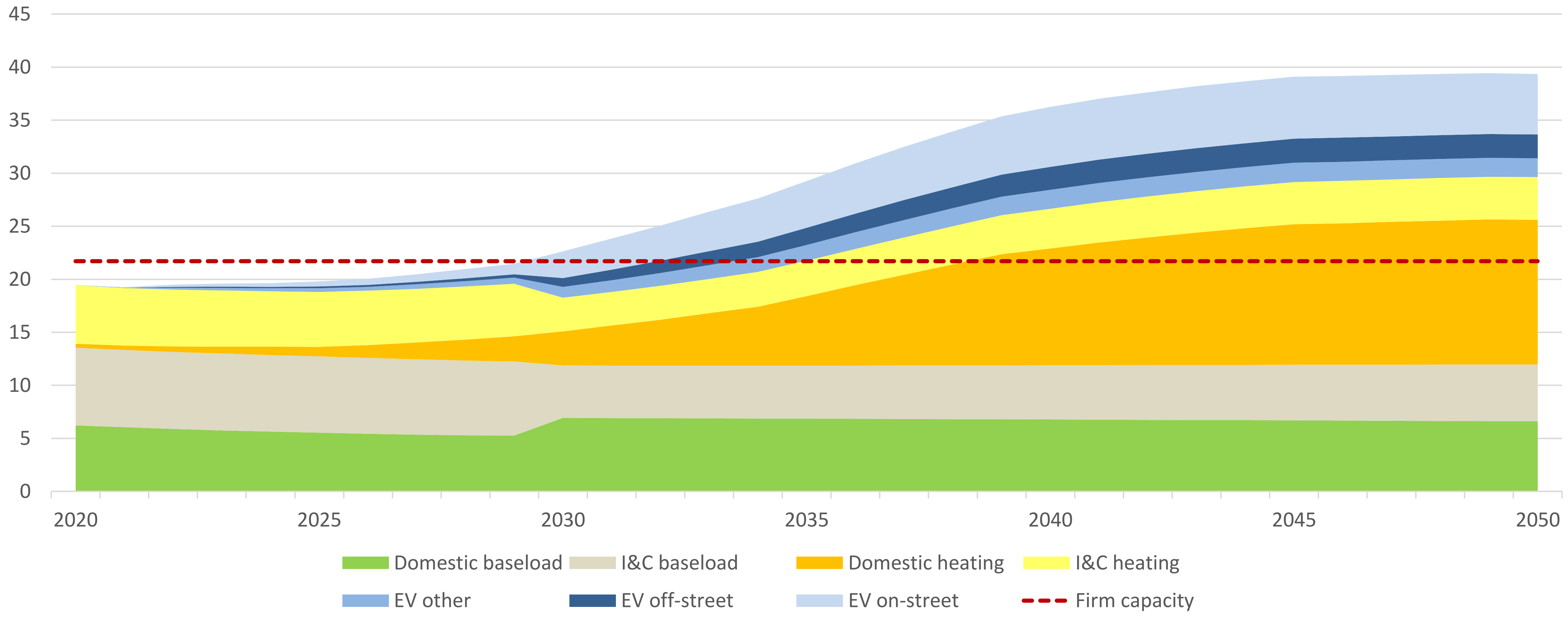
Brighton Central primary
Urban, mostly domestic
Consumer Transformation scenario

Key Outputs On and off-street charging flexibility could be capable of mitigating constraints up to 2033

Task 3

Primary

Peak half hour winter load each year split by demand class (MW)



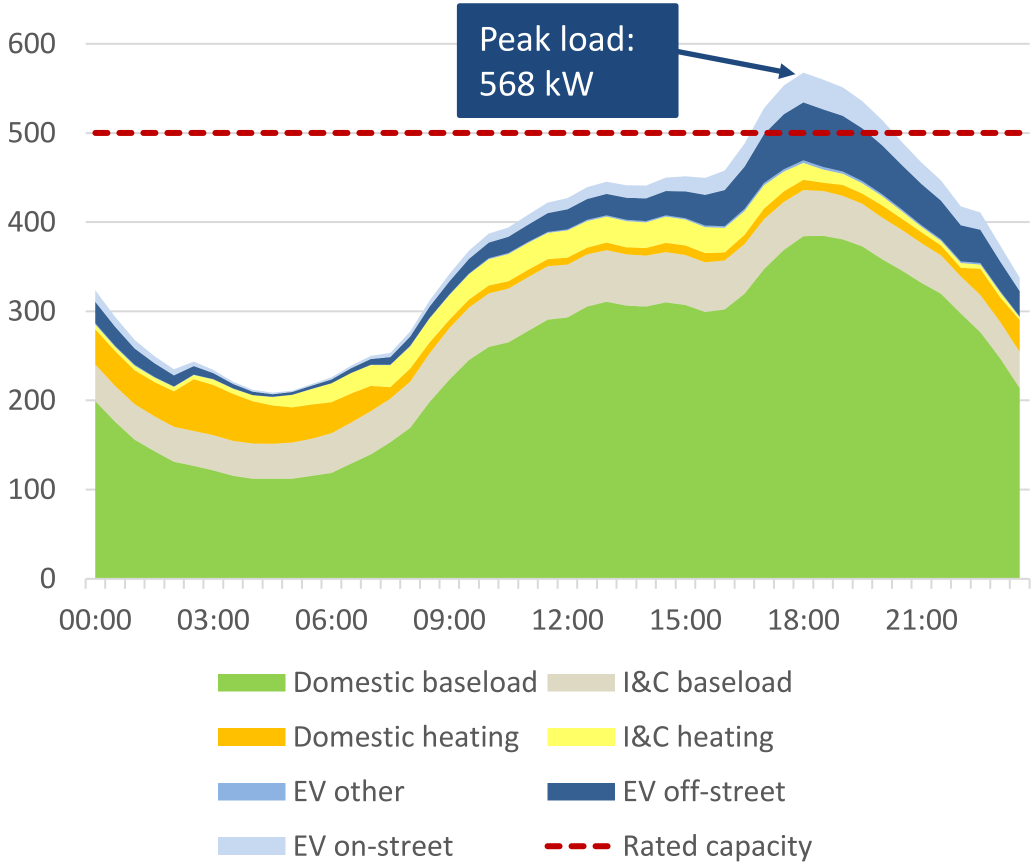
Brighton Suburban secondary
Suburban, mostly domestic
System Transformation scenario, 2030

Key Outputs | **Off-street charging flexibility could help to mitigate constraints (on-street flexibility alone is not viable)**

Task 3

Secondary

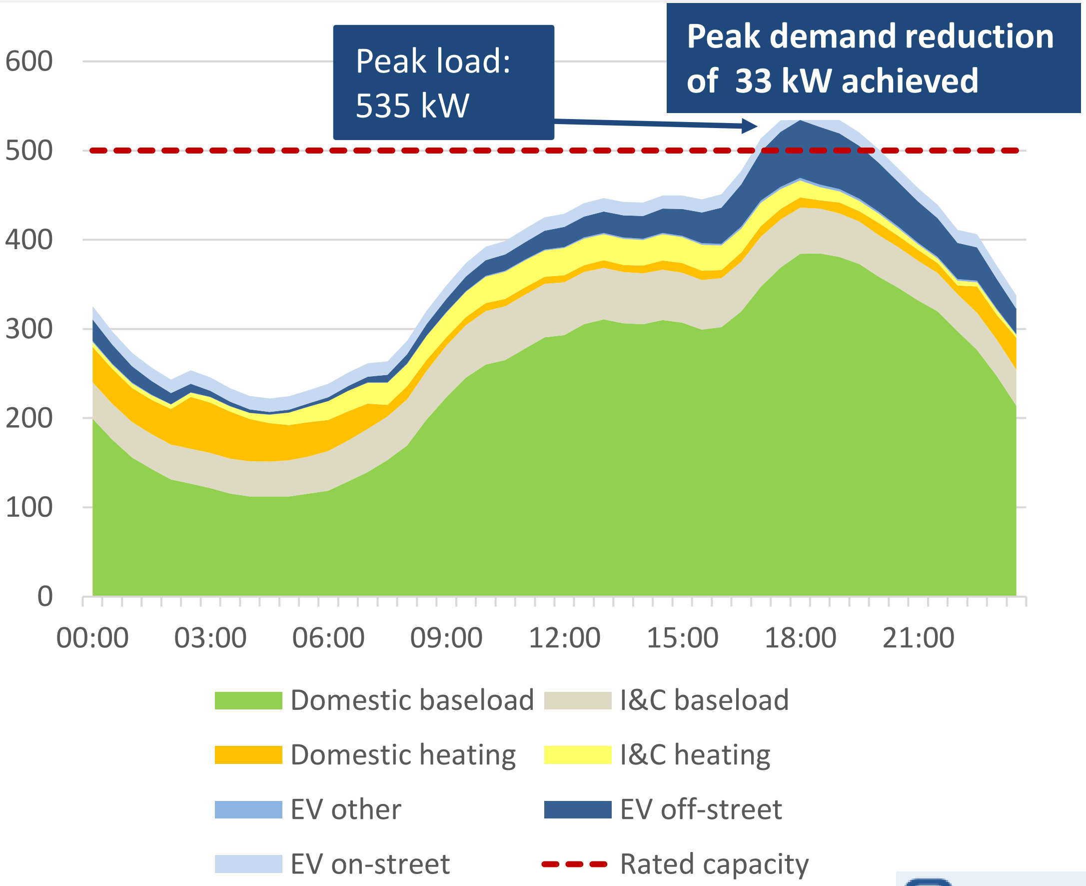
Demand profile with unmanaged on-street charging (kW)



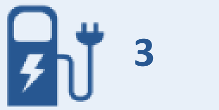
Maximum unmanaged on-street charging demand: 34 kW

Rated capacity: 500 kVA

Demand profile with smart on-street charging (kW)



Maximum smart on-street charging demand: 15 kW



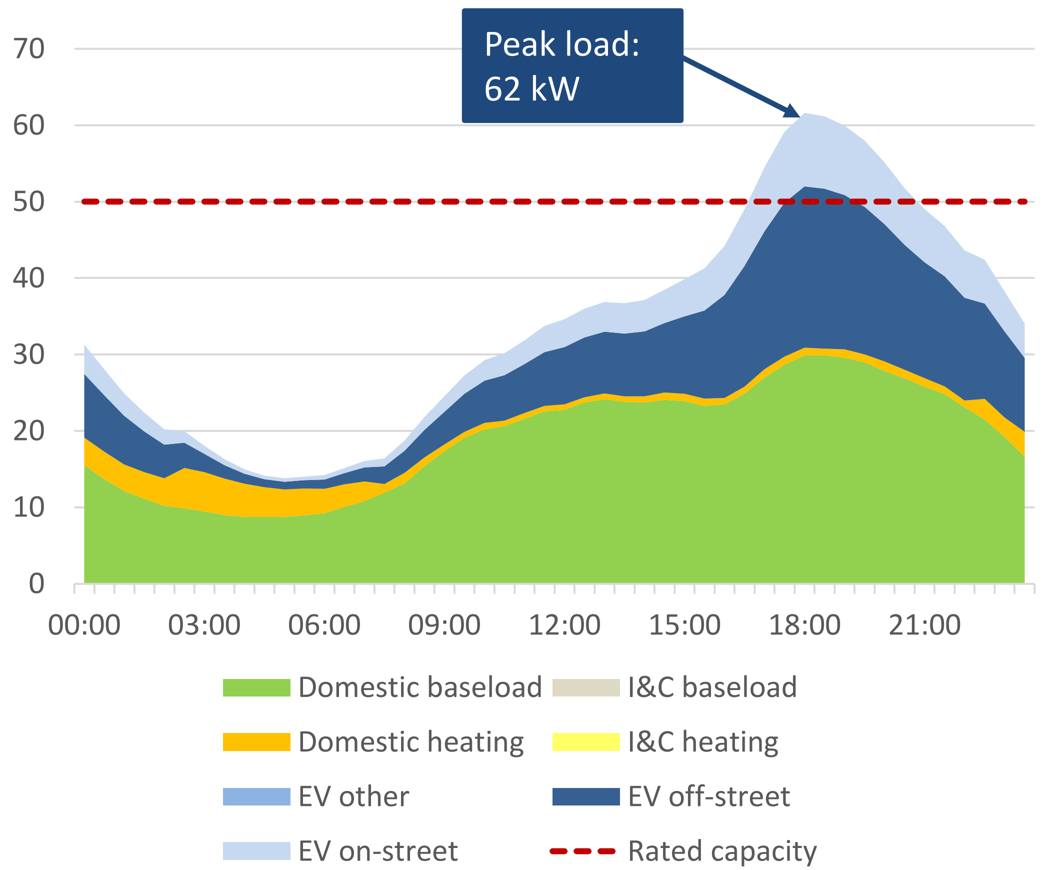
Rural secondary
Rural, mostly domestic
System Transformation scenario, 2030

Key Outputs High share of off-street parking means there is much more off-street than on-street charging demand

Task 3

Secondary

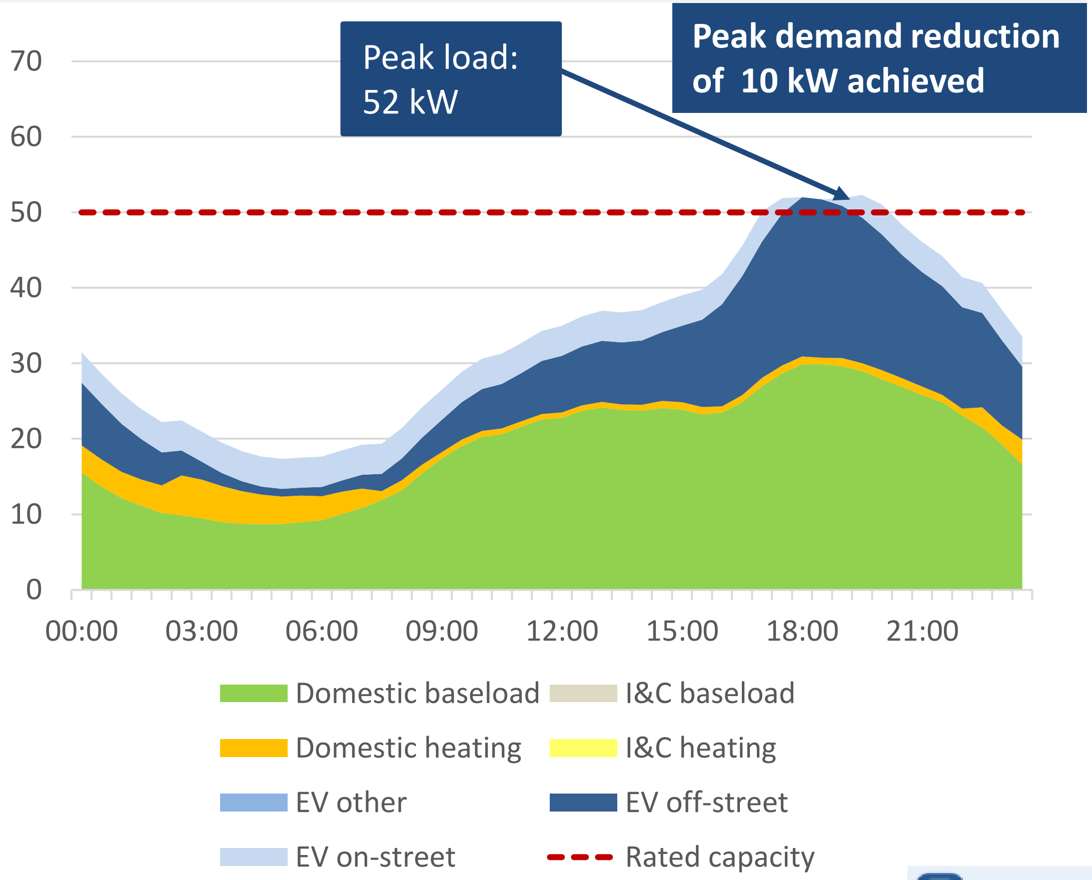
Demand profile with unmanaged on-street charging (kW)



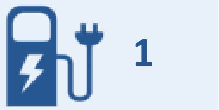
Maximum unmanaged on-street charging demand: 10 kW

Rated capacity: 50 kVA

Demand profile with smart on-street charging (kW)



Maximum smart on-street charging demand: 4 kW



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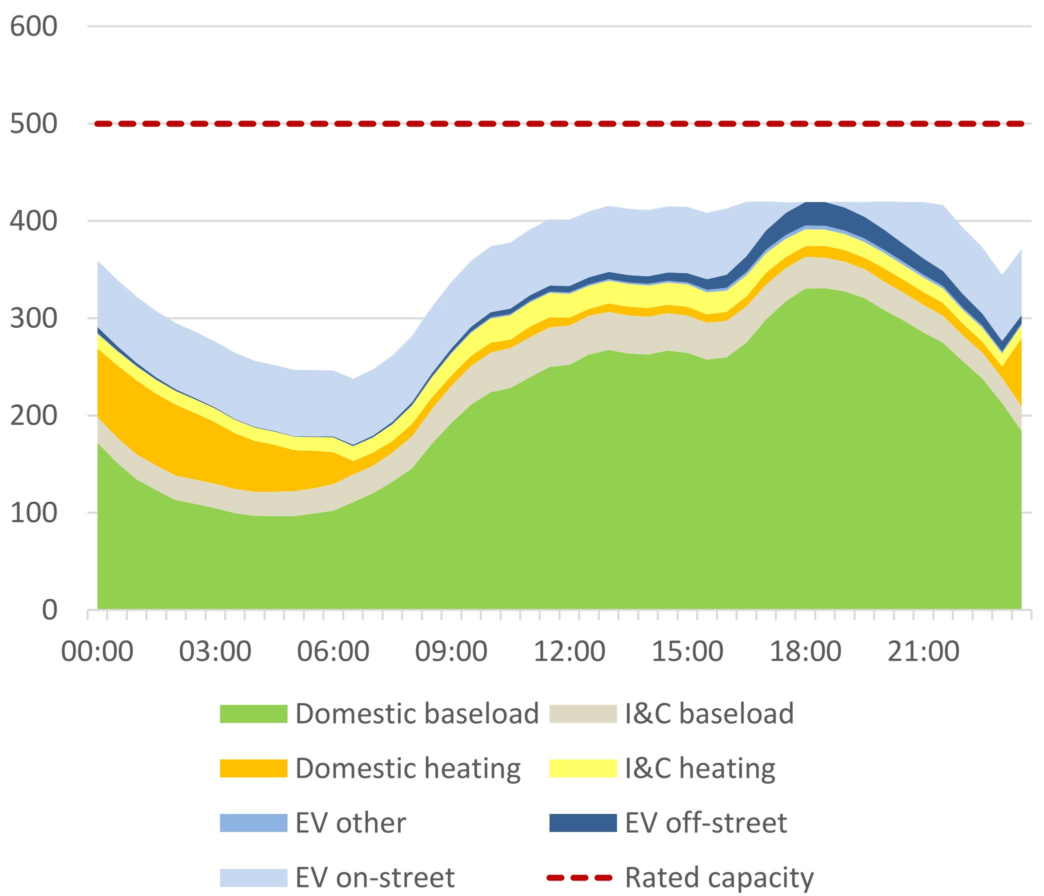
Constraints can be mitigated for the foreseeable future in locations where peak load growth is dominated by on-street charging and there is capacity to shift this demand

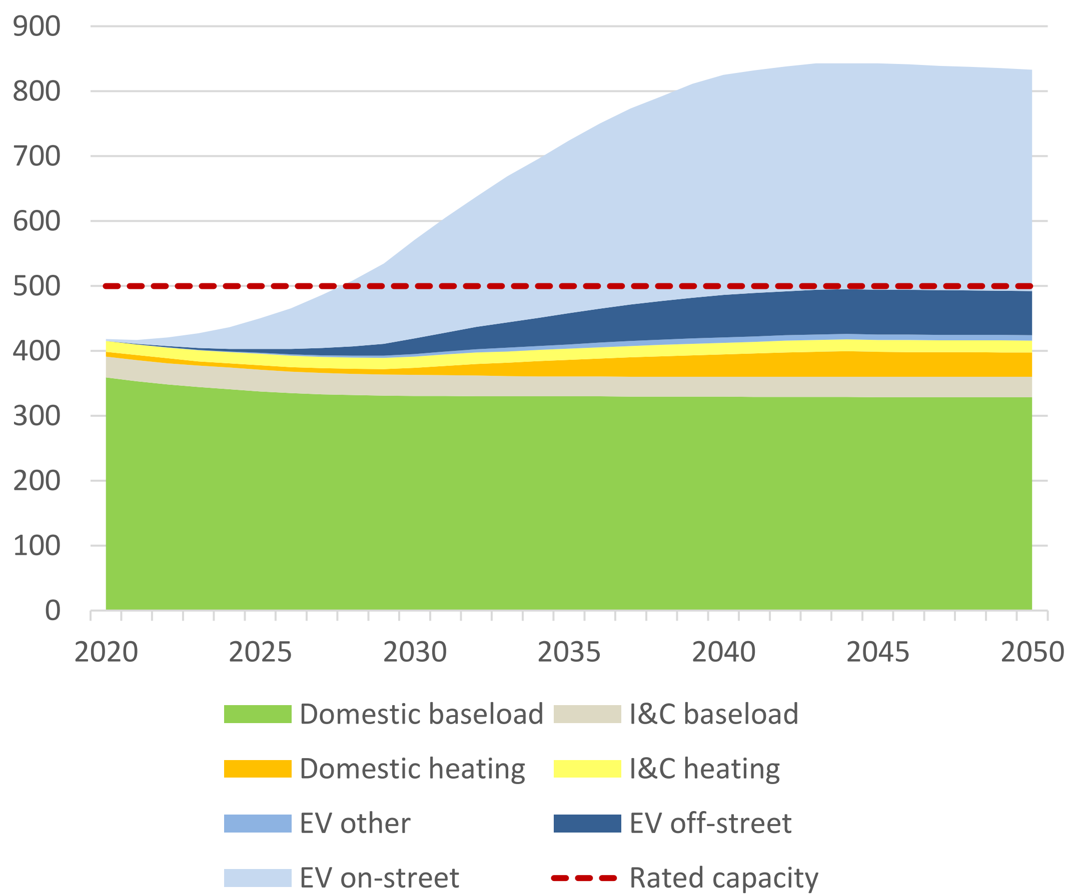
Task 3

Profiles shown for London Central secondary substation

Peak day demand profile with smart on-street charging, 2030 (kW)



Peak half hour winter load each year split by demand class (kW)



- Peak demand reduction of 152 kW achieved using smart on-street charging
- Smart on-street charging avoids network constraints

- As peak load growth is dominated by on-street charging, constraints can be managed by on-street smart charging into the future

Constraint mitigation from on-street smart charging is most feasible in urban locations with high levels of on-street charging demand

Task 3

Use case Substation archetype	Firm / rated capacity	Peak demand reduction in 2030	Key outputs	Potential for on-street smart charging
Brent secondary **Urban, mostly I&C**	500 kVA	9.2 kW	• Dominated by I&C loads • Very little potential for on-street smart charging in 2030	
London Central secondary **Urban, mostly domestic**	500 kVA	152 kW	• Large peak in on-street charging in the evening • Spare capacity available overnight so demand can be shifted • Demand growth is dominated by on-street EV charging • On-street smart charging has potential for long-term constraint mitigation	
London Central primary **Urban, mostly domestic**	57.3 MVA	5.7 MW	• Overnight EV bus charging limits how much on-street charging demand can be shifted into overnight period • On and off-street charging flexibility could be capable of mitigating constraints up to 2035	
Brighton Central secondary **Urban, mostly domestic, night heating**	500 kVA	70 kW	• Night storage heating demand limits shifting of on-street demand into the overnight period	
Brighton Central primary **Urban, mostly domestic**	21.7 MVA	4.2 MW	• On-street smart charging can delay capacity breaches but underlying demand is high • On and off-street charging flexibility could be capable of mitigating constraints up to 2033	
Brighton Suburban secondary **Suburban, mostly domestic**	500 kVA	34 kW	• Off-street charging flexibility could help to mitigate constraints	
Rural secondary **Rural, mostly domestic**	50 kVA	9.3 kW	• High share of off-street parking means there is much more off-street than on-street charging demand	

I&C: industrial and commercial

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Several filtering stages have been used to determine the most suitable sites for commercial model analysis of flexibility services

Shortlist filtering process

10 shortlisted substations

↓

Can constraints be managed using smart charging?

No → **Substation is not considered further**

Yes ↓

Qualitative assessment of on-street smart charging potential:

- Parking suitability
- Charger power and availability
- Smart charging acceptance required
- Likely size of benefits

↓

Most suitable substations are selected for commercial model analysis of flexibility services

Due to low potential of on-street smart charging the Brent use case has not been considered

Can constraints be managed using smart charging?

Task 3

Use case Substation archetype	Constraints in 2030 mitigated by on-street smart charging?	Constraints in 2030 mitigated by on and off-street smart charging?	How long is mitigation expected to last?	Potential value of on-street smart charging
Brent secondary **Urban, mostly I&C**	✗	✗	N/A	
London Central secondary **Urban, mostly domestic**	✓	✓	2050	
Brighton Central secondary **Urban, mostly domestic, night heating**	✓	✓	2050 (but capacity is breached by overnight heating from 2020-2028)	
Brighton Suburban secondary **Suburban, mostly domestic**	✗	✓	2050	
Rural secondary **Rural, mostly domestic**	✗	✓	2050	
Brent primary **Urban, mostly I&C**	✗	✗	N/A	
London Central primary **Urban, mostly domestic**	✓	✓	2035	
Brighton Central primary **Urban, mostly domestic**	✓	✓	2033	
Brighton Suburban primary **Suburban, mostly domestic**	Constraints not expected in 2030	Constraints not expected in 2030	2034	
Rural primary **Rural, mostly domestic**	✗	✓	2033	

I&C: industrial and commercial

4 selection criteria have been used to assess the potential value of on-street smart charging

Parking suitability

- Number of public car parks within 500m of a secondary substation, or within the service area of a primary substation (source: OpenStreetMap data)
- Predominant housing type in the local area

Charger power and availability

- Total number of existing charge points within 500m of a secondary substation, or within the service area of a primary substation (source: UK Power Networks licence area charge point data from ZapMap)
- Further broken down by number of slow (3 kW), fast (7-22 kW) and rapid (50+ kW) connectors

Penetration rate required

- Share of on or off-street charging demand that needs to be shifted away from the peak to avoid constraints in 2030 (source: SFS data). These rates are deemed feasible based on the outputs from consumer research

Likely size of benefits

- Reduction in peak demand achievable with on-street charging in 2030 (source: SFS data)
- Year constraints can be delayed to assuming all on and off-street charging demand can be shifted away from peak

Predominant vehicle type was also assessed but all substations had mostly cars and vans registered, so this information has not been included in following slides

Urban secondary substations rank highly across all categories

Qualitative assessment of on-street smart charging potential

Task 3

Secondary

Highest ranking

Substation	Number of car parks	Predominant housing type	Charge points	Slow connectors	Fast connectors	Rapid connectors	Penetration rate required	Peak demand reduction / kW	Mitigation possible until
London Central secondary Urban, mostly domestic	21	Terraced	3	2	2	0	47% of peak on-street charging demand	152	2050
Brighton Central secondary Urban, mostly domestic, night heating	14	Terraced, semi-detached	18	9	18	0	72% of peak on-street charging demand	70	2050
Brighton Suburban secondary Suburban, mostly domestic	0	Terraced, semi-detached	0	0	0	0	68% of peak **on and off-street charging demand**	34	2050
Rural secondary Rural, mostly domestic	0	Detached	0	0	0	0	37% of peak **on and off-street charging demand**	9.3	2050

Connector power: slow = 3kW, fast = 7-22 kW, rapid = 50+ kW

Urban primary substations rank highly across all categories

Qualitative assessment of on-street smart charging potential

Task 3

Primary

Highest ranking

Substation	Number of car parks	Predominant housing type	Charge points	Slow connectors	Fast connectors	Rapid connectors	Penetration rate required	Peak demand reduction / MW	Mitigation possible until
London Central primary Urban, mostly domestic	145	Terraced	58	42	32	5	23% of peak on-street charging demand	5.7	2035
Brighton Central primary Urban, mostly domestic	18	Terraced	45	42	6	0	49% of peak on-street charging demand	4.2	2033
Brighton Suburban primary Suburban, mostly domestic	24	Terraced, semi-detached	38	35	10	2	No constraint expected in 2030	2.0	2033
Rural primary Rural, mostly domestic	9	Detached	1	0	2	0	78% of peak **on and off-street charging demand**	0.18	2033

Connector power: slow = 3kW, fast = 7-22 kW, rapid = 50+ kW

Based on assessment of previous slides we are selecting the London Central and Brighton Central secondary and primary substations for commercial model analysis

Most suitable substations are selected for commercial model analysis

Task 3

Decision to take forward substations were based on the following factors

Factors

Parking suitability

- Higher numbers of public car parks
- More residents have to rely on on-street parking in urban areas

Charger power and availability

- Charge points are currently much more prevalent in urban areas

Penetration rate required

- In many cases constraints can be avoided by solely deploying on-street smart charging
- A lower level of acceptance is needed than in rural areas, where on-street charging demand is lower

Likely size of benefits

- As urban substations are more costly to upgrade, mitigating constraints is significantly more valuable
- For both the London Central and Brighton Central use cases, constraints at both secondary and primary level can be mitigated for a long time using smart on-street charging

Selected Use Cases

- London Central – Secondary
- London Central – Primary
- Brighton Central – Secondary
- Brighton Central - Primary

Due to the different load profiles on each of the four selected substations, they are also suitable for trialling the benefits that can be achieved by different commercial models

Executive Summary

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Report Conclusions

Appendix

Urban substations with high on-street demand were found to be the sites with most opportunities to benefit from on-street flexibility

Task 3

Shortlist filtering process

10 shortlisted substations

Can constraints be managed using smart charging?

No → **Substation is not considered further**

Yes ↓

Qualitative assessment of on-street smart charging potential:

- Parking suitability
- Charger power and availability
- Smart charging acceptance required
- Likely size of benefits

Most suitable substations are selected for commercial model analysis

Summary of findings

- Constraint management using smart charging was not possible on the Brent substations, as these were dominated by I&C loads
- The remaining urban substations (London Central and Brighton Central primaries and secondaries) consistently performed better on qualitative assessment than the more rural substations
- As a result 4 substations were chosen for commercial model analysis:
  - London Central secondary
  - London Central primary
  - Brighton Central secondary
  - Brighton Central primary

Executive Summary

Task 1: Evaluating Barriers and Opportunities

Task 2: Analysis of Charging Behaviour + Substation Selection

Task 3: Assessment of Flexibility Opportunities

Task 4: Development of Use Cases and Commercial Models

Conclusions

Report Conclusions

Appendix

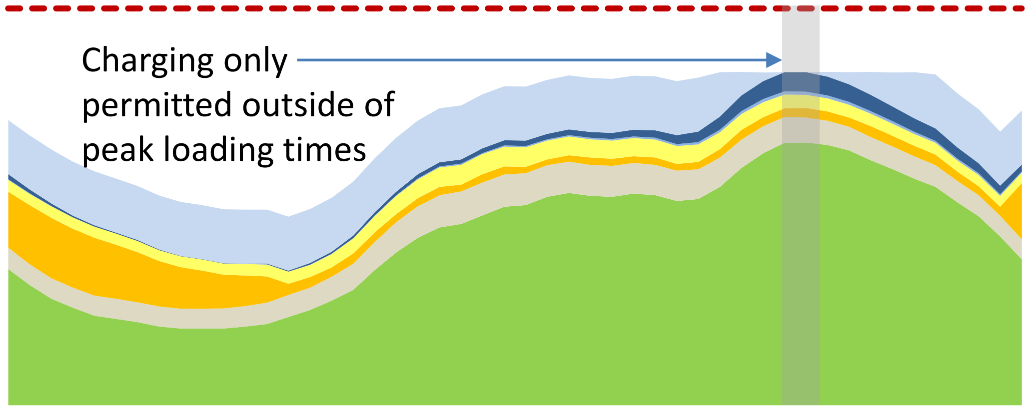
Possible commercial models – DNO perspective

Task 4

Timed connection

- CPOs agree to only load the network at specific times
- Consists of a one-off payment from CPO to DNO for the connection
- Analysis of charging event data has shown that demand can be shifted so this option is feasible
- Only connecting at certain times means peak demand can be guaranteed to not increase
- If there is insufficient headroom or vehicles are only plugged in during peak times then not all demand can be shifted in this way

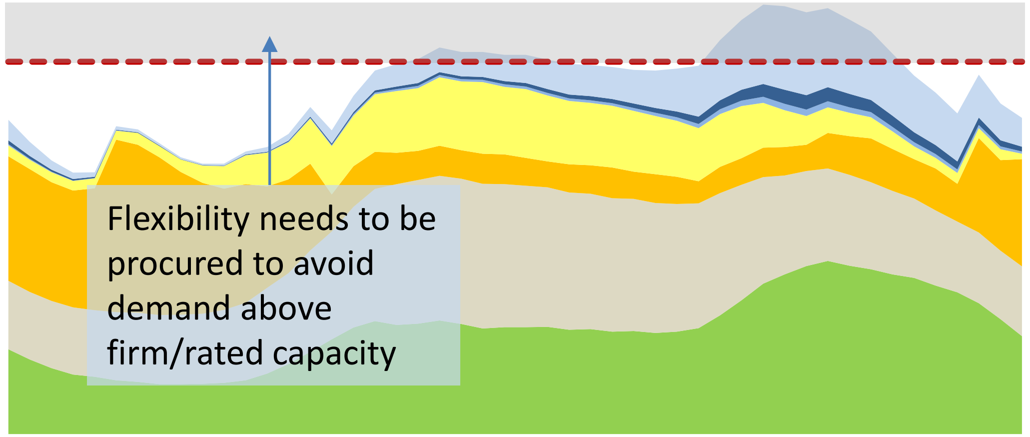
London Central secondary substation



Flexibility procurement

- UK Power Networks may agree to connect CPOs even if this could lead to a breach in firm/rated capacity, and procure flexibility services from a variety of sources to create sufficient headroom
- In addition to the initial connection cost paid by CPO to DNO, the DNO has to actively manage the network to prevent constraints and pay for any flexibility procured to achieve this
- Can be used in cases where demand cannot be completely shifted away from peak times
- May need to over-procure flexibility to ensure constraints can be avoided

Brighton Central secondary substation



Possible commercial models – CPO perspective

Task 4

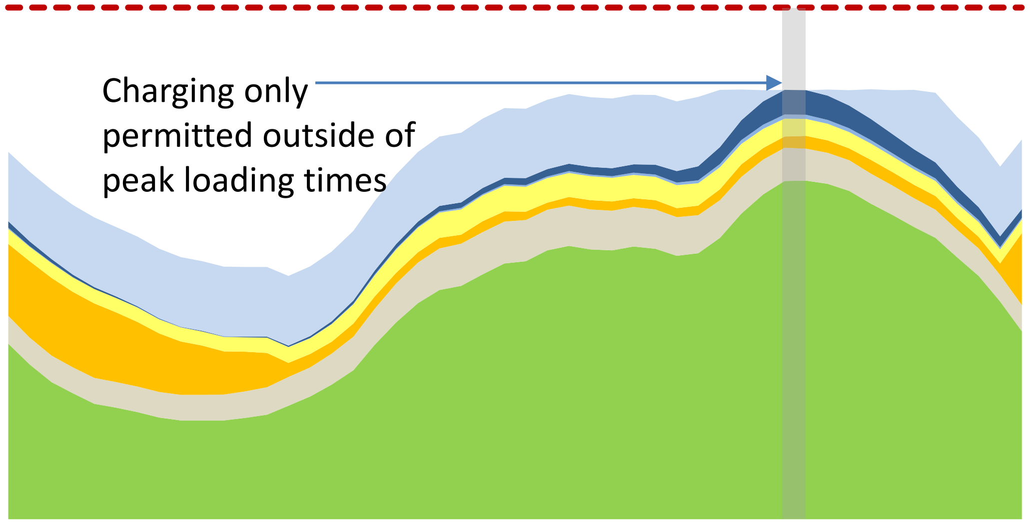
Timed connection

- Timed connections are likely to be cheaper than unrestricted connections due to the value provided to the DNO
- The CPO must manage its load to avoid producing demand outside of agreed connection times
- Reduced connection costs could be passed on to the consumer via lower charging tariffs

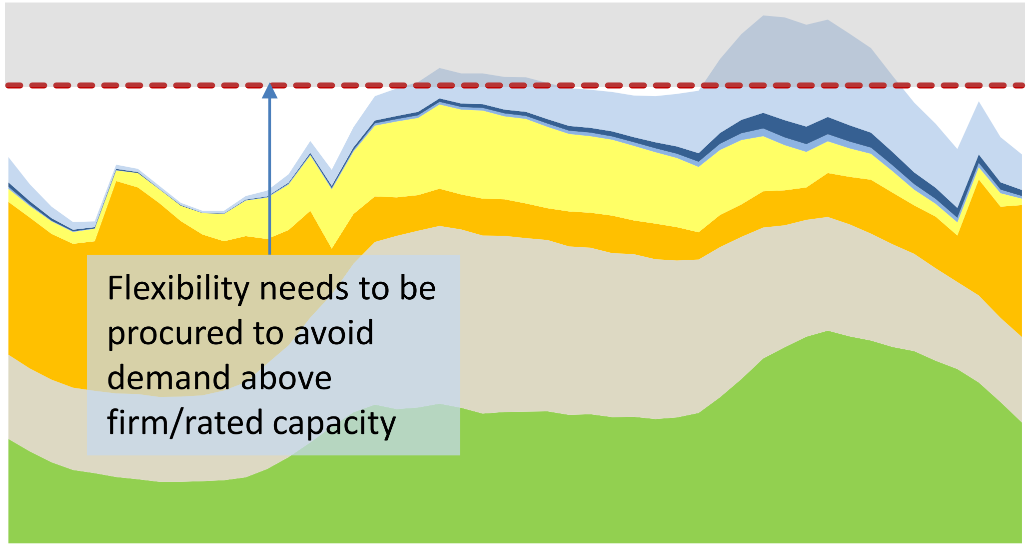
Flexibility procurement

- Allows consumers to charge with fewer restrictions – charging still possible during peak time but possibly at a reduced rate
- Only viable if DNO can ensure that flexibility can provide security; CPO may have to offer flexibility as part of connection agreement
- Due to the need for DNO to procure flexibility services this will likely be a more expensive connection for CPOs

London Central secondary substation



Brighton Central secondary substation



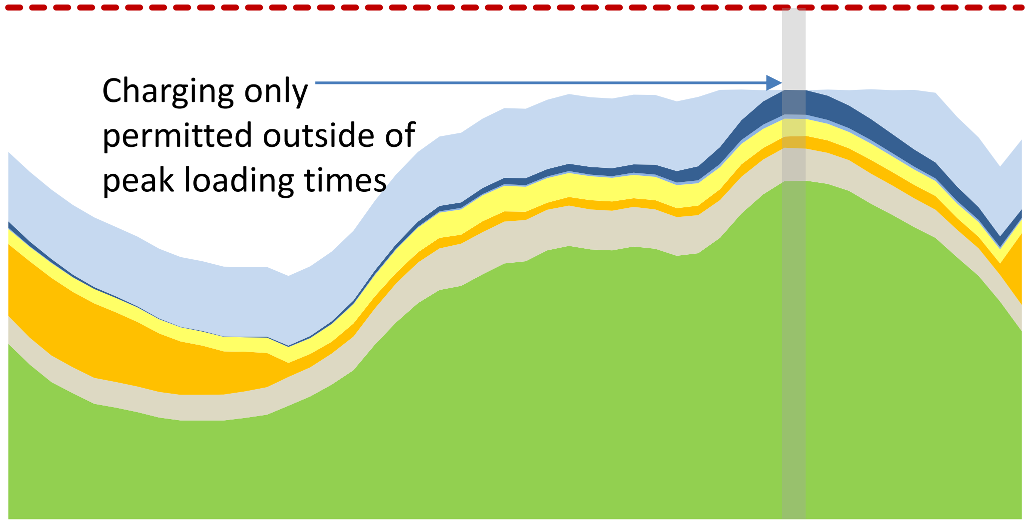
Possible commercial models – Consumer perspective

Task 4

Timed connection

- As connection costs are reduced this could lead to a lower charging tariff for consumers
- Unlikely to affect users who charge overnight as start of charge event can simply be delayed until after evening peak in demand, leaving sufficient time to charge until the morning
- Consumers who need to charge urgently during the evening peak will be unable to do so if only timed connections are used

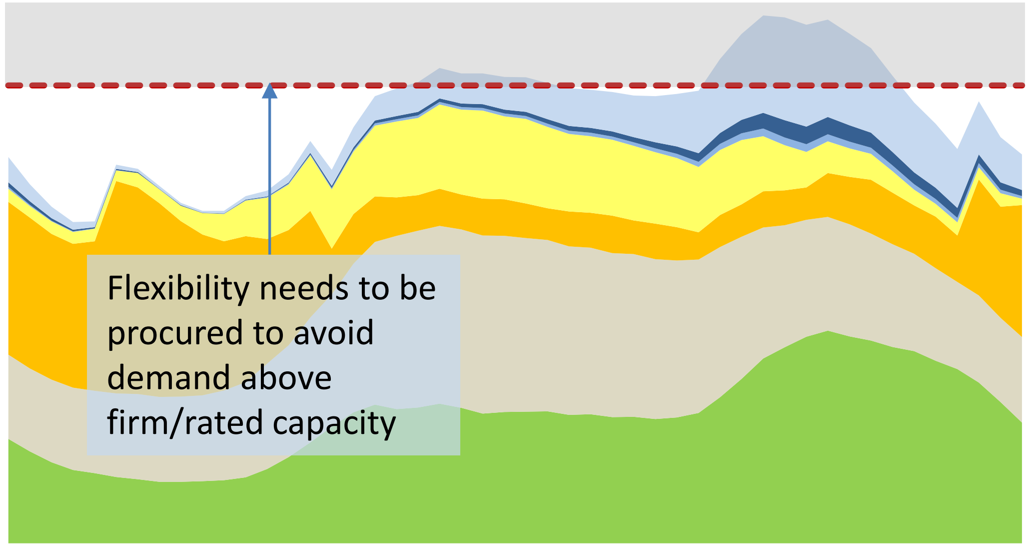
London Central secondary substation



Flexibility procurement

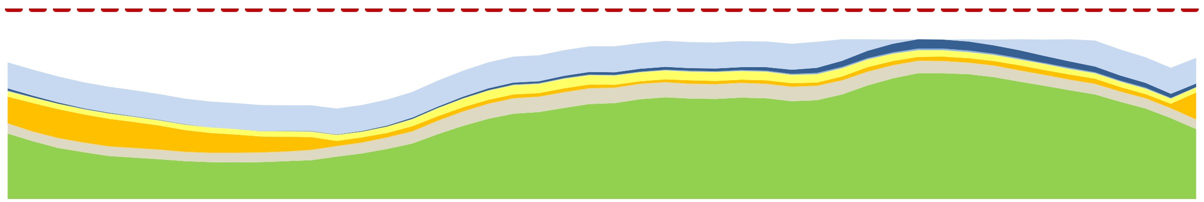
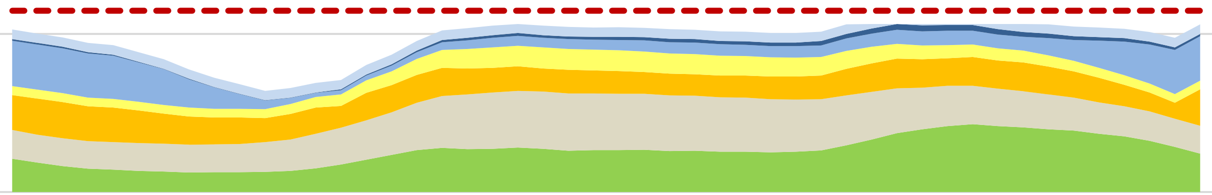
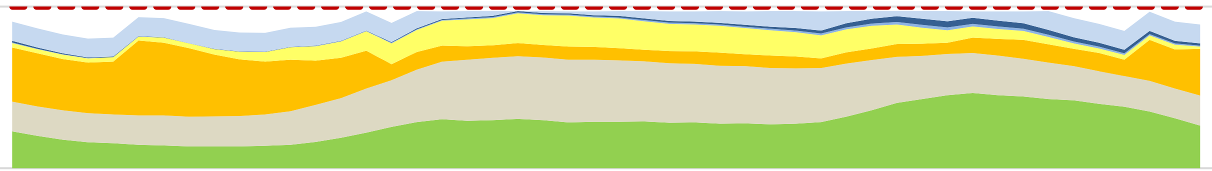
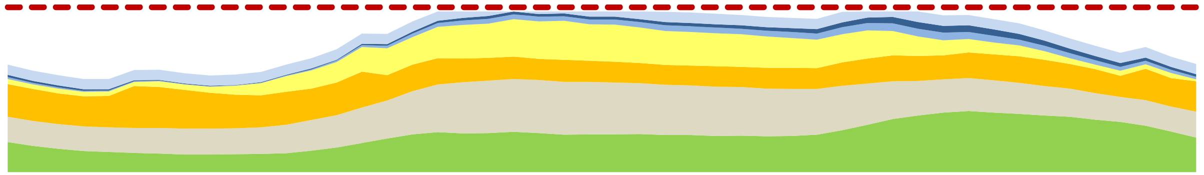
- Consumers will be able to charge with fewer restrictions than with timed connections
- Charging rate may be limited at peak times
- Price signals may be used to discourage consumers from charging at peak times, making charging less affordable
- More expensive connection costs for CPOs may result in higher charging tariffs for consumers

Brighton Central secondary substation



Commercial models that UK Power Networks currently use for other technologies can be employed at high-value use cases for on-street charging – the optimal model depends on the characteristics of the substation and what loads are connected to it

Task 4

Use case	Level	Load profile assuming smart on-street charging	Commercial model
London Central	Secondary		**Timed connection** due to high capacity in overnight period
London Central	Primary		Some capacity in overnight period (limited by EV bus charging) so a combination of **timed connection and flexibility procurement** may be suitable
Brighton Central	Secondary		Overnight storage heating means timed connections are not feasible so **flexibility procurement** would be required
Brighton Central	Primary		**Timed connection** due to high capacity in overnight period

Note that different commercial models are optimal at the different voltage levels for both use cases – one solution might work for secondary but lead to constraints on primary, so the competing needs of the primary and secondary substations would need to be balanced in this case

Executive Summary

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Task 2: Analysis of Charging Behaviour + Substation Selection

Task 3: Assessment of Flexibility Opportunities

Task 4: Development of Use Cases and Commercial Models

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Different commercial models are applicable to the substations being studied

Task 4

Commercial models considered

- **Timed connection**: CPOs can apply to only load the network at specific times. Analysis of charging event data has shown that demand can be shifted so this option is feasible, however if there is insufficient headroom or vehicles are only plugged in during peak times then not all demand can be shifted in this way
  - **Need to consider the minimum viable scale:** While CPOs can benefit from a timed connection on a single charger, network constraint mitigation is only feasible if many chargers are connected on a timed basis. This should be assessed in area planning, in line with the Charge Collective overall approach
- **Flexibility procurement**: UK Power Networks may agree to connect CPOs even if this could lead to a breach in firm/rated capacity and procure flexibility services to create sufficient headroom. Requires active management of the network by the DNO and paying for flexibility services when load would otherwise exceed capacity, but can be used in cases where demand cannot be completely shifted away from peak times

These results could be used to complete a cost-benefit analysis of managing constraints using commercial models considered compared to upgrading network capacity

Commercial models recommended for each substation

Use case Substation archetype	Level	Commercial model
London Central **Urban, mostly domestic**	Secondary	**Timed connection** due to high capacity in overnight period
London Central **Urban, mostly domestic**	Primary	Some capacity in overnight period (limited by EV bus charging) so a combination of **timed connection and flexibility procurement** may be suitable
Brighton Central **Urban, mostly domestic, night storage heating**	Secondary	Overnight storage heating means timed connections are not feasible so **flexibility procurement** would be required
Brighton Central **Urban, mostly domestic**	Primary	**Timed connection** due to high capacity in overnight period

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Findings from this study can be used to inform further trials of on-street smart charging

Task 4

Recommendations from these findings

- **Assess ability to use on-street smart charging to shift load in the real world**
  - This could be done in the SmartSTEP trial[1]; even if shifting of load in the trial does not lead to mitigation of constraints this could prove its practical feasibility as a peak load reduction technique
  - Include assessment of the level of scale required/coordination of infrastructure needed to provide smart charging value
  - Further consumer research to understand what tariff is required in order for consumers to participate
  - The financial incentive needed for CPOs/aggregators to shift on-street charging load could also be assessed
  - Cost and effort of alternative options (e.g. flexibility procurement, active network management) could be assessed to determine possible cost savings from on-street smart charging
- **Further study of the London Central and Brighton Central use cases**
  - If expected substation upgrade costs are available, the value of avoiding constraints could be assessed

This Charge Collective work, combined with the SmartSTEP trial[1] will enable the value of flexibility services to be assessed from all stakeholder perspectives – see matrix on the right

Key Themes	Charge Collective	SmartSTEP Trial	Post-SmartSTEP
Consumer Acceptance	Researched	Will be trialled	*Should SmartSTEP be successful from a consumer and commercial perspective, the focus of future work will be on the technical aspects for wider rollout and adopting industry standards*
Opportunity for Peak Load Shift	Modelled	Can be analysed based on real data	
Financial Value (for DNO, CPO, Consumer)	Commercial models explored	Can be analysed based on real data	
Technical Feasibility[2]	No work completed	Aspects will be trialled	More work needed

1 – The SmartSTEP trial is a BEIS funded project, where 90 chargers deployed in Brent will be upgraded to a smart charging solution in Autumn 2021

2 – DNO communication to DCC

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Appendix

Glossary of Terms and Abbreviations

Glossary of Terms

SFS: Strategic Forecasting System, a load forecasting model currently being developed by Element Energy for UK Power Networks

DFES: Distribution Future Energy Scenarios, a set of scenarios developed by each DNO to represent growth of their network, in line with National Grid's Future Energy Scenarios

Smart charging: a form of EV charging where charging rate can be controlled to lower the load on the distribution network from charging at times of high electricity demand

V2G: Vehicle to Grid charging, where energy stored in a charging EV's battery can be provided to the distribution network at times of high electricity demand. Note: V2X refers to Vehicle to Everything, where the EV's battery may discharge to a home, building, or grid

Rebound peak: Peak in demand after avoided demand period

Abbreviations

DNO	Distribution network operator
EVCP	Electric vehicle charge point
EV	Electric vehicle
PHV	Private hire vehicle
I&C	Industrial and commercial
DFES	Distributed Future Energy Scenarios
CPO	Charge point operator

UK Power Network's Charge Collective project

- **Low provision of public charging infrastructure** has been identified as a significant barrier to growth in the EV market
  - Companies that invest in new charge points face barriers, including high up-front investment costs combined with market, policy and regulatory failures
- UK Power Network's Charge Collective project aims to develop a **framework to overcome barriers to investment in public charging infrastructure**
  - A **network solution** will be developed that cuts through **market, co-ordination, social** and **upfront connection cost challenges** for on-street charging
- The project will be completed in three steps, as detailed below
  - The research in this report forms part of **Step 3**

Step 1: Optimising charge point locations	**Step 2:** Trial an intervention to promote investment	**Step 3:** Research opportunities for flexibility services
Develop a framework to decide where charge points should be installed, and the level of support payments required to reduce upfront connection costs and incentive investments.	Design, road-test and evaluate a method for taking the outputs of this framework to tender the support payments to charge point investors.	Investigate opportunities for flexibility services from public charging infrastructure, with an opportunity to trial specific use cases if there is justification for them.

Source: UK Power Network's Charge Collective Project-on-a-Page summary

Questions asked during focus groups – willingness to smart charge

Task 1

1. **Would you be willing to smart charge at a public charge point? Why/why not?** *Open question, participants asked to discuss or write on the whiteboard*

2. **Tick the box if you would be willing to participate in smart charging at a public charge point, under the following circumstances:**
    - After plugging your car in at a public charge point, your car is automatically smart charged. You can't opt out.
    - After plugging your car in at a public charge point, your car is automatically smart charged, but you can choose to opt out.
    - After plugging your car in at a public charge point, your car does not automatically smart charge, but you can choose to opt in to participate in smart charging.
    - You personally choose to plug your car in to a public charge point at an off-peak time to make use of cheaper electricity.
    - None of the above

    Participants asked to respond to a poll

3. **Tick the box if you would be willing to smart charge at a public charge point, under the following circumstances:**
    - In most cases you were assured that your car would have the same charge at the end of a smart charging session as at the end of an unmanaged session
    - You were guaranteed that your car would have a minimum percentage charge by the end of your charging session
    - You were guaranteed that your car would charge to a minimum level before starting to participate in smart charging
    - You could not be guaranteed a minimum percentage charge by the end of your charging session
    - None of the above

    Participants asked to respond to a poll

Questions asked during focus groups – motivations and concerns

4. **Which of the following advantages of smart charging are important to you?**
   - I would like to use renewable energy in an optimal way
   - I would like to save money on the cost of charging
   - I would like to prevent a future increase in electricity prices (including domestic electricity)
   - I would like to contribute to a stable energy grid
   - Other (please discuss)
   - None of the above

   Participants asked to respond to a poll

5. **What about smart charging would potentially be a concern to you?**
   - I am concerned that my battery will not be fully charged when I want to leave
   - I am concerned that it will lead to a higher cost of charging
   - I am concerned that it will be too much hassle
   - I am concerned that it will lead to longer charging times
   - I am concerned about sharing my charging data
   - I am concerned that it would damage my car
   - Other (please discuss)
   - None of the above

   Participants asked to respond to a poll

Questions asked during focus groups – billing and information

Task 1

6. **What concerns do you have around payments or billing at smart public charge points?** *Open question, participants asked to discuss or write on the whiteboard*

7. **What information would you like to have access to while participating in public smart charging?**
    - Estimate of current state of charge of EV
    - Estimate of time at which EV will be fully charged
    - Current charging power
    - Calculated savings/rewards from smart charging participation
    - Quantification of renewable energy consumed
    - Other (please discuss)
    - None of the above

    Participants asked to respond to a poll

8. **When using a smart public charge point, what concerns would you have around sharing your charging data?** *Open question, participants asked to discuss or write on the whiteboard*

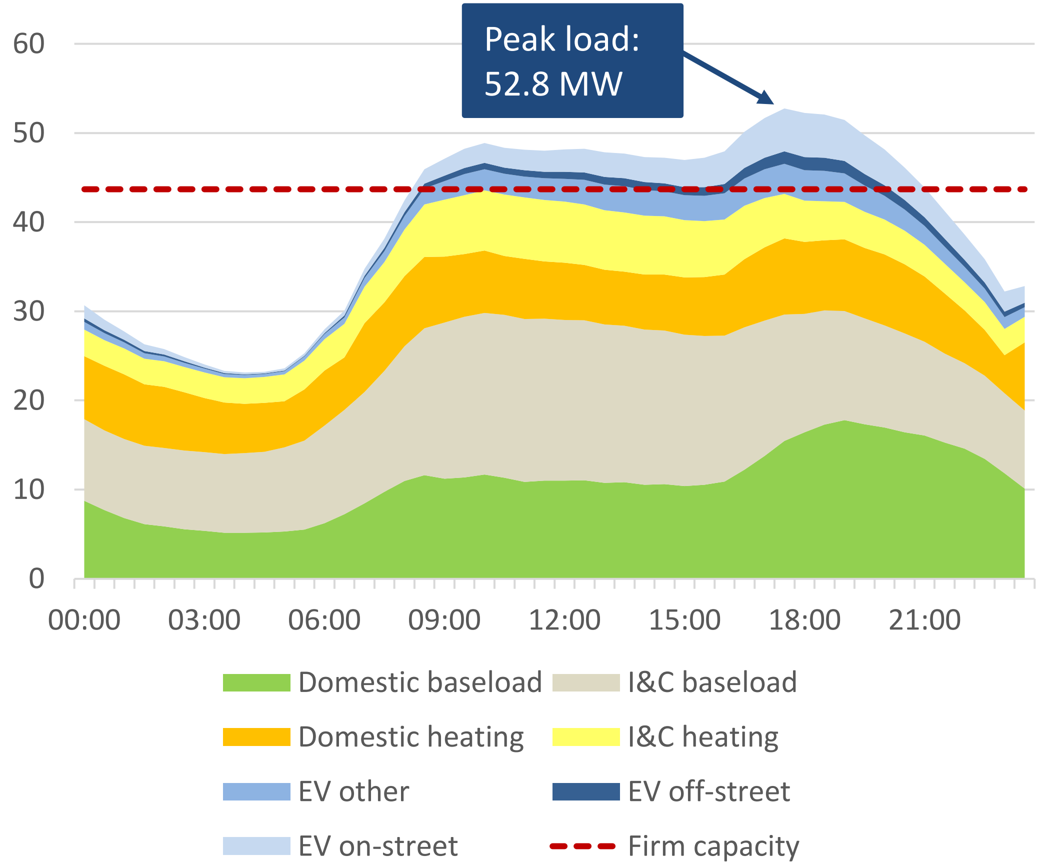
Brent primary
Urban, mostly I&C
Consumer Transformation scenario, 2030

Key Outputs Demand is dominated by large I&C loads

Task 3

Primary

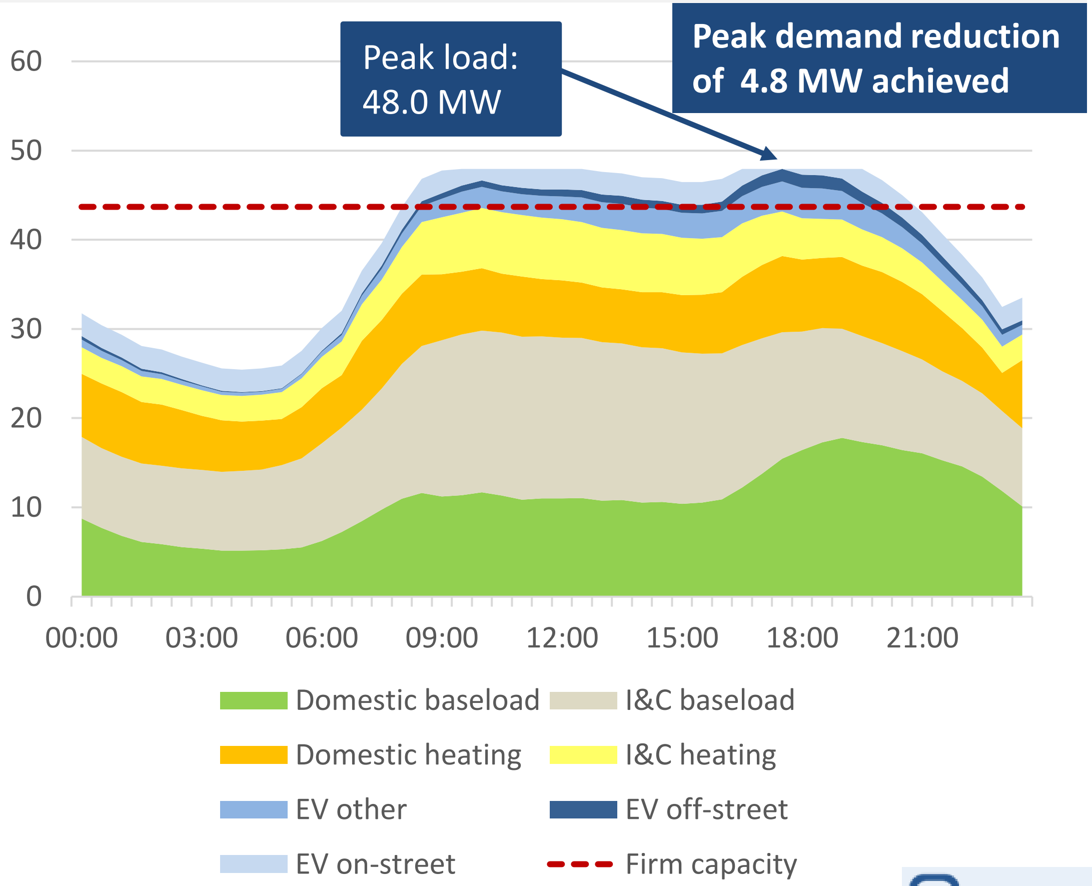
Demand profile with unmanaged on-street charging (MW)



Maximum unmanaged on-street charging demand: 4.9 MW

Firm capacity: 43.7 MVA

Demand profile with smart on-street charging (MW)



Maximum smart on-street charging demand: 2.5 MW → 360

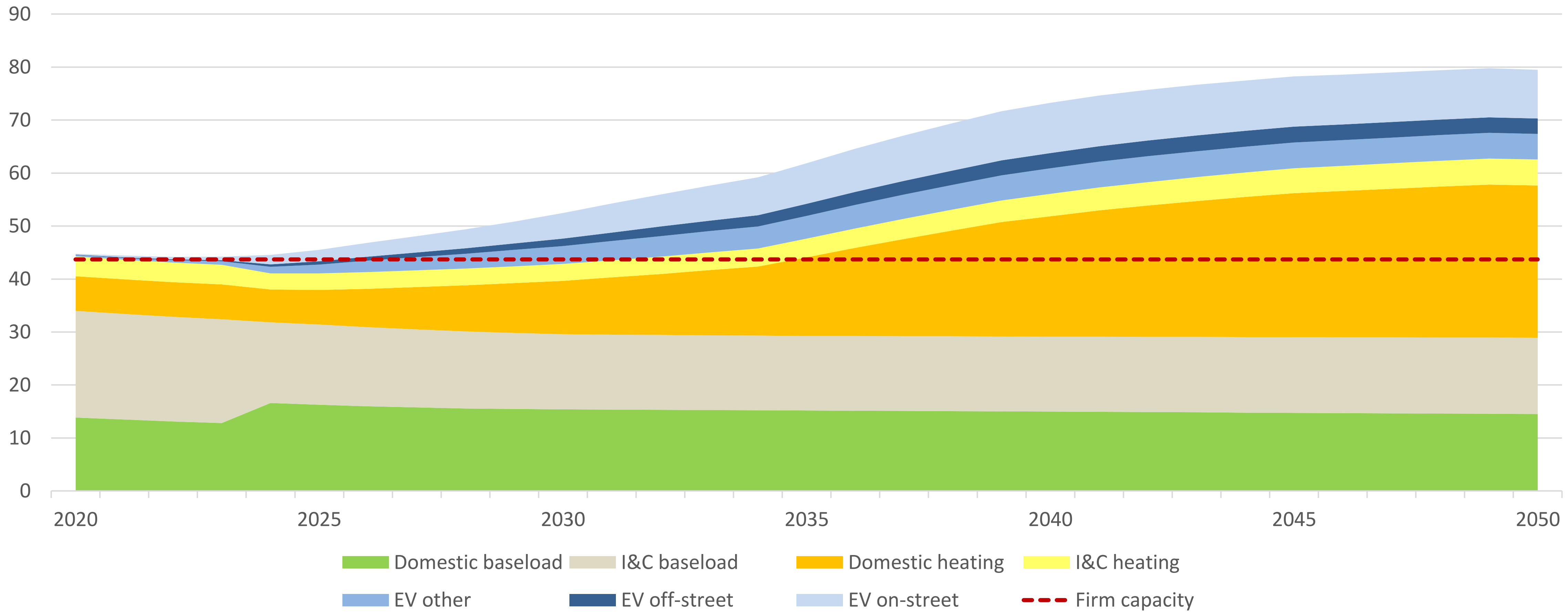
Brent Primary
Urban, mostly I&C
Consumer Transformation scenario

Key Outputs Low potential for on-street flexibility to mitigate constraints in the long term

Task 3

Primary

Peak winter annual load split by demand class (MW)



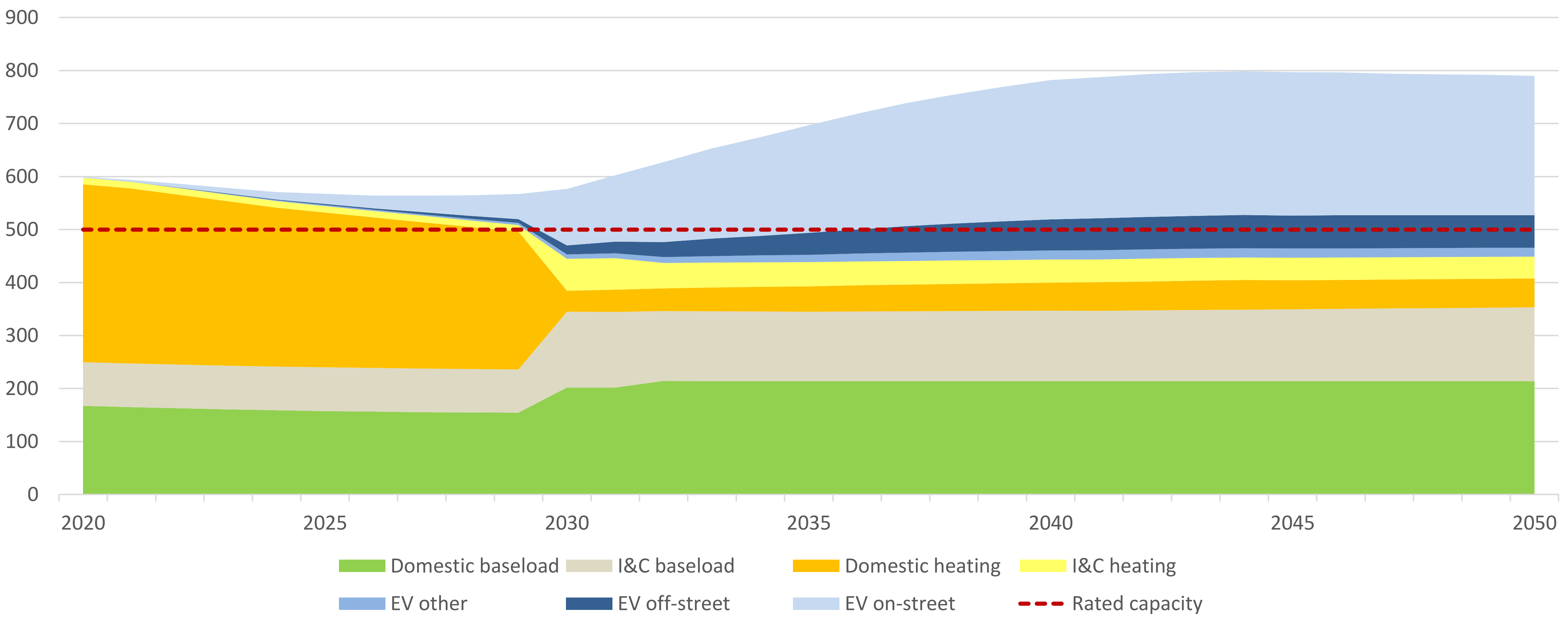
Brighton Central secondary
Urban, mostly domestic, night heating
System Transformation scenario

Key Outputs | **Near-term peak demand is dominated by night storage heating**

Task 3

Secondary

Peak winter annual load split by demand class (kW)



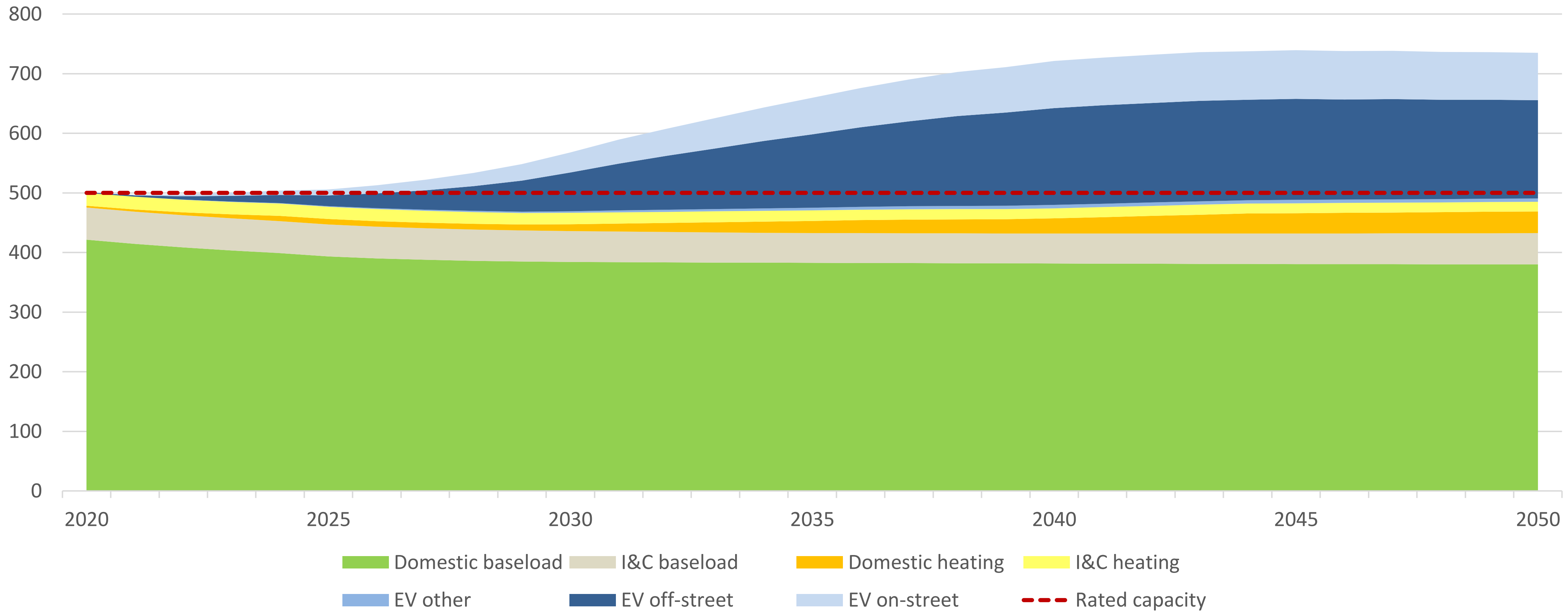
Brighton Suburban secondary
Suburban, mostly domestic
System Transformation scenario

Key Outputs Smart on-street and off-street charging may be able to mitigate constraints in the long term

Task 3

Secondary

Peak winter annual load split by demand class (kW)



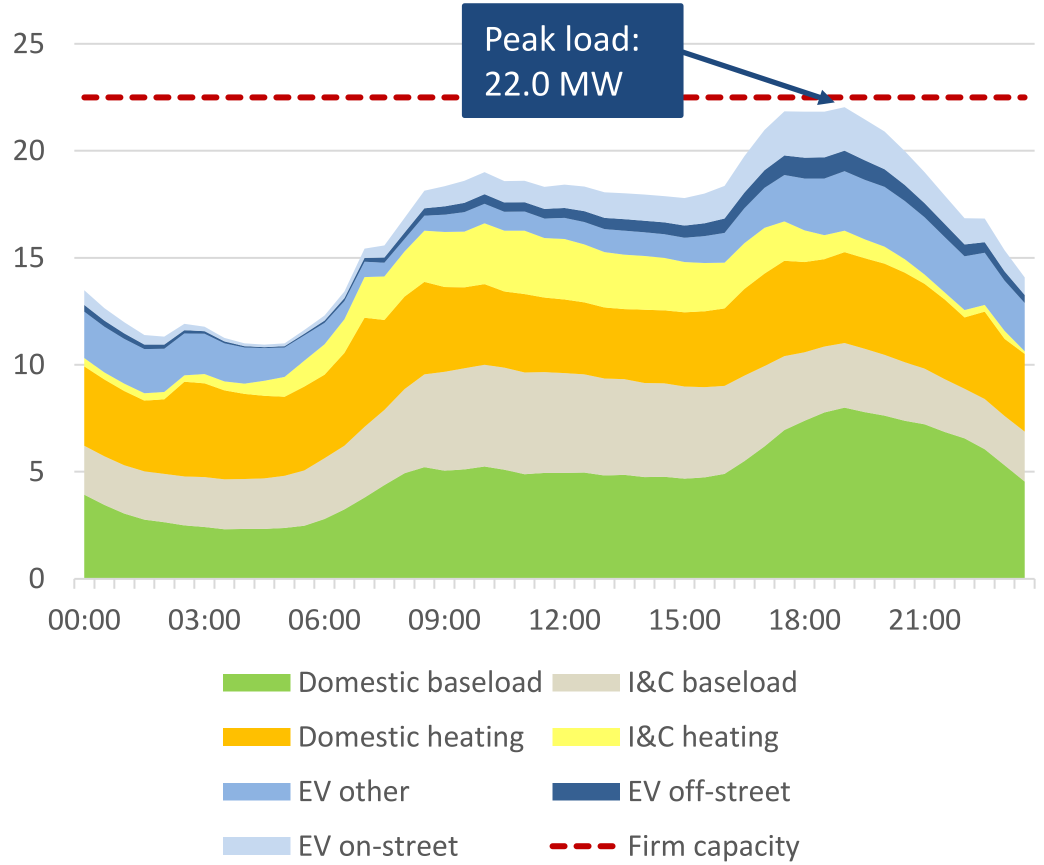
Brighton Suburban primary
Suburban, mostly domestic
Leading The Way scenario, 2030

Key Outputs EV bus demand partly responsible for evening peak – this may be flexible

Task 3

Primary

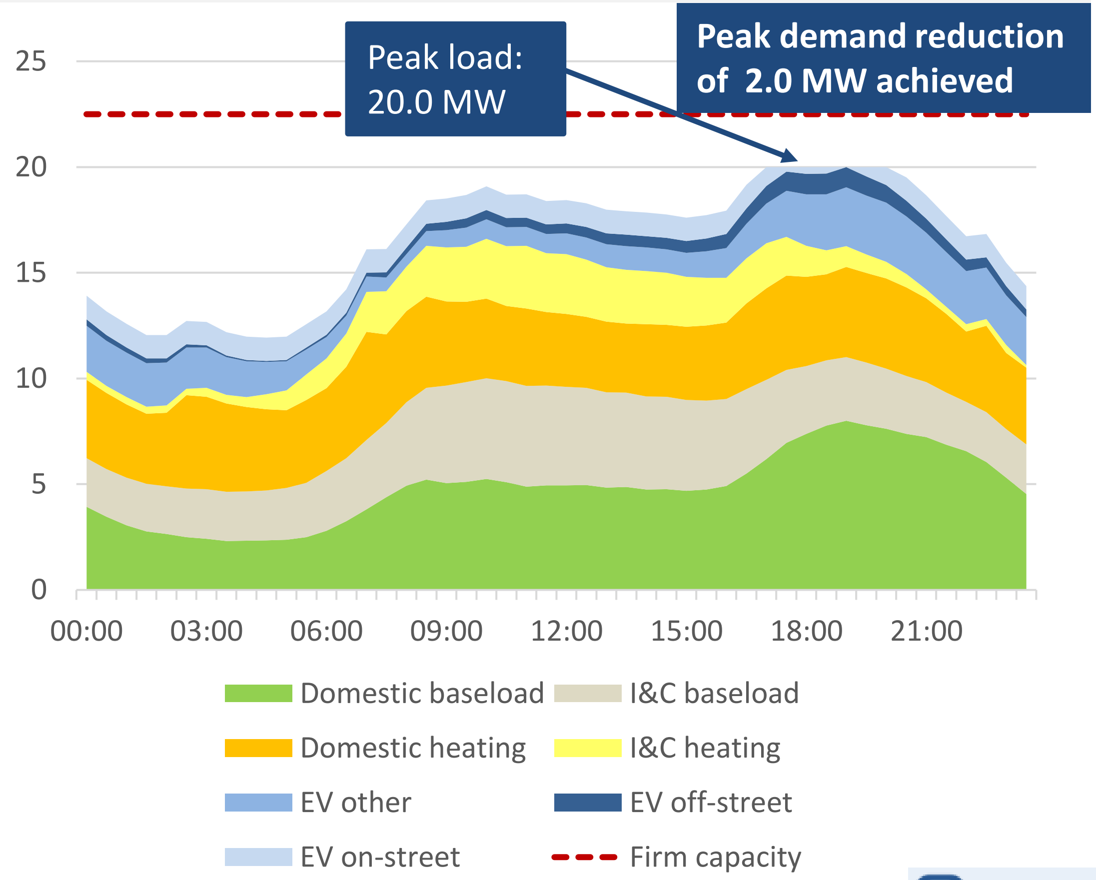
Demand profile with unmanaged on-street charging (MW)



Maximum unmanaged on-street charging demand: 2.1 MW

Firm capacity: 22.5 MVA

Demand profile with smart on-street charging (MW)



Maximum smart on-street charging demand: 1.1 MW

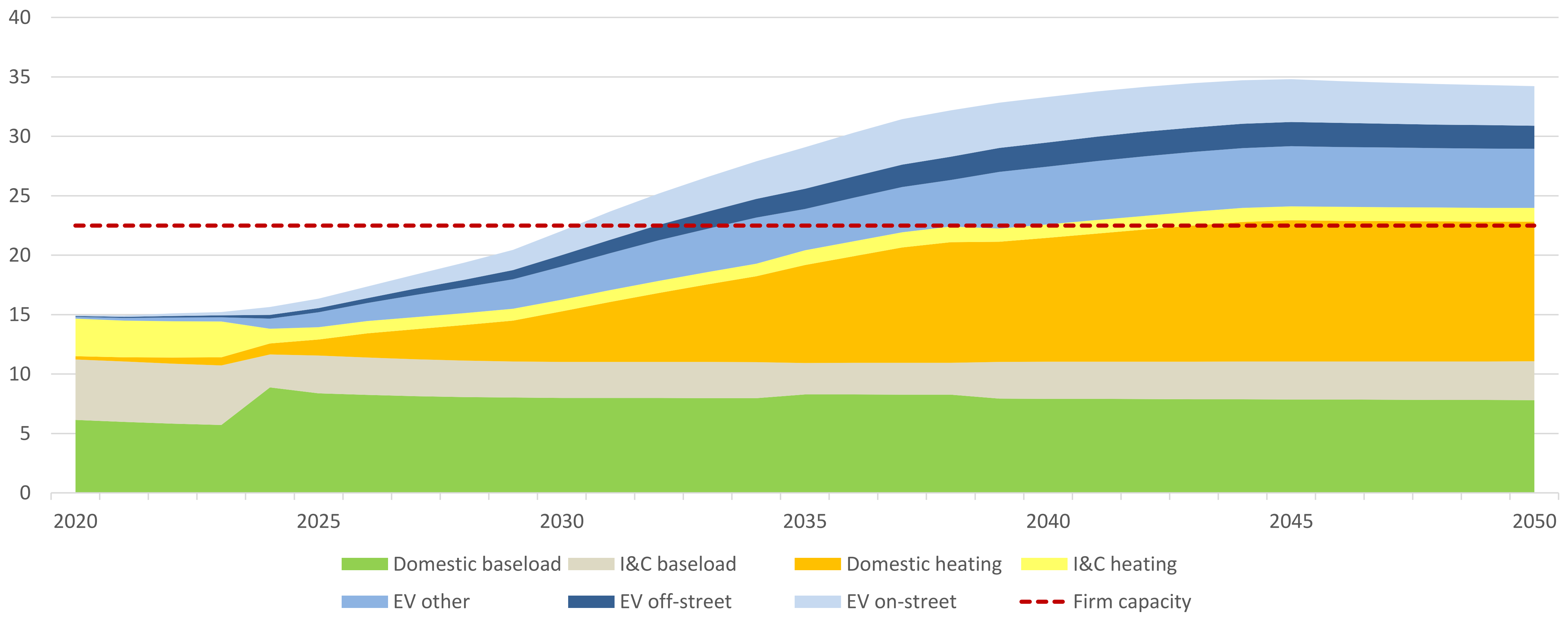
Brighton Suburban primary
Suburban, mostly domestic
Leading The Way scenario

Key Outputs On and off-street charging flexibility could be capable of mitigating constraints up to 2033

Task 3

Primary

Peak winter annual load split by demand class (MW)



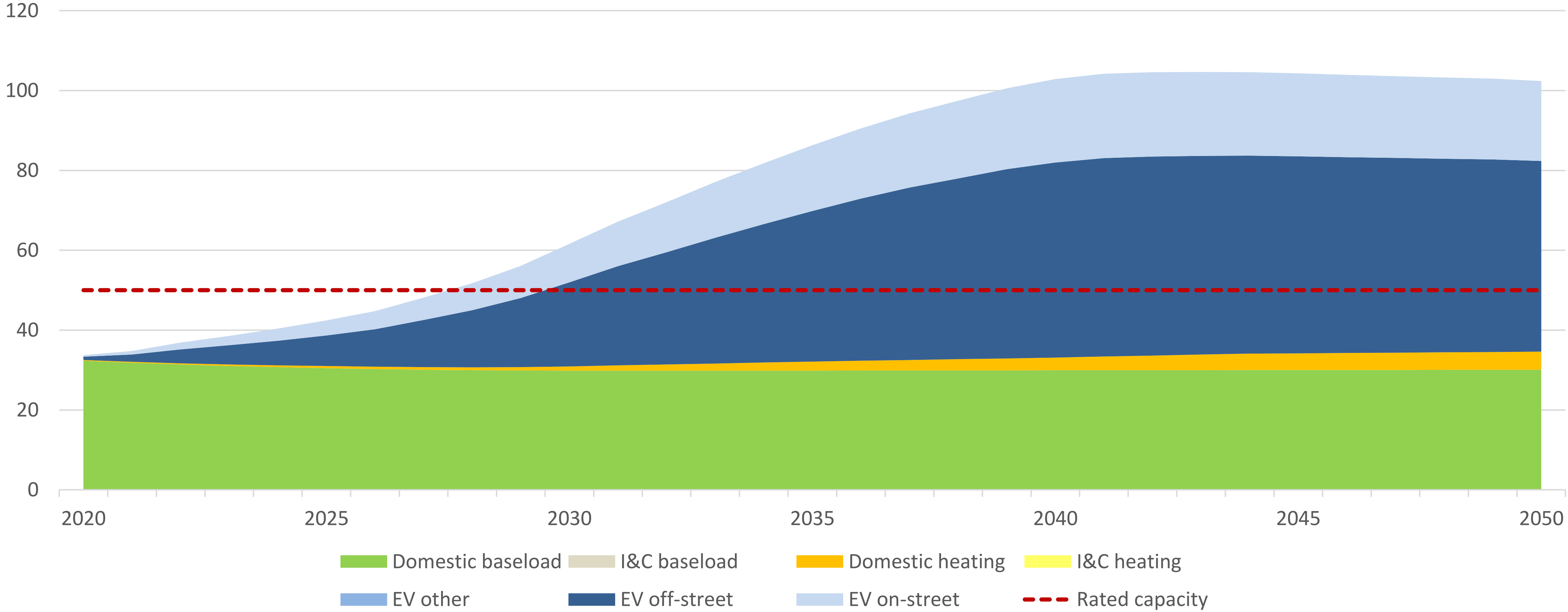
Rural secondary
Rural, mostly domestic
System Transformation scenario

Key Outputs Smart on-street and off-street charging may be able to mitigate constraints in the long term

Task 3

Secondary

Peak winter annual load split by demand class (kW)



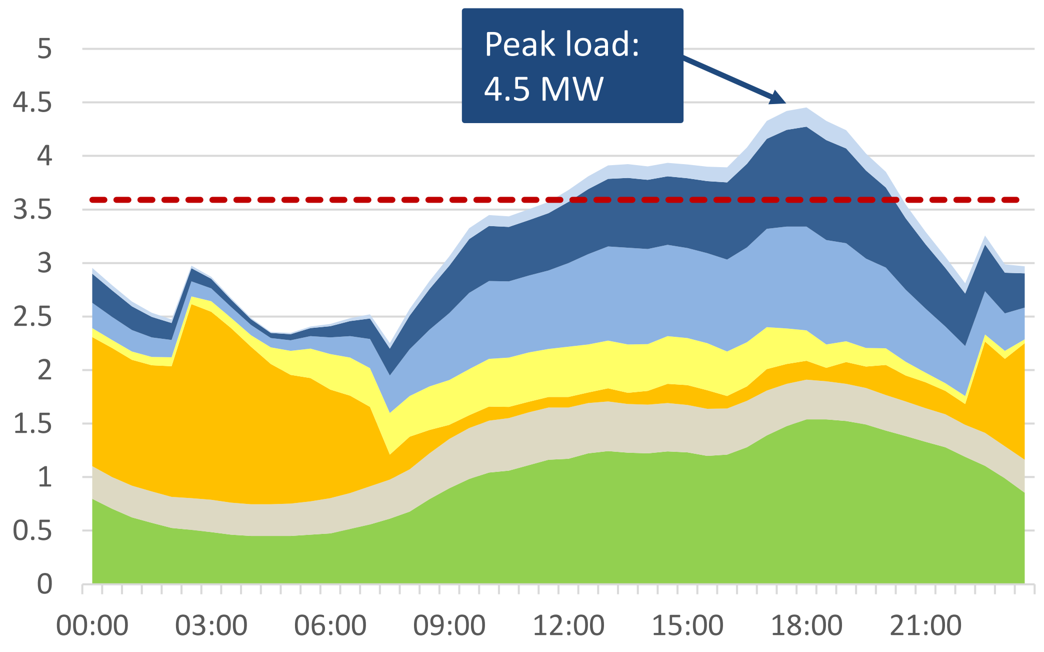
Rural primary
Rural, mostly domestic
System Transformation scenario, 2030

Key Outputs High share of off-street parking means there is much more off-street than on-street charging demand

Task 3

Primary

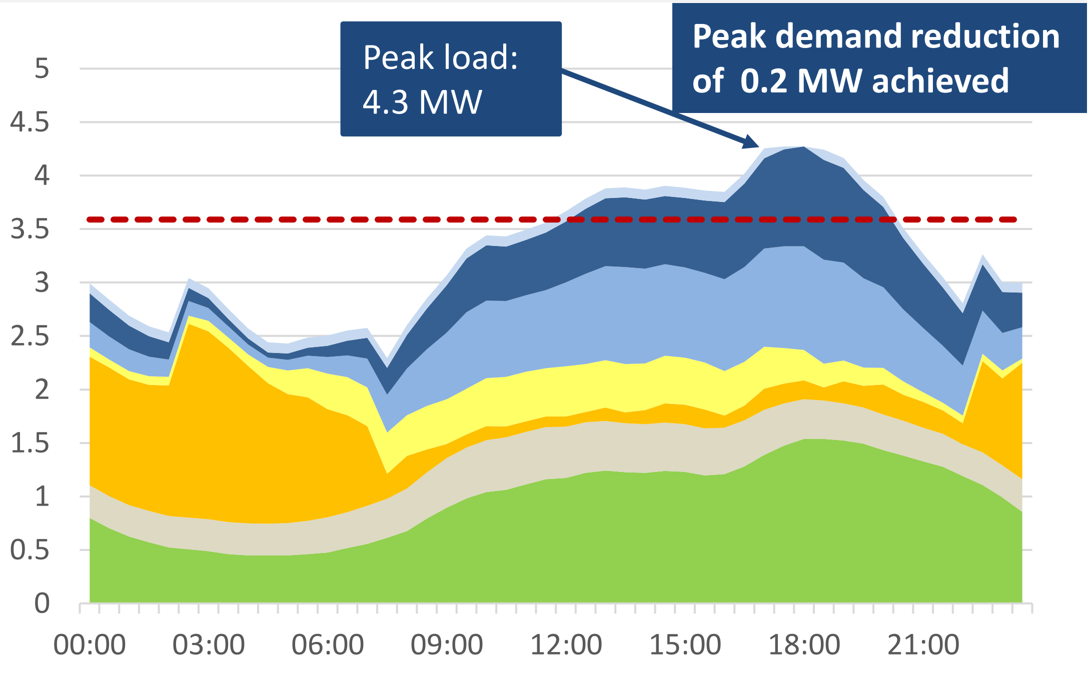
Demand profile with unmanaged on-street charging (MW)



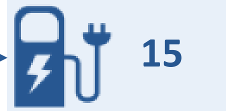
Maximum unmanaged on-street charging demand: 0.2 MW

Firm capacity: 3.6 MVA

Demand profile with smart on-street charging (MW)



Maximum smart on-street charging demand: 0.1 MW



Rural primary
Rural, mostly domestic
System Transformation scenario

Key Outputs On and off-street charging flexibility could be capable of mitigating constraints up to 2033

Task 3

Primary

Peak winter annual load split by demand class (MW)